DEVELOPMENT OF AN AUTOMATED OPTICAL DETECTOR FOR MICROPLASTICS IN ROCK SALT FOR RETAIL AND DOMESTIC USE

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Requirements for the Degree

Bachelor of Science in Electronics Engineering

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APPROVAL SHEET

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The Researchers

DEDICATION

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ABSTRACT

This study developed an automated optical detection system using artificial intelligence (AI) to identify, classify, and quantify microplastics in rock salt for retail and domestic use. Given the increasing health concerns regarding microplastic contamination in food, this research provides an accessible alternative to time-consuming laboratory methods. Utilizing the ADDIE (Analysis, Design, Development, Implementation, and Evaluation) model, the researchers engineered a prototype integrating a Raspberry Pi 5 processing unit, an optical imaging setup with controllable illumination (white and UV LEDs), and a YOLOv11 object detection model. The system evaluates rock salt samples to determine microplastic count, estimate particle size, and classify contamination levels.

To validate performance, the system's detection accuracy and processing speed were compared against conventional laboratory procedures using samples from various local sources. Additionally, system effectiveness covering functionality, usability, performance efficiency, and safety was assessed by users based on ISO/IEC 25010 standards. Findings revealed that the AI-integrated system successfully and consistently quantified microplastics, achieving a mean detection accuracy of 67.89% compared to laboratory results, with near real-time processing averaging 18 milliseconds per sample.

The system demonstrated a balanced trade-off between precision (82.6%) and recall (73.3%) during model training. Furthermore, user evaluations yielded a "Highly Acceptable" rating across all ISO/IEC 25010 criteria, confirming the system's operational capability and user-friendliness. The developed automated optical detector serves as a highly efficient, reliable, and viable tool for the preliminary screening of microplastics in commercial rock salt. By minimizing reliance on manual microscopic inspection, this AI-driven approach significantly improves food safety monitoring. Future enhancements recommend expanding the AI dataset and conducting broader laboratory validations to further refine detection accuracy.

Keywords: Microplastics, Artificial Intelligence, Optical Detection, Rock Salt, YOLOv11

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter consists of the background of the study, statement of the problem, significance of the study, and the scope and delimitations of the study.

BACKGROUND OF THE STUDY

Microplastics are tiny bits of plastics ranging from one micrometer to five millimeters that originate either from the fragments of larger plastics or the intentional made of microscopic beads. Because of their size and does not breakdown in nature, they stay in the environment for a long period and eventually build up in oceans, on land, and even in food consumed by humans. According to recent studies, microplastics have been found in all sorts of food, including bottled water and salt. Rock salt has become a problem because its natural extraction from nature makes it more prone during the harvest, process, and the package itself. Detecting microplastics is not easy because microplastics are frequently colorless, transparent, and visually like salt crystals, making microscopic detection harder (Kim et al., 2018).

In recent scientific studies have found out that people can eat up to five grams of microplastics per week because of contaminated food, water, and air. This kind of exposure has been associated with inflammation, oxidative stress, immune system disruption, DNA damage, metabolic issues, neurotoxicity, and reproductive and developmental toxicity, raising an urgent concern about long-term health risks such as organ damage and disease proneness (Li et al., 2023).

This concern increases because of the country's environmental conditions; the Philippines produces of plastic waste annually with an estimated 2.7 million tons, and it results in most of the contributors to the marine plastic pollution (Plastic Education, 2025). Because of this, it rises the probability that salt produced in the Philippines, particularly rock salt, will be put at risk to microplastics during the harvest, production, and in transport.

Even confirming globally that the salt has microplastics, and still the country still lacks an affordable optical detection system that integrates with AI capable of detecting microplastic particles from salt crystals under controlled imaging processes. The present lab techniques still remain inaccessible for local researchers, and limited national data comparing microplastic existence to commonly bought salt types or origin. This gap underscores the need for an effective detection tool and a comparative assessment of the nation's widely used salts.

This research aims at creating an automated rock salt optical detector with artificial intelligence (AI) that can detect rock salt's microplastic content. The system aims to employ image processing to detect, and quantify rock salt's microplastic particles from physical properties. The artificial intelligence is going to inspect size and microplastic counts from the images taken through the camera. In this research, the researchers aligned it with United Nation Sustainable Development Goals (SDG 3: Good Health and Well-being, SDG 9: Industry, Innovation, and Infrastructure, and SDG 12: Responsible Consumption and Production).

COMMUNITY PROFILE

Every Filipino home uses salt daily for food preparation, cooking, and preservation, regardless of income level. Only roughly 33.2% of households consume enough iodized salt (salt with ≥ 15 ppm iodine), even though 64.9% of households are aware of iodized salt and 55.7% report using it, according to the 2021 DOST-FNRI Expanded National Nutrition Survey (ENNS). (FNRI-DOST, 2023)

Many homes continue to use unpackaged salts like coarse rock salt or locally made salts despite the push for iodized salt. 52.2% of households report using rock salt, 37.9% use fine salt, and 9.9% use both rock and fine salt, according to the same national dataset. (FNRI-DOST, 2023)

Consumers buy salt from a variety of retail establishments. While fine, commercially packaged iodized salt is typically found in supermarkets or grocery stores, rock salt is typically found in public markets, talipapa, and sari-sari stores. (DOST-FNRI, 2023) This indicates that salt purchasing patterns involve both traditional and commercial supply chains, but many households purchase salt in small quantities, without really caring about the brand, source or origin, process, or packaging integrity.

Due to the variety of salt types (rock salt, fine salt, local coarse salt, and commercially packaged iodized salt) and sources (supermarkets, sari-sari stores, public markets, talipapa, and local producers), salt consumption in the nation varies greatly. This variety is critical for researchers investigating microplastic contamination of salt. To accurately describe microplastic exposure in Filipino homes, researchers should collect data from various popular types and sources, not just a few.

The study's initial survey results support these national conclusions. The overwhelming majority of respondents stated that they purchase salt solely on the basis of accessibility, affordability, and availability rather than source purity or brand credibility. Furthermore, a large number of consumers were unaware that the salt might contain microplastics, demonstrating a general lack of knowledge about this kind of contamination. This trend aligns with previous local observations, indicating that household salt consumption is habitual and driven by environmental factors rather than safety concerns. Consuming salt on a regular basis may have a non-invasive impact on microplastic exposure.

Since everyone consumes salt on a daily basis and most households don't think about where it comes from or how it is made, it's possible that tiny amounts of microplastics are being exposed through salt (though this is frequently undetected). Such reality makes our study more relevant and urgent. Data from commonly bought salt types, along with the addition of an automated microplastic detection system, will allow the researcher to establish information on risks of salt consumption. The results of this study can potentially assist consumers in informing them, improve public policy for public health and safety, and contribute to safer and more intelligent purchasing decisions for Filipino households.

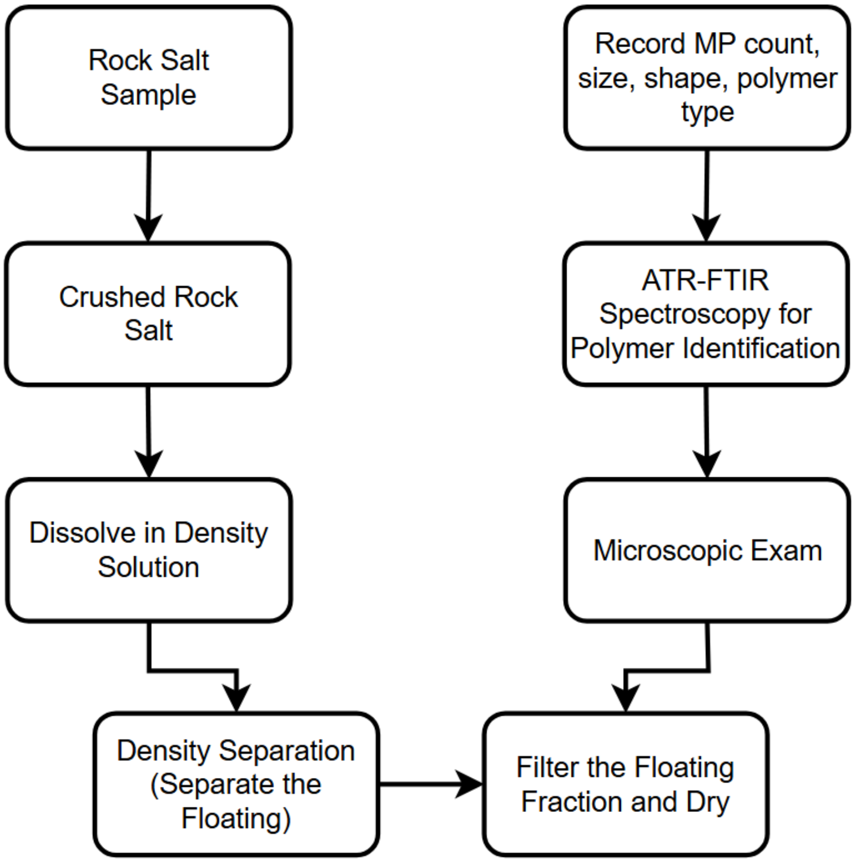

Figure 1.1 Laboratory Detecting Method For Microplastics In Rock Salt

In Figure 1.1. Crutchett, T. W., & Bornt, K. R. (2024): The first step is you need a rock salt sample and turn it into smaller pieces. Once you turn it into smaller pieces, you need a density solution for you to dissolve and at the same time separate those tiny bits that are floating, usually the microplastics. Those particles will filter out and dried before being put to microscopic examination.

In other laboratories, there is an optional step for them to confirm after the microscopic examination. The process of it is they will use hot needle or so-called ATR-FTIR spectroscopy for them to know the properties of the polymer that they are dealing. Lastly, the technician will take note of the data gathered like microplastic count, size, shape, and the classification of polymer based on the findings.

Even so, this lab technique causes challenges that provides to the general problem of the current study. Distinguishing microplastic visually or manually is hard and not only that, but it also takes a lot of time and money, and it is dependent on experts. Relying to the human judgment can cause errors and inconsistencies and it may lead to inaccuracy and the same time repeatability. Because of these challenges, the researchers suggest for a better and fully automated optical detection system integrated with AI and advanced image processing techniques to classify and quantify microplastics in rock salt with a better accuracy, speed, and reliability.

CONCEPTUAL FRAMEWORK

INPUT	PROCESS	OUTPUT
• Rock salt sample • Reference microplastic images • Existing analysis procedure • Expert guidance • Questionnaire • Respondent	• Analyze existing detection methods • Review of existing microplastic detection methods • Asses weakness in current methods • Identification of gaps and needs for an automated detection method	• Identified limitation of current manual microplastic detection methods • Defined needs and requirements for an automated detection approach • Refined and focused problem statement for the proposed study

Figure 1.2. Conceptual Model of the Needs Assessment of the Automated Optical Detector for Microplastics in Rock Salt Using AI

The conceptual framework of this study follows the input-process-output (IPO) model. The first is the input, it includes the rock salt samples, microplastic images, existing analysis procedure, advice from the expert, questionnaire, and respondents. These inputs will be used to go through a series of process. In the process, these are the analyzed and review of the existing detection method, identifying gaps and needs for the automated detection system. Because of these processes, the researchers were able to produce an output like the identified limitations of manual detection method, defined needs and requirements for automated approach, and lastly a refined and focused problem statement that will serve as the backbone of the proposed system.

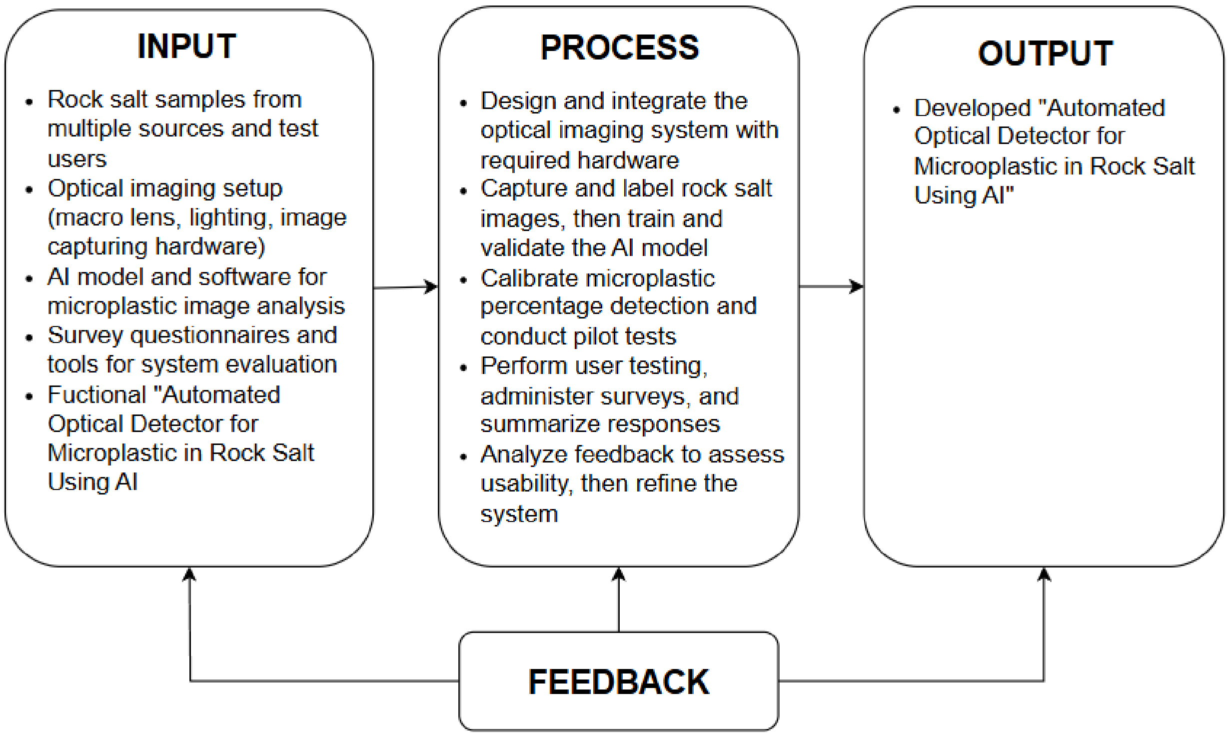

Figure 1.3. Conceptual Model of the Development of Automated Optical Detector for Microplastic in Rock Salt Using AI

The conceptual framework of this study follows the input-process-output (IPO) model. The model starts with an input, this includes the rock salt sample from different sources, optical imaging set up, AI model and software for microplastic image analysis, survey questionnaires and tools for system evaluation, and the functional propose system. These inputs are necessary because it sets the foundation, these are the resources, data, samples, and materials that will be needed to produce valid and reliable output.

Next is the process, the first process is to design and integrate the optical imaging system with the required hardware to make a decent image acquisition of the rock salt. After that, the researchers capture and name rock salt images and use it to train and validate the AI model for microplastic detection. Next, once the system is up and ready, to inspect its performance on how accurate it quantifies microplastic it needs to go through in calibration mode and initial testing. Then, perform a user testing, administer surveys, and summarize its responses. Finally, analyze feedback to assess usability, accuracy, and acceptability. If improvements are needed, refine the system accordingly. The study's output is the "Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use," which aims to detect microplastics in rock salt samples efficiently and reliably.

The framework has a feedback component that allows the system to be continuously improved by using the evaluations and results from the output to improve the input and process.

GENERAL PROBLEM

The main focus of this study is the absence of a precise, dependable, and effective method for identifying microplastics in rock salt. The majority of laboratory techniques used today are costly, time-consuming, and require skilled workers with a high degree of manual inspection, which can lead to human error and produce results that are not repeatable. Considering these disadvantages, developing a novel and fully automated optical detection system integrated into various advanced artificial intelligence and image processing Technologies are urgently needed to identify, classify, and analyze microplastics in rock salt samples with speed, precision, and consistency.

STATEMENT OF THE PROBLEM

The issue of increasing prevalence of microplastic contamination worldwide in various food products has become a growing environmental and health concern. Rock salt, a common ingredient in industrial and household use, is not exempt from this issue. It aims to answer the following questions:

1. How can an automated optical detection system integrate with artificial intelligence be developed to accurately identify microplastic in rock salt samples?
2. What are the recorded data that resulted from image processing and AI-based classification techniques for effective detection of microplastic that are utilized to analyze particles such as:
    a. Microplastic Count, and
    b. Size
3. How is the system more efficient compared with the laboratory procedures for microplastic analysis in terms of:
    a. Microplastic Count, and
    b. Size

4. How effective is the developed system based on the evaluation from the respondents based on ISO 25010 evaluation instrument results?

4.1 Functionality;

4.2 Usability;

4.3 Performance Efficiency; and

4.4 Safety and Maintenance.

OBJECTIVE OF THE STUDY

The objective of the study is to develop and implement of an automated optical detection system with the ability to identify microplastics in rock salt using artificial intelligence. This will ensure efficiency and accuracy in the identification and classification of microplastic particles through image analysis and hence minimize manual observation under microscopes. Specifically, the study aims to achieve the following goals:

1. To develop an automated optical detection system integrated with artificial intelligence to accurately identify microplastics in rock salt samples.

2. To determine the recorded data resulting from image processing and AI-based classification techniques for the effective detection of microplastics utilized to analyze particles such as:

   a. Microplastic Count, and

   b. Size

3. To compare the efficiency of the developed system with laboratory procedures for microplastic analysis in terms of:

    a. Microplastic Count, and

    b. Size

4. To evaluate the effectiveness of the developed system based on the evaluation from the respondents using the ISO 25010 evaluation instrument results in terms of:

    4.1 Functionality;

    4.2 Usability;

    4.3 Performance Efficiency; and

    4.4 Safety and Maintenance.

SCOPE AND DELIMITATIONS OF THE STUDY

This research focuses solely on the "Development of an Automated Optical Detector for Microplastic in Rock Salt for Retail and Domestic Use" using AI and image processing technology.

The following claims, according to the researchers, must be met.

1. **Compared to the Laboratory Method:** The study will assess the performance of an AI-based optical detector in detecting microplastics in rock salt, evaluating its precision, accuracy, and efficiency to the standard laboratory method.
2. **Evaluation of the System:** The study will assess how effective, reliable, and consistent the AI-based system is compared to laboratory methods, with a focus on its ability to deliver accurate and repeatable results.
3. The project has the following features:

    3.1 **AI Implementation:** AI Algorithms are used to analyze images and find microplastics in rock salt by looking at their visual features.

    3.2 **Image Processing Technology:** High resolution pictures of rock salt samples will be captured using optical imaging, and AI will analyze these images to detect and measure microplastics.

    3.3. **Automatic:** Capable to process the rock salt samples in real time to detect microplastics with minimal to no human interaction.

    3.4. **Real-Time Data Display:** Screen will be able display the data gathered once detection has been completed in rock salt samples.

This study is limited to:

1. **Focus on Rock Salt Only:** This study will focus on rock salt samples only and no other types of foods or materials that might also be impacted by microplastics will be cover.
2. **Detection through Imaging:** The system uses optical imaging assist with AI to detect microplastics. It only focuses on visual detection, meaning that other technique like chemical analysis or spectroscopy are not included in this study.
3. **What the AI looks for:** The AI is only trained to recognize microplastics by their appearance, specifically their size and count, and other microplastics that do not clearly show these visual features might not be detected in the system.
4. **Training Data Limitations:** The AI system was trained using images of microplastics specifically found in rock salt only, so it might not work on samples that look different or come from other sources.
5. **Not Yet Industrial Scale:** Although the goal is full automation and real-time processing, the study is focused on building and testing a working prototype. More work and testing would be needed before the system.

SIGNIFICANCE OF THE STUDY

This project was developed to respond to growing concerns over salt contamination with microplastics. This has been achieved through developing an Automated Optical Detection System for Microplastics, which includes camera-based sensing and uses artificial intelligence (AI).

Furthermore, this study promotes food safety and environmental sustainability through innovation in detection technology. It will be of significant help to the following:

Community.

This project will help raise awareness about microplastic contaminants in common foods that people eat every day (e.g., salt), and how a low-cost, quick way of detecting microplastics could support community health and provide consumers with contaminant-free, safe products.

Researchers.

As a basis for future development, this project will allow researchers to improve on and adapt the model used for detecting microplastics in several types of food; as well as the dataset generated from this project and the prototype created during this research.

Manufacturers and Food Safety Authorities.

This project supports salt manufacturers and regulatory bodies (food safety agencies) in assessing their product's quality. The device is intended as a first screening device before the product reaches the marketplace and helps protect against potential harm from ingesting microplastics and ensures adherence to all applicable health and safety regulations.

Engineers and Technicians.

This project represents a good example of integrating machine learning (ML) with optical sensors and provides a useful resource for engineers and technicians looking to create smart monitoring devices.

Environmental Advocates and Policy Makers.

This project contributes to the larger effort to preserve our environment globally and aligns with the United Nations' Sustainable Development Goals (SDG 3: Good Health and Well-being, SDG 9: Industry, Innovation, and Infrastructure, and SDG 12: Responsible Consumption and Production). The project also encourages sustainable practices and raises awareness of the impact of microplastic pollutants.

DEFINITION OF TERMS

Artificial Intelligence (AI): A sub-specialty of computer science that allows machines to carry out human-level computation, such as pattern recognition, classification and decision making. AI is used to classify microplastics based on visual characteristics such as size and count.

Automated Optical Detection System: It is a system that utilizes cameras, optical lenses, and image-based sensors to automatically capture, process, and analyze images on its own. Instead of manual microscopic detection, artificial intelligence and image processing techniques are employed in this study based on an automated optical detection system for identifying microplastics in rock salt samples.

Contamination Level / Percentage Detection: It is the measured amount of microplastic particles shown compared to the total rock salt sample. This percentage is computed automatically to understand how much contamination is present in rock salt.

Density Separation: This is a laboratory method where you isolate microplastics from salt by dissolving samples in a solution. This method was used in previous studies but replaced in this research by a faster AI-based optical method.

Digital Microscope/Macro Imaging Module: It is a camera or lens capable of capturing intense zoom images of small objects. It enables clear visualization of microplastics that are otherwise too tiny to see using standard cameras.

Fourier Transform Infrared Spectroscopy (FTIR): A laboratory technique used to identify what microplastic are made of. Although reliable, FTIR is time-consuming and not included in the automated system developed in this study.

Image Processing: A method of converting images into digital data for enhancement, detection, segmentation, and analysis. This study use image processing to identify microplastic particles from captured images by determining their size and count.

Microplastics: These are plastic smaller than 5 millimeters in size, formed when larger plastic materials are destroyed or from microbeads used to make plastics. They are usually invisible to the naked eye, and can contaminate salt, water, air, and food products. This paper aims at characterizing microplastics in rock salt with optical detection.

Object Classification: A process in AI where objects within an image are classified based on their respective characteristics. This research classifies particles in rock salt as microplastics or non-microplastics.

Optical Imaging: This takes advantage of light and lenses to obtain very clear pictures of small objects along with their detail. This serves as the basis of detection, enabling the camera to disclose microplastics in salt samples.

Prototype: This provides the first working model of the automated optical detection system designed for testing and improvement. It includes hardware, software, imaging, and AI components.

Raspberry Pi 5: It is the small single-board computer running the AI model and optical detection system. It is the main processing part that takes images, processes images, implements AI classification, and outputs detection results.

Rock Salt: A naturally occurring, unrefined form of salt typically harvested from underground deposits. Because rock salt undergoes minimal processing, it is more prone to contamination during extraction, handling, and packaging, making it a relevant sample for microplastic detection in this study.

Validation: A method used to contrast the system-created images with lab-measured ones to establish that they are correct, reliable, and consistent.

YOLOv11 (You Only Look Once Version 11): A deep learning-based object detection model that can quickly detect small objects. In this work, microplastic detection will be accelerated using YOLOv11, which uses a real-time recognition of particulate elements in the salt sample.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDY

This chapter provides a comprehensive review of relevant literature and previous studies to introduce the present study. The researchers gathered distinctive articles from various sources to improve the understanding of the presence and detection of microplastics in salt products and their connections for human health. These collected studies offer a wide range of insights into the subject matter.

Related Literature

EMERGING MICROPLASTIC CONTAMINATION IN THE FOOD INDUSTRY: THE CASE OF COMMERCIAL TABLE SALTS IN ILIGAN, PHILIPPINES

Capangpangan, R. Y., et al. (2024) Table salts within the food industry have become a serious concern because of the microplastic contamination in the manufacturing and distribution of commercial table salts. The research further contributed because the salt found in every food, there is a higher chance that individuals will consume sea salt containing microplastics. The researchers found out that the human body can absorb polycaproamide and polypropylene, through eating salt. They also highlighted that poor packaging and processing conditions on food processing has negative affect. These results support the need to develop a means of monitoring the risk of microplastic contamination in food salts and the increasing concern regarding its occurrence in the food chain.

MICROPLASTICS CONTAMINATION IN SELECTED STAPLE CONSUMER FOOD PRODUCTS

According to Amparado, A. F. V., et al. (2024), in table salt, fish, and rice microplastics are now present in these foods, this indicates the rampant plastic pollution in the food supply chain of the human. The researchers discovered the reasons that introduce microplastics into edible products, it includes production, transportation, and packaging. Due to the exposure to seawater and airborne human caused particles during processing, salt products have the higher chance to contaminate, specifically rock salt. For product safety and to enable consumers to make a wise decision, the study highlights the need to implement advanced detection systems, including optical imaging and spectroscopic measurement.

ASSESSMENT OF MICROPLASTICS IN HIGHLAND ROCK SALTS

In accordance with Viswanathan, P. M., et al. (2024), the researchers reported that even mineral based salt sources remain at risk to pollution because of the existence of microplastics in rock salts from onshore and highland areas. The results of their study showed microplastic particles measuring 20 to 200 micrometers, with fibers and films being the dominant forms. The study linked the contamination to improper waste management, atmospheric fallout, and salt processing packaging. The researchers note that the microplastics may also serve as a vehicle for chemical additives that pose long-term risks to human health. To enhance contamination tracking, they suggested systematic monitoring using AI-assisted detection.

A REVIEW OF MICROPLASTICS IN TABLE SALT, DRINKING WATER, AND AIR: DIRECT HUMAN EXPOSURE

. According to Zhang, Q., et al. (2020), microplastic exposure is a global issue because it occurs in critical resources such as table salt, drinking water, and air. The review found that microplastics smaller than 150 micrometers can pass through conventional filtration systems and enter the human body via ingestion and inhalation. The authors believe that the accumulation of such particles can cause chronic toxicity and inflammation. The researchers discovered that sea and rock salts carry plastic pollutants from contaminated marine environments. They recommended ongoing development in analysis detection and public education about microplastic sources in order to reduce exposure.

MICROPLASTIC POLLUTION: A GLOBAL ENVIRONMENTAL CRISIS IMPACTING MARINE LIFE, HUMAN HEALTH, AND POTENTIAL INNOVATIVE SUSTAINABLE SOLUTIONS

According to Karak, P., et al. (2025), the global spread of microplastic pollution constitutes an environmental and health emergency. Their research revealed that plastics degrade into tiny pieces that settle in the sea and eventually find their way into seawater-derived products like table salt. The authors explained how consuming microplastics can expose humans to harmful chemicals such as bisphenol A and phthalates. They proposed the use of innovative technologies, such as artificial intelligence and autonomous optical detection, as long-term approaches to detecting and preventing microplastic contamination in consumables.

THE IMPACT OF MICROPLASTICS IN FOOD AND DRUGS ON HUMAN HEALTH: A REVIEW OF THE MENA REGION

Alziny et al. (2025) examined how microplastics entered food and medications throughout the Middle East and North Africa (MENA) and could have negative effects on the heart, digestive system, and hormones. Due to inadequate monitoring and a lack of detection technologies, microplastics frequently enter the body through common foods like salt. This growing concern about human health highlights the importance of developing better tools for detecting these particles. In this regard, the current research focuses on developing an AI optical detector for microplastics in rock salt, which will provide a faster, more accessible, and non-destructive method for detecting contamination in one of the most commonly used food ingredients.

MICROPLASTIC CONTAMINATION OF SALT INTENDED FOR HUMAN CONSUMPTION: A SYSTEMATIC REVIEW AND META-ANALYSIS.

Danopoulos, Jenner, and Rotchell (2020) conducted a systematic review and meta-analysis of microplastic contamination in human-grade salt. The study emphasizes that salt, particularly sea salt, is a significant vector for introducing marine microplastics into the human diet, and it collects data from around the world to establish baseline contamination levels. This study demonstrates the direct connection between food safety and marine pollution and emphasizes how widespread microplastic contamination is. The study looks at the amount and form of microplastics in a variety of salt types, such as sea, lake, and rock salt. The results demonstrate that sea salt concentrations are consistently higher when directly extracted from contaminated marine environments. Furthermore, quantitative analysis provides important statistical evidence of the widespread presence of microplastics by revealing potential human exposure through salt consumption. This literature, which establishes global contamination standards and emphasizes the importance of reliable detection methods, serves as the foundation for the current research. It emphasizes the importance of evaluating regional products by framing microplastic contamination in salt as a global concern.

MICROPLASTICS IN SALT: A CRITICAL REVIEW OF CONTAMINATION.

De Nido et al. (2025) emphasized the recent rise in pollutants in their review of microplastic contamination in commercial salts. The study describes a number of contamination pathways, including seawater evaporation, industrial processing, and atmospheric fallout during drying and packaging. It draws attention to the production chain's multi-source contamination and highlights the fact that environmental sources are not the only ones.

The review evaluates discrepancies in microplastic concentration data and links them to non-standard detection methods. Source-tracking investigations benefit from the identification of common polymer contaminants, such as polyethylene and polypropylene. To ensure precise measurement, which is required for determining contamination levels, the authors advocate for standardised procedures. This review supports the current study's use of accurate detection techniques and helps to distinguish contamination from raw materials. It increases the validity of local assessments and encourages methodological rigor.

A DETAILED REVIEW STUDY ON POTENTIAL EFFECTS OF MICRO- AND NANOPLASTICS ON HUMAN HEALTH

Campanale et al. (2020) investigate the toxicological effects of micro- and nanoplastics, focusing on ingestion as a major route of exposure via foods such as salt. The review summarizes evidence on how microplastics cross biological barriers, potentially causing inflammatory responses and oxidative stress. It also emphasizes the importance of chemical additives in plastics, including endocrine disruptors such as BPA and phthalates.

According to the study, the physical presence of microplastics is only one aspect of the risk; chemical leaching raises the prospect of long-term effects on human health. The review recommends more in vivo research to better understand chronic exposure and bioaccumulation. This literature emphasizes the significance of detecting microplastics in salt for public health. It provides scientific support for the current research by linking contamination to actual risks to human health and highlighting the importance of monitoring and regulation.

MICRO/NANOPLASTICS AND HUMAN HEALTH: A REVIEW OF THE EVIDENCE

Singh et al. (2025) provide a comprehensive review on the impact of micro and nanoplastics on human physiology. Nanoplastics' small size allows for easier penetration of cell membranes, resulting in increased cytotoxicity. The article discusses the "Trojan Horse" phenomenon, where microplastics can carry toxins and pathogens to evade immune responses.

The study highlights the need for highly sensitive detection systems capable of identifying even the smallest particles. Plastics pose a double threat, both biologically and physically. Such properties are required for everyday foods like salt, which must be protected from long-term exposure. This review supports the technical design of detection techniques, highlighting the importance of thorough screening for micro and nanoplastic contamination.

IMPACTS OF MICRO AND NANOPLASTICS ON HUMAN HEALTH

Jayavel et al. (2024) explore the health impacts of microplastics, focusing on metabolic and reproductive toxicity. The study used animal models to suggest that chronic exposure can disrupt gut microbiota, which plays a crucial role in processing food and salts.

The study highlights regulatory gaps, including the lack of safe limits for microplastic consumption under food safety regulations. The researchers recommend urgent policy updates and the establishment of acceptable daily intake limits based on toxicological evidence. This study provides a strong rationale for microplastic monitoring, which is relevant to current research. The detection of microplastics in salt contributes to scientific knowledge and regulatory frameworks, highlighting its practical significance.

IS THERE EVIDENCE OF HEALTH RISKS FROM EXPOSURE TO MICRO- AND NANOPLASTICS IN FOODS?

Molina and Bené (2022) investigated whether there is clear evidence linking microplastics in food to human diseases. The review acknowledges microplastics' widespread presence in the food supply, but highlights limitations in current toxicity studies. This article discusses how plastics may break down in the stomach and release additives.

Although salt has low acute toxicity, it is still a significant source of exposure due to its direct consumption. The authors emphasize the importance of addressing chronic and cumulative exposure. This study provides a balanced perspective that emphasizes the importance of measuring and monitoring microplastic contamination without causing unnecessary alarm. It supports the current study's approach of monitoring salt as a major dietary source of microplastics.

CAN THE IMPACT OF MICRO- AND NANOPLASTICS ON HUMAN HEALTH REALLY BE ASSESSED USING IN VITRO MODELS? A REVIEW OF METHODOLOGICAL ISSUES

Forest et al. (2023) investigated in vitro toxicity testing for micro- and nanoplastics. The study highlights challenges like particle aggregation, which can impact toxicity results, and suggests advanced protocols to improve accuracy. This work examines the vulnerability of various cell types to microplastic exposure, highlighting the challenge of accurately simulating human reactions in lab settings.

This study offers methodological guidance for our current research. Toxicology testing requires reliable detection and characterization, making this literature relevant for experimental design and results interpretation.

A COMPLETE GUIDE TO EXTRACTION METHODS OF MICROPLASTICS FROM COMPLEX ENVIRONMENTAL MATRICES

Rani et al. (2023) provide a comprehensive review of microplastic extraction techniques, such as density separation, filtration, and chemical digestion. The paper assesses the effectiveness of these methods in recovering various polymer types without damaging the particles. The difficulties caused by salt crystallization, which can trap microplastics, are addressed. The study recommends using spectroscopy for confirmation and proper dissolution procedures prior to filtration.

This review supports the use of validated extraction and detection methods for accurate microplastic quantification in the current study.

MICROPLASTICS IN EUROPEAN SEA SALTS

Thiele et al. (2023) analyzed microplastic concentrations in European sea salts from various coastal regions. The study found that fibers and fragments were present regardless of price or brand, indicating a systematic rather than product-specific issue. Strict laboratory controls were used to ensure reliability. The study establishes geographic benchmarks for pollution levels in source waters and contamination in salt.

This study provides useful information by comparing local contamination levels to international findings, indicating relative pollution severity.

SYSTEMATIC REVIEW ON THE IMPACT OF MICRO-NANOPLASTICS EXPOSURE ON HUMAN HEALTH AND DISEASES

Krishnan et al. (2022) reviewed existing proof the connection between exposure to micro and nanoplastic on human health and diseases, such as inflammatory, respiratory, digestive, and brain effects. Due to their high surface are, nanoplastics can stick biomolecules forming protein coronas that can influence how these tiny bits interact with biological systems.

It can increase toxic burden by regularly eating contaminated condiments like salt. It highlights the lack of long-term data on organ accumulation. This literature gives a fundamental health focused justification for the current research by highlighting the societal and medical significance of microplastic monitoring in food products.

FROM SEA WATER TO SALT CRYSTALS: AN ONSITE INVESTIGATION OF MICROPLASTICS

Chanpiwat et al. (2024) observe the journey of microplastics starting from the seawater to crystallized salt in real-time evaporation ponds. Standard mesh filter frequently fail to capture finer fibers, and the microplastics in salt become concentrated as the water evaporates according to their study. The study gives practical suggestions for improving the filtration to reduce contamination. This study helps clarify the mechanisms behind salt contamination and offers production level strategies that are relevant to the present study.

MICROPLASTIC CONTAMINATION IN TABLE SALT SOLD IN SELECTED LOCAL MARKETS IN SAMAR, PHILIPPINES

Pajarillo, Panganoron, and Arcales (2024) evaluate the contamination and the cause of microplastic for table salt sold in markets in Samar. They confirmed the presence of microplastic in both microscopic and in chemical in distinct local brands. In Philippine setting, this study is valuable because it provides local data. The study said, polluted seawater, environmental condition, and packaging practices in wet markets are also the reason of contamination.

Researchers discovered microplastic in urban industrial centers, indicating that the issue extends beyond a specific region in the Philippines. This research provides a basic level of comparison to the current study. Determine whether microplastic contamination is widespread or unique based on local handling practices and environmental conditions.

DETECTION AND IDENTIFICATION OF MICROPLASTICS IN SEA SALT PRODUCTS SOLD AT DASMARIÑAS CITY WET MARKETS

Besana et al. (2019) studied sea salt products in Dasmariñas City, specifically the wet market supply chains. The study found microplastic fibers, particularly blue and red particles, and linked them to plastic sacks used for transportation and storage. Microscopic analysis confirmed the widespread presence of secondary contamination.

The study provides information on post-production pollution by distinguishing between contamination from packaging and the ocean. The results highlight useful intervention points and imply that appropriate packaging could greatly reduce contamination.

It provides data for the analysis of contamination sources; this source is critical to the ongoing investigation. Knowing whether plastics are sourced from the sea or from storage practices can help guide appropriate salt safety mitigation strategies.

ABUNDANCE OF MICROPLASTICS IN A RIVER-ESTUARY ALONG A RURAL-URBAN GRADIENT PATHWAY

Quindao and Sargado (2023) investigated microplastic transport in river estuaries with rural-urban gradients. Their research shows how plastic concentrations rise when rivers flow through densely populated areas, significantly contributing to coastal pollution.

The study highlights the importance of estuary proximity in determining contamination levels and identifies residential and agricultural sources of plastics. As a result, salt farms close to river mouths are more vulnerable. This research informs site selection analysis for salt sampling in the current study. It provides a hydrological explanation for contamination patterns, helping researchers interpret variations in local salt microplastic content.

OVER 90% OF SAMPLED SALT BRANDS GLOBALLY FOUND TO CONTAIN MICROPLASTICS

According to a global study summarized by Greenpeace Philippines (2018), microplastics are present in more than 90% of sampled salts. The Philippines and other Asian nations have high levels of contamination, according to the report.

The advocacy piece raises public awareness and demonstrates the global scope of the issue while localizing it for Filipino audiences. It encourages corporate responsibility and public participation in reducing plastic pollution. This study, which provides relevant statistics to support the study's importance and urgency, is useful as an introduction to current research.

VISUAL DETECTION OF MICROPLASTICS BY RAMAN SPECTROSCOPIC IMAGING

Liu, Pang, Chen, and Jiang (2023) used Raman spectroscopic imaging to visually detect polymers using color-coded chemical maps. Their approach demonstrates how optical systems can distinguish between plastics based on their spectral fingerprints. This shows that optical detection combined with image processing can successfully identify microplastics, which validates our findings. But because Raman imaging is slower and more costly, the researchers want to create an automated detector that is easier to use and more affordable.

RELATED STUDIES

MICROPLASTIC CONTAMINATION IN TABLE SALT SOLD IN THE SELECTED LOCAL MARKETS IN SAMAR, PHILIPPINES

According to Monteza, I. D. C., et al. (2024), the microplastic load of locally sold table salts in Samar varied from 20 to 400 particles per kilogram. These were mostly fibers and fragments less than 100 micrometers. The researchers determined the sources of contamination as seawater pollution, salt drying operations, and packaging materials. Their results highlight the necessity of local monitoring systems and emphasize the potential for cumulative human exposure through daily consumption. The research was one of the initial regional assessments of salt contamination in the Philippines and provided baseline information for risk assessments.

DETECTION AND IDENTIFICATION OF MICROPLASTICS IN SEA SALT PRODUCTS SOLD AT DASMARIÑAS CITY WET MARKETS

According to Besana, A. S., et al. (2019), microplastics were also identified successfully in commercial sea salts by microscopic and spectroscopic techniques. Their work noted that microplastic fragments were found in all the tested samples, indicating the omnipresence of plastic contamination in marine-derived products. Their research also suggested standards for sample collection and analysis protocols to provide comparable results across laboratories. The identification of chemical polymers like polyethylene terephthalate (PET) also supported the proposition that salts are susceptible to environmental pollution.

IDENTIFICATION AND QUANTIFICATION OF COMMON MICROPLASTICS IN TABLE SALTS BY A MULTI-TECHNIQUE-BASED ANALYTICAL METHOD

According to Li, H., et al. (2022), a combination of Fourier-transform infrared spectroscopy (FTIR), Raman spectroscopy, and visual imaging was utilized to accurately identify and quantify microplastics in table salt samples. The study demonstrated that integrating multiple detection techniques provides greater precision in differentiating polymer types and estimating particle sizes. Their results revealed widespread contamination across tested brands, confirming that microplastics persist even in refined salts. The researchers emphasized the potential application of automated systems and AI-based image recognition to enhance detection speed and accuracy.

QUANTITATIVE AND QUALITATIVE EVALUATION OF MICROPLASTICS IN DIFFERENT SALTS FROM IRAN

Sharifi, H., and Attar, H. M. (2021), stated that microplastics were present in all salt samples including rock, sea and lake salts, all studied. Their report mentioned that the registered particles ranged from 50 to 280 microplastics per kilogram, and the whole sample consisted of polypropylene and polyethylene. Extraction and packaging of salt was also noted as a point of contamination in the industry where it could be handled through improved industrial processes. Third, the authors provided recommendations that countries implement regular inspection systems to ensure commercial salt products are secure to sell in the market.

CONSUMING MICROPLASTICS? INVESTIGATION OF COMMERCIAL SALTS AS A SOURCE OF MICROPLASTICS IN DIET

Kuttykattil, A., et al. (2023), a global large-scale survey confirms that table salts are a major indirect source of dietary microplastics. It analyzed samples obtained from different countries and revealed polymeric fragments and fibers dominate across brands. Exposure rates estimated from humans suggest that an average adult might ingest as many as thousands of microplastic pieces annually based on salt consumed alone. Health implications of continued consumption are yet to be determined, but the accumulation of human tissue is something to be constantly monitored, the authors note. This paper suggests that monitoring systems with artificial intelligence assisted systems for automatic detection and quantification of microplastics in food items should be recommended.

MICROPLASTIC CONTAMINATION IN COMMERCIAL SALT: AN ISSUE FOR THEIR SAMPLING AND QUANTIFICATION

A study by Di Fiore et al. (2022), evaluated the microplastic content of Italian sea salts and observed that individual samples tested contained a certain amount of contamination. Polyethylene, polypropylene, and polyamide fibers were the most common microplastics found. They assess the samples by using ATR-FTIR, Raman spectroscopy and microscopy. They're techniques that are accurate, but time-consuming and expensive and require specialized expertise. This highlights the absolutely crucial need for improved detection technologies to make detection techniques more efficient.

Building on these findings, the present study aims to develop an AI-based automatic optical detector for microplastics in rock salt. The system uses image processing and AI to detect, identify, and measure small plastic particles in terms of their appearance, such as color, size, and shape. This methodology aims to deliver a quicker, increased ease and value, cheaper, better and more reliable substitute for lab methods, allowing food safety checks and environmental monitoring to be more effective and accessible for food quality and environmental monitoring of food safety.

MICROPLASTICS IN THE WATER OF BATANG ANAI ESTUARY, PADANG PARIAMAN REGENCY, INDONESIA: ASSESSING EFFECTS ON RIVERINE PLASTIC LOAD IN MARINE ENVIRONMENT

A study by Suparno et al. (2024) investigated microplastic pollution in the Batang Anai Estuary, located in Padang Pariaman Regency, Indonesia, and found out that the concentrations ranged from 37 to 77 particles per liter of water. The researchers identified and classified these microplastics using filtration and ATR-FTIR spectroscopy.

Although this study is focused on a river, it highlights how extensive microplastic pollution has become and underscores the limitations of laboratory techniques. These findings address a clear gap: lab-based methods are too slow and resource intensive. That is why the present research, "Development of an Automated Optical Detector for Microplastics in Rock Salt," is well justified by prior findings.

PRESENCE OF MICROPLASTICS CONTAMINATION IN TABLE SALT AND ESTIMATED EXPOSURE IN HUMANS

Syamsu et al. (2024) explored the presence of microplastics in table salt and discovered that all 21 salt samples tested from Indonesia contained traces of these tiny plastic particles, ranging from 33 to 313 pieces per kilogram. Most of the detected microplastics were small fragments and fibers, measuring between 100 and 300 micrometers in size and made mostly of common plastic materials like polyethylene (PE), polypropylene (PP), and polyethylene terephthalate (PET). Their study revealed that something as ordinary and widely used as table salt can be a hidden source of microplastic ingestion, potentially exposing people to hundreds or even thousands of these particles every year depending on how much salt they consume. The researchers relied on manual inspection and Fourier Transform Infrared (FTIR) analysis to identify these microplastics, methods that, while precise, can be slow, expensive, and require specialized laboratory tools.

By doing this, the present study aims to develop an automated optical detector that can identify microplastics specifically in rock salt. Unlike table salt, rock salt comes directly from natural mineral deposits, giving it different textures and optical properties that may influence how microplastics appear under light. Through AI and optical detection, this project aims to create a faster, simpler, and more accessible way to detect microplastics without the need for complex laboratory procedures. By focusing on rock salt, the study also fills a gap in current research, which has mostly concentrated on sea and table salts. Overall, this study hopes to support safer food quality monitoring and raise awareness about the unseen microplastics that may be present in everyday food or ingredients like salt.

MICROPLASTICS POLLUTION IN SALT PANS FROM THE MAHESHKHALI CHANNEL, BANGLADESH

Rakib et al. (2021) shows that all unrefined sea salt samples from the Maheshkhali Channel, Bangladesh, contained microplastics, mostly fragments, films, and fibers made of PET, PP, and PE. Their study shows that even everyday salt can carry invisible contaminants that can enter the human body and that can cause health issues. Inspired by this, our study develops an AI-powered optical detector for rock salt aiming for faster, accurate, and easier detection to keep this common ingredient safe for consumption.

QUANTIFICATION AND CHARACTERIZATION OF MICROPLASTIC CONTAMINATION IN COMMERCIALLY AVAILABLE SALT IN PARTIDO AREA, PHILIPPINES

A study by Garcia et al. (2024) shows that all salt samples collected and tested from the Partido District in Camarines Sur contained microplastics, with levels ranging from 15.56 to 369.07 particles per kilogram. Most of the microplastics were tiny fibers and filaments, coming from fishing gear, fabrics, and packaging. Most of the particles of salt were smaller than 10 micrometers, which makes them more harmful since they are microscopic and can enter the human body easily. This study shows the need for improvement in detecting microplastics in salt; present methods still use Laboratory inspection under a microscope, which can be slow and unreliable. Developing an AI optical detector can help make the process faster, more accurate, and more consistent. This technology can play a key role in improving salt quality control and protecting consumer health.

ASSESSMENT OF QUANTITY AND QUALITY OF MICROPLASTICS IN THE SEDIMENTS, WATERS, OYSTERS, AND SELECTED FISH SPECIES IN KEY SITES ALONG THE BOMBONG ESTUARY AND THE COASTAL WATERS OF TICALAN IN SAN JUAN, BATANGAS

A study by Deocampo et al. (2020) discovered microplastics in water, sediments, oysters, and fish in the Bombong Estuary and Ticalan waters of Batangas. Most of the particles were small fragments, fibers and films of polyethylene. This demonstrates that MPs are ubiquitous and may be present in food chains via marine organisms. Detection of microplastics in lab, such as filtration and density separation, and FTIR spectroscopy, are slow to perform, costly and require trained experts to perform. As a result, these methods for detection need to be simpler and more rapid. An AI detector developed could assist effectively in detection and monitoring of microplastics more efficiently in rock salt. The findings could make tests easier, cheaper, and more accessible to food safety and environmental monitoring.

FROM SEA WATER TO SALT CRYSTALS: AN ONSITE INVESTIGATION OF MICROPLASTICS IN A CONVENTIONAL SEA SALT FARMING SYSTEM.

Chanpiwat, P., & Damrongsiri, S. (2024) reported that microplastic concentrations increase stepwise as salt is synthesized to and by concentrations rise with increasing from the salt production process to the maximum concentrations in crystallization ponds. The researchers detected large quantities of microplastic loads in salt water, macroalgae, pond-floor clay and crude salt from seawater, including fragments, amounting to 99% of all particles found. All microplastic types are similar across sample sources, suggesting that contamination is due to degraded plastics and farm equipment. This investigation confirms the current study by proving that microplastics in salt already exist even before refining, showing the need for efficient and reliable detection methods, so it strengthens the case to develop an automated optical detector.

FIRST EVER STUDY UNCOVERS MICROPLASTIC CONTAMINATION IN NEPALESE TABLE SALT.

Recent research on table salt from Nepal Hussain, S., Bano, N., Sarwar, F., Iqbal, W., Shrestha, S. B., & Shrestha, S. S. (2024) used Fenton's reagent digestion and manual microscopy to identify 219–381 microplastics per kg, mainly fibrous particles made from polymers e.g. polyethylene and polypropylene. However, the process taken was time-consuming and necessitated extensive chemical pre-treatment, necessitating faster methodologies for direct automated optical detection. In line with the objective of our study, we design an automatic optical detector for microplastics detection in rock salt.

DETECTION OF MICROPLASTIC IN SALTS USING TERAHERTZ TIME-DOMAIN SPECTROSCOPY.

Choi, Hong, and Bahk (2021) investigated the detection of microplastics in table salt using terahertz time-domain spectroscopy. They found that optical characteristics like refractive index and dielectric constant could be used to non-destructively identify embedded plastics. Their results demonstrate the value of the noninvasive optical approaches and a baseline idea for automation of detection in solid salt matrices.

MATERIAL-SPECIFIC DETERMINATION BASED ON MICROSCOPIC OBSERVATION OF SINGLE MICROPLASTIC PARTICLES STAINED WITH FLUORESCENT DYES.

Aoki (2022) proposed a fluorescence microscopy technique for material-specific identification of microplastics via selective dyes such as Nile Red. Through visualizing changes in fluorescence intensity, the study indicated that microplastics could be efficiently categorized according to the polymer type. This is an approach that has direct application for automated optical detection systems. Dye assisted fluorescence is suggested for the enhancement of both identification and quantification in rock salts.

BIG DATA, TINY TARGETS: AN EXPLORATORY STUDY IN MACHINE LEARNING-ENHANCED DETECTION OF MICROPLASTIC FROM FILTERS

Miclea et al. (2025) showed machine learning models (e.g., YOLO) to scan electron microscope images for automated microplastic detection. Whilst their study was limited to filter samples, it shows the potential of image-based AI to be used to automate the recognition and classification of particles which already is an applicable concept in use in fact a concept that can be immediately applied to the optical detection of the rock salt.

AN AUTOMATED LEPTOCORISA ORATORIUS ERADICATION DEVICE BASED ON YOLOV8 FOR DETECTION AND MITIGATION OF PESTS IN RICE FIELDS OF SAN RAFAEL, BULACAN.

Agbay et al. (2025) invented an automated Leptocorisa oratorius eradication system for rice fields with YOLOv8 object detection system to detect and control pests on the spot. This was a computer vision with automatic responses which demonstrated the potential of image-based AI system in fine target detection and actions in complex environments. This approach is applicable to microplastic detection in rock salt, highlighting the potential for incorporating automated optical imaging and machine learning, as used in this study to detect and quantify small particulate matter on an automated detection model with little or no human involvement in a conceptual manner.

SYSTEM PERFORMANCE AND RESPONSE TIME

In real-time systems, response time is a critical factor in determining performance efficiency and usability. Real-time systems require minimal delay to ensure responsive and effective operation. Studies indicate that human perception of system response can occur within milliseconds, and faster processing speeds improve overall system interaction and user experience (PubNub, 2024). These findings highlight the importance of maintaining low latency in AI-based systems, particularly in applications involving real-time detection and analysis.

NILE RED STAINING TECHNIQUE FOR MICROPLASTICS DETECTION IN SARDINES (SARDINELLA LEMURU).

Sargadillos et al. (2024) demonstrated the effectiveness of fluorescence-based detection using Nile Red staining agent mixed with UV illumination at 365 nm. Their study indicates that microplastic particles showing a significant brightness when exposed to UV light, which enables them to be easily distinguished from non-plastic materials in food samples. This approach supports the use of UV illumination in the current study, as it increases contrast between microplastics and rock salts, making it suitable for integration with automated image processing and AI-based classification systems.

A LOW-COST FLUORESCENCE MICROSCOPE FOR THE DETECTION OF MICROPLASTICS. METHODS AND PROTOCOLS, 5(1), 4.

Prata et al. (2022) investigate the use of low-cost fluorescence microscopy system using a 365 nm UV light source for detecting microplastics in environmental samples. Their findings showed that plastic particles emitted significantly higher intensity of brightness compared to natural materials, improving visibility and detection accuracy. This supports the current study, employing a UV-based optical detection as efficient method for identifying microplastics in rock salt samples.

IDENTIFICATION OF MICROPLASTICS USING THEIR INTRINSIC FLUORESCENCE. ENVIRONMENTAL POLLUTION.

Araujo et al. (2022) explored the optical properties of microplastics and found that certain polymers show natural brightness when exposed to UV light, even without staining agents. This autofluorescence behavior depends on polymer type and environmental exposure but can still be utilized as a basis for detection. The study features that UV-based imaging can identify microplastics, that provides a non-destructive and simplified detection approach. This supports the current study using UV illumination, even in the absence of chemical staining.

GLOBAL PATTERN OF MICROPLASTICS (MPS) IN COMMERCIAL FOOD-GRADE SALTS: SEA SALT AS AN INDICATOR OF SEAWATER MP POLLUTION

Kim et al. (2018) conducted a global assessment of microplastics in commercial food grade salts and analyzed 39 salt brands from different regions worldwide. They reported a wide range of microplastic concentrations, their findings showed that rock salt contained up to 148 microplastic particles per kilogram. The study confirmed the widespread presence of microplastics in salt and provided baseline data for estimating human intake. In relation to the current study, these values serve as a reference for developing threshold levels in evaluating microplastic contamination in rock salt samples and assessing potential intake risks.

RESEARCH GAP

Microplastic contamination in food products, particularly in rock salt, has become an increasing environmental and public health concern. Existing methods for detecting microplastics primarily rely on laboratory-based techniques such as microscopic examination and spectroscopic analysis, which require specialized equipment, trained personnel, and lengthy processing times. While these methods provide accurate results, they are often costly and inaccessible for routine screening in retail and household settings.

Several studies have explored the use of image processing and artificial intelligence for microplastic detection; however, most have focused on environmental samples such as water, sediments, and marine ecosystems. Limited research has been conducted on the application of AI-based optical detection systems for identifying microplastics in rock salt samples. Furthermore, existing studies often emphasize particle detection alone and do not provide an integrated system capable of automatically analyzing microplastic count, estimating particle size, and classifying contamination levels in a user-friendly and portable platform.

Therefore, a gap exists in the development of an accessible, automated, and AI-integrated detection system specifically designed for the preliminary screening of microplastics in rock salt. This study addresses this gap by developing an automated optical detection system that combines image processing, artificial intelligence, and real-time analysis to detect microplastics, estimate particle size, determine microplastic count, and provide contamination assessments for retail and household rock salt samples.

SYNTHESIS OF THE STUDY

System Features	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Microplastic particles in rock salt are precisely identified and categorized by the system using artificial intelligence.	✕	✕	✕	✕	✕	✕	✕	✓	✕	✓	✕	✓	✓	✕
It uses image processing methods to improve captured photos and extract characteristics required for the detection of microplastics.	✕	✕	✓	✓	✓	✓	✕	✓	✕	✓	✓	✓	✕	✓
The system automatically captures, analyzes, and detects images without the need for human assistance.	✕	✕	✕	✕	✓	✕	✕	✓	✕	✓	✕	✕	✕	✕
Measures particle size for classification	✕	✕	✓	✓	✕	✓	✕	✓	✕	✕	✕	✕	✕	✕
Automatically detects and counts microplastic particles	✓	✕	✓	✕	✓	✓	✕	✓	✕	✕	✕	✕	✓	✕

It displays measurements and detection results instantly via a real-time data display interface.	✕	✕	✕	✕	✓	✕	✕	✕	✕	✓	✓	✓	✕	✓
Domestic and retail handling of rock salt increases the risk of microplastic contamination due to exposure during storage and sale.	✕	✕	✕	✕	✕	✕	✕	✕	✓	✓	✕	✕	✓	✕
Enhances visibility of microplastics using UV light	✕	✕	✕	✕	✕	✕	✕	✕	✕	✕	✓	✓	✓	✓

System Legend Table:

Source	Name	Description
A	De nido et al., 2025	Microplastics in salt: a critical review of contamination. A detailed review study on potential effects of micro and nanoplastics on human health
B	Rani et al., 2023	A Complete Guide to Extraction Methods of Microplastics from Complex Environmental Matrices
C	Besana et al., 2019	Detection and Identification of Microplastics in Sea Salt Products Sold at Dasmariñas City Wet Markets
D	Liu, Pang, Chen, & Jiang, 2023	Visual Detection of Microplastics by Raman Spectroscopic Imaging
E	Agbay et al., 2025	An Automated Leptocorisa oratorius Eradication Device Based on YOLOv8 for Detection and Mitigation of Pests in Rice Fields of San Rafael, Bulacan
F	Garcia et al., 2024	Quantification and Characterization of Microplastic Contamination in Commercially Available Salt in Partido Area, Philippines
G	Choi, Hong, & Bahk, 2021	Detection of Microplastic in Salts Using Terahertz Time-Domain Spectroscopy
H	Miclea et al., 2025	Big Data, Tiny Targets: An Exploratory Study in Machine-Learning-Enhanced Detection of Microplastic from Filters
I	Pajarillo et al., 2024	Microplastic Contamination in Table Salt Sold in Selected Local Markets in Samar, Philippines
J	Suparno et al. (2024)	Development of an Automated Optical Detector for Microplastics in Rock Salt.
K	Prata et al., 2022	Showed that UV fluorescence (365 nm) makes microplastics appear brighter than natural materials, enabling simple and low-cost detection.
L	Aoki, 2022	Used fluorescent dyes (e.g., Nile Red) to classify microplastics by polymer type via fluorescence intensity differences.
M	Sargadillos et al., 2024	Found that Nile Red and UV light make microplastics glow, enabling easy detection.
N	Araujo et al., 2022	Showed that some microplastics naturally fluoresce under UV light, allowing simple, non-destructive detection without staining.

CHAPTER III

RESEARCH METHODOLOGY

This chapter discuss the method of research used, the sources of data, the data-gathering instrument, the data-gathering procedure, the ethical considerations, the statistical treatment of data, the technical feasibility, the project design, as well as the software methodology.

METHODS OF RESEARCH

The research methodology will adopt both quantitative and experimental approaches to comprehensively test the accuracy, functionality, and reliability of the prototype for detecting microplastics present in rock salt samples.

Quantitative methods will be utilized through collecting and processing quantifiable data on the performance of the proposed optical detecting system as well as the extent of use of the proposed device. Questionnaire method will be employed in this work to collect quantitative data showing the participants' thought process, opinion, and acceptability of the implemented prototype. A questionnaire prepared by the International Organization for Standardization (ISO 25010) can be employed periodically to assess the reliability of the prototype. Data will be analyzed with different quantitative methods for data analysis, including pattern-recognition analysis, percentage computation, simulation, and comparison testing to measure the accuracy and efficiency of the system in detecting microplastic fragments. A different experimental method will be applied to selectively adjust different variables such as light intensity, camera capture settings, and AI-based optical system detection ranges to align with the quantitative approach. Dependent factors such as accuracy, precision, and detection speed can be assessed in terms of their impact. The prototype designed to detect microplastics in rock salt samples will be tested in several controlled trials to confirm its accuracy and reproducibility. With this process, researchers intend to establish the cause and to measure effects of diverse factors on its detection performance. Apart from the testing methods, survey research methods will be employed to harvest further information from the experts and the assessors on the accuracy and implementation of the prototype in the laboratory and business operations. A range of measuring methods will be employed in this survey research to reach insights into the effectiveness of the project from the perspective of the respondents. Moreover, it will offer better quantitative data based on evaluation and test results. A combination of the quantitative analysis and experimental way of investigation would yield a complete picture with regard to how the optical detector works and will potentially assist further developments in environmental monitoring systems and research on microplastic detection technology. The aim of this work is to demonstrate that environmental testing with AI-based optical analysis can be accomplished using a Raspberry Pi 5 and YOLOv11. It offers a digital and fully automated alternative to the traditional laboratory techniques that rely on filtration and microscopy for detection of microplastics.

SOURCES OF DATA

This research utilizes both primary and secondary data for a potential automated optical detector for microplastics in rock salt. Among the main sources, the researchers surveyed consumers to learn the kinds of salt used and their sources. These findings provided important information on how consumers select rock salt to eat. The researchers also collected salt samples from various production regions and tested them under laboratory conditions through optical detection techniques to find and measure microplastics. Experimental results formed the basis of the design and improvement of the automated detection system. Some sources used were published articles, reports, online sources, and other information referring to microplastic contamination and optical detection systems of this topic. In combining firsthand observations with a solid body of research, this study targets a system of identifying microplastics in rock salt that could offer a credible solution as well as a method to enhance the overall safety of the food process and environmental well-being.

DATA GATHERING INSTRUMENTS

The researchers used a variety of tools to gather data in the previous study that contributes to the discovery of microplastics in rock salt by artificial intelligence. Informed observations were collected at selected sites and times to observe the properties and environment of the salts, which showed much information regarding the production of salt and contamination. Surveys and interviews with salt farmers, local community groups, and salt producers had also indicated useful information on the process of processing salt and, therefore, the amount of salt and contaminants. They provided experts validation to the tools for reliability. The weights and size of the salt samples were measured to compare to an outcome of AI testing. Samples will also be taken from multiple areas including Pangasinan, Batangas, Cavite, Navotas/Malabon, and Zambales, and from sellers such as markets, grocery stores, and locally repacked salts. The careful collection assists the training of AI in ferreting out microplastics, thus providing consumers with a safer option for salt.

DATA GATHERING PROCEDURE

Data collection for the research followed a concise and structured plan. Research planning initiated this autumn (early October), where researchers concentrated on developing methodology, background information generation, key informants, and focused and honed research questions. Late autumn, beginning in October and into early November, the researchers studied existing literature/similar studies, and the scientists utilized internet and library resources to research extant methods for detection of microplastics and the technologies they adopted. The researchers continued investigating other related study work throughout November, and as they designed and improved the methods presented for the study. The draft of the research presentation was ready by the end of the month. With the proper permits obtained, the researchers re-visited and refined the questionnaire instruments, which already had been reviewed and validated by the experts.

For the purposes of this research, data was collected solely from the planning and assessment process, and not from any device. The primary data was collected with the aid of questionnaires to explore its awareness among the population toward microplastics, salt purchasing habits, and what it expected to be on the automated optical detector. These observations were valuable to understand what the consumer did, whether or not they were worried about health and what were the "hands-on" needs that the future system needs to cover. Moreover, the primary data consisted of laboratory testing of rock salt samples at a variety of sources to assess their content of microplastics. These laboratory results support testing and optimization of the identified system of detection. Relevant literature and prior studies were the secondary data in the previous work, and this gave technical basis for “Development of an Automated Optical Detector for Microplastics in Rock Salt.

ETHICAL CONSIDERATION

This study adhered to the ethical standards prescribed by the Research and Development Office (RDO) of MPC to ensure the protection of participants’ rights, safety, privacy, and welfare throughout the conduct of the research.

Social Value. The study was conducted to address concerns regarding the efficiency and reliability of object detection systems through the proposed automated optical detector. The findings are expected to contribute to technological advancement and provide relevant information that may improve safety, accuracy, and operational effectiveness in applicable environments. The study design, methodology, and data collection procedures were aligned with the research objectives to ensure meaningful and practical outcomes.

Voluntary Participation and Informed Consent. Participation in the study was entirely voluntary. Participants were informed of the purpose of the study, the significance of their responses, and how the collected information would be utilized in relation to the proposed automated optical detector. Prior to participation, informed consent was obtained, and participants were given the opportunity to ask questions. They were also informed of their right to refuse participation or withdraw from the study at any stage without penalty.

Vulnerability of Research Participants. The study did not intentionally involve vulnerable populations, such as individuals incapable of independently providing informed consent due to physical, cognitive, or mental limitations. Participants included in the study were capable of understanding the nature and purpose of the research and voluntarily agreed to participate.

Risk, Benefits, and Safety. The study involved minimal risk to participants, as data collection was limited to questionnaires and did not expose respondents to physical, psychological, or social harm. The anticipated benefit of the study includes contributing knowledge and practical insights that may support the improvement of automated optical detection systems.

Privacy and Confidentiality of Information. In compliance with the provisions of the Data Privacy Act of 2012, no personal or identifying information was collected from participants. Responses were treated with strict confidentiality, securely stored, and used solely for academic purposes. Only the researchers had authorized access to the collected data, and participant anonymity was maintained throughout the study.

Justice. The selection of participants was conducted fairly and without discrimination to ensure equitable participation. No participant was unfairly burdened or excluded, and the research was designed to avoid reinforcing social inequalities.

Transparency and Results Communication. The researchers maintained transparency regarding all aspects of the study that could influence participants' decisions to participate. Relevant details about the objectives, procedures, and intended use of data were clearly communicated. Furthermore, findings and information presented in the questionnaires were explained in a clear, understandable, and respectful manner, recognizing the contributions of the participants.

Frequency and Percentage Distribution

This was used to determine how many respondents selected each option for awareness, purchasing behavior, and perception-based questions.

$$Percentage = \frac{f}{N} \times 100 \qquad (1)$$

Where:

$\boldsymbol{f}$ = frequency of a response

N = total number of respondents

This helped identify trends such as the most common sources of salt, levels of awareness about microplastics, and preferred salt types.

2. Mean (Average)

For questions using Likert-scale responses (level of concern, perceived need for a detector), the mean was computed to understand the general sentiment of participants.

$$Mean = \sum \frac{X}{N} \tag{2}$$

3. Weighted Mean

Used for evaluating respondents' agreement levels on statements regarding microplastic risks, trust in salt brands, or their perception of the importance of the proposed detector

$$Weighted\ Mean = \sum \frac{(W \times X)}{N} \tag{3}$$

Where:

W = weight assigned to each response

X = frequency of each response

This allowed the researchers to rank perceptions and identify which concerns or expectations were most significant to respondents.

Likert Scale. For the assessment of the survey. The researchers use the likert scale, a numerical scale for its evaluation. The collected data will be computed for the mean and over-all mean. The interpretation of data will use descriptive rating.

Table 3.1. Likert Scale for the Evaluation Survey

Scale	Verbal Interpretation
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

System Performance Evaluation

In addition to the survey data, the performance of the developed microplastic detection system was evaluated using computational and statistical methods. The following formulas were used to analyze system outputs and compare them with laboratory results.

1. Microplastic Concentration (MP/g)

Microplastic concentration was computed as the number of detected microplastic particles per gram of sample. This standardizes the results and allows comparison across different samples regardless of their weight.

$$MP/g = \frac{Number\ of\ Detected\ Microplastic\ Particles}{Sample\ Weight\ (g)} \tag{4}$$

2. Absolute Difference

The absolute difference was used to determine the deviation between the system-generated results and the laboratory reference values. Smaller values indicate closer agreement between the two.

$$Absolute\ Difference = |Measured\ Value - Reference\ Value| \tag{5}$$

3. Percentage Error

Percentage error was used to evaluate the accuracy of the developed system by comparing its results with laboratory values. Lower percentage error indicates higher accuracy and reliability.

$$Percentage\ Error = \left(\frac{Reference\ Value - Measured\ Value}{Reference\ Value}\right) \times 100 \tag{6}$$

4. Mean Absolute Error (MEA)

The Mean Absolute Error (MAE) was used to measure the average difference between the system output and laboratory results. It provides an overall indicator of the system's accuracy.

$$MAE = \frac{1}{n}\sum|Laboratory\ Value - System\ Value| \tag{7}$$

5. Processing Time (Average)

The average processing time was computed to evaluate the performance efficiency of the system. It represents the average time required to analyze each sample.

$$\bar{T} = \frac{\sum T}{n} \tag{8}$$

In addition to computed average processing time, latency-based classification was used to assess system responsiveness. Processing times less than 0.010 seconds were classified as "very fast," between 0.010 and 0.030 seconds as "fast," and greater than 0.030 seconds as "acceptable." This classification is based on real-time system performance and human perception thresholds.

TECHNICAL FEASIBILITY

The aim of the capstone project, "Automated Optical Detector for Microplastics in Rock Salt" will allow the user to deliver an efficient, convenient, reliable, accurate and relatively cheap means of locating microplastics isolated from rock salt samples. It is composed of a Raspberry Pi 5, digital microscope, LED lighting system, display, and monitor for the real time view of the microscope etc. The components were selected for their convenience, availability, strength, processing efficiency and usefulness for AI-based optical detection for AI applications.

The digital microscope allows high-magnification imaging by picking up microplastics larger than 50 μm, and the LED lighting system provides a stable and even illumination that prevents any reflections or shadows if these are to impair detection. The Raspberry Pi 5 scans microscope images with the YOLOv11 AI model which can quickly identify and sort microplastics by size and count.

It displays the contamination percentage and calculation real time detection results on a monitor connected to Raspberry Pi. The system does run on a software interface so users can look at processed images and generate high level reports. All of this is suitable for laboratory, field and educational applications.

Testing confirmed the prototype to be a reliable system for detection of microplastics across a variety of specimen types and environmental conditions. The prototype offers a practical, automated and user-friendly alternative to laboratory methods, which are time-consuming, expensive and require specialized personnel. It would be portable and simple and would be suitable for local salt producers, researchers and educators to accurately monitor microplastic contamination.

ENGINEERING DESIGN PROCESS

Adopting the ADDIE model (Analysis, Design, Development, Implementation, and Evaluation) based on the systematic steps outlined in Figure 3.1, the advocates emphasize the transformation possibilities introduced by technology in microplastics detection which supports decision-making and a new era of food safety surveillance and environmental awareness.

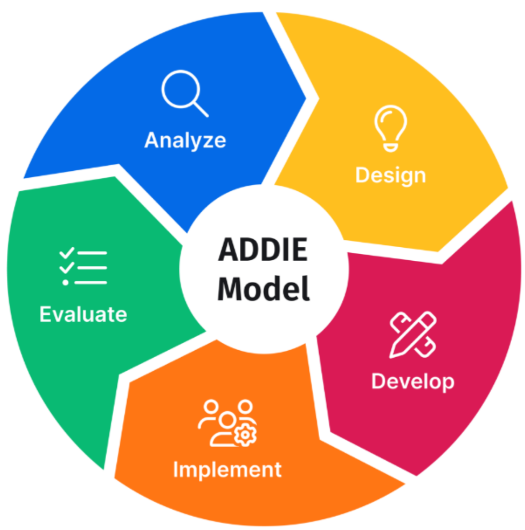

Figure 3.1 ADDIE Model

According to the ADDIE model, the process starts with the Analysis phase, where the researchers identified the key elements and requirements for real-time microplastic detection in rock salt with an automated optical system and YOLOv11 AI model. To optimize image acquisition and determine the necessary dataset to train the AI model under different lighting conditions, the team assessed the physical properties of salt samples, including crystal size and environmental sources of contamination. Furthermore, specific hardware and software requirements were specified, ensuring that the Raspberry Pi 5 could efficiently handle image pre-processing, AI inference, and real-time visualization, while providing an easy-to-use experience and robust data logging capabilities.

During the Design phase, we developed a comprehensive plan for the seamless integration of the physical and software architectures. Digital microscope, white and UV LED lighting, and touchscreen monitor were physically planned for the uniform illumination and high-quality image capture. At the same time, the software workflow was planned which includes the image acquisition module, the pre-processing techniques such as resizing and noise reduction, the detection engine YOLOv11 and the Graphical User Interface (GUI). The GUI was developed to allow real time detection results (bounding boxes and contamination percentages) and data storage organized into raw image, processed file and detection log.

Development was the stage where the prototype was actually put together and coded according to the design plans. The hardware parts consisting of the Raspberry Pi 5, microscope, LED lighting, load cell, and monitor were physically integrated. From the software point of view, the image acquisition module and the preprocessing algorithms were implemented to prepare the inputs for the AI. At the same time, the YOLOv11 The GUI was developed for live visualization, and automated scripts were written to handle data saving, model inference, and detection logging. model was trained on a curated dataset of labeled microplastic images to optimize model weights for efficient inference on Raspberry Pi. GUI was developed for visualization live. Scripts were automated to save data, run the model inference and log detections. written for data saving, model inference and detection logging.

During this phase internal testing and debugging were conducted to ensure smooth communication between the physical hardware modules and the software system. In the Pilot Implementation phase, the complete prototype and software were deployed for real-time testing with real rock salt samples from retail and domestic sources. The system was tested under different conditions to tune preprocessing algorithms, detection thresholds and GUI responsiveness based on live feedback. Performance was closely monitored to ensure that all functionalities, from image capture to AI inference and visualization, operated effectively. End-users and evaluators were trained on the proper operation of the system, the interpretation of detection results via the GUI, and the management of stored data. All findings and operational changes were documented.

The Evaluation phase evaluated the overall performance of the automated optical detector in terms of detection accuracy, model speed, GUI usability and system stability. The results of AI-based detection were validated with standard laboratory analyses to confirm the reliability of AI-based detection.

Users and evaluators provided feedback on the system’s ease of use, clarity of the interface, and the performance of the hardware in trials conducted in real-world settings. This thorough evaluation enabled the researchers to identify specific areas for improvement, such as AI model refinement, preprocessing enhancements, and lighting adjustments, while also making recommendations for future system upgrades and dataset expansion.

CHAPTER IV

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

This chapter focuses on presenting, analyzing, and interpreting the data collected for the study.

The following are the processes in constructing the “Development of an Automated Optical Detector for Microplastics in Rock Salt for Retail and Domestic Use”.

1. **Development of the Automated Optical Detection System Integrate with AI for Microplastic in Rock Salt.**

The “Development of an Automated Optical Detector for Microplastic in Rock Salt for Retail and Domestic Use” was carried out using the ADDIE model, which consists of Analysis, Design, Development, Implementation, and Evaluation phases. This structured approach ensured that the system was systematically planned, developed, tested, and evaluated.

Phase 1: Analysis

Figure 4.1 Conducting the Needs Assessment Within the Marikina Public Market

During this phase, the researchers not only establish the technical and system requirements for real-time microplastic detection in rock salt using AI or YOLOv11, but also incorporates field validation through interviews with salt vendors from the public market of Marikina City. Based on the interview's finding, it was learnt that the common salt that has been used and sold from the public market is rock salt, as well as most of the vendors are not aware or does not have any knowledge about microplastic or the microplastic contamination in rock salt. These insights from these validators help the researchers conceptualize the problem, ensure the system aligns with real-world practices. Additionally, the study recognizes the conventional laboratory method expensive and time consuming, which strengthen the need for a faster, cost effective, and automatic solution. Through this analysis, plans are set for a system that responds to the needs and requirements of detecting microplastic in rock salt.

In addition, the researchers clarified the major requirements of the proposed system. These requirements included the high-quality camera, controllable illumination, for focusing the sample visibility, a touchscreen graphical user interface for easy operation, an AI model for automated optical detection, a sample weight measurement component for concentration calculation, and a result management module that can automatically generate report and store it. Based on the needed requirements, the system was planned as a compact embedded prototype that can support the analysis of the sample as well as to document the data in a single platform.

Moreover, the analysis phase formed the expected outcome of the system. These outcomes included the estimated particle size, microplastic count per grams, stored records, and automatically generated reports. The findings from this phase will be served as a basis for design and develop phase.

Phase 2: Design

Before the “Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use” system is developed, the design phase is included. At this step, efforts are made to develop the system in a structured way that matches the researchers expected outcome.

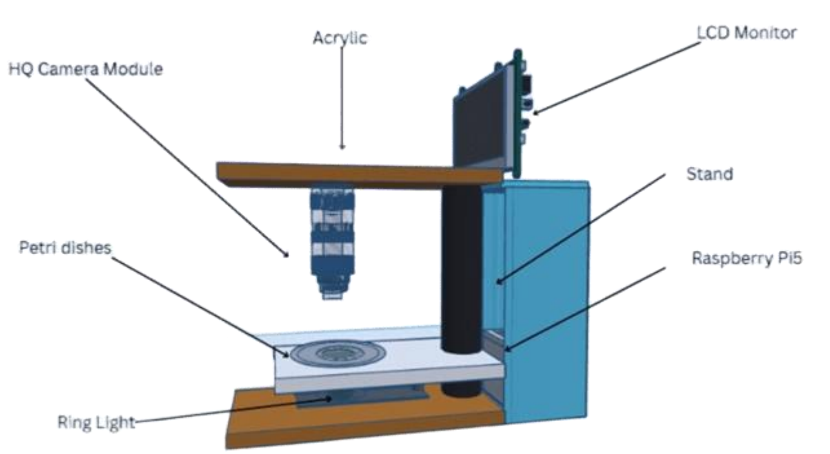

Figure 4.2 The 3D Design of the System Prototype

This figure 4.2 reflects the first layout and main parts of the system prototype in a 3-dimensional form. This phase focused on determining how the major hardware and software components could be integrated so that the system can operate in a firm, organized, and user-friendly manner. The researcher’s conceptual design envisions a portable device that integrates various components to deliver a smooth user experience. At its core, the prototype will utilize a monitor as the main display, that will give a visually appealing platform for users to interact with and will provide real time feedback and information of rock salts, enhancing the overall user experience. A Raspberry Pi HQ camera that will be integrated into the design, allowing users to initiate the test of the rock salt. The Raspberry Pi 5 is the cpu or the brain of the system that is responsible for the operation of all components and to ensure a connection efficient. It also includes white and UV illumination modules installed to support different observation conditions, and a load cell integrated for sample mass measurement. Together, these components will create a cohesive and functional prototype for detecting microplastics in rock salt for retail and home use.

Besides, a first design layout was prepared for the software, though this design changes as the development continues to fit what the project needs and makes the software user-friendly. The design phase is significant because it translated the needs identified during the analysis phase into a visual appealing, organized, and implementable system structure.

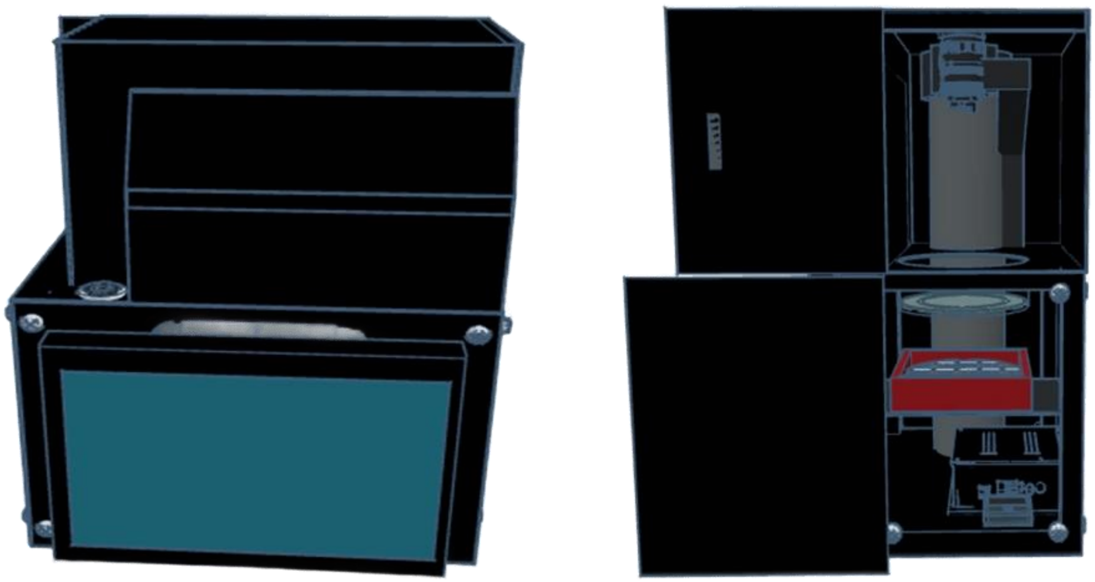

Figure 4.3 Final Design of the Prototype

Phase 3: Developed System

During the development phase, this is where the researchers assemble and test the hardware and also to write and check the code. For hardware, connecting the tubular for the skeletal of the system came first. Following that, the researchers built the system around the Raspberry Pi 5, which served as a central processing unit of the device. A Raspberry Pi HQ Camera based on the IMX477 sensor was integrated as the main image acquisition device of the system. This camera takes the sample images needed for detection and analysis. A 7-inch LCD touchscreen with a resolution of 800x480 pixels was also integrated to provide a user-friendly interface for operating the prototype, this will serve as a navigation to the major functions of the system. The white LED and UV LED supports multiple viewing conditions and lets the software to run the appropriate lighting mode based on the current analysis condition. The hardware also includes a load cell integrated with an HX711 amplifier module which responsible for measuring sample mass and converts the analog load cell output into a digital signal readable by the Raspberry Pi.

Figure 4.4 The Developmental Stage of the Hardware

These hardware components make sure that they are in the right connections, and that each component is working properly and delivering correct readings. This phase established the physical infrastructure needed for the system to support image capture, controlled illumination, sample weight measurement, user-interactive, and safe embedded operation. The actual hardware assembles of these components allowed the prototype to function as a complete microplastic detection rather than as an isolated software application.

Table 4.1 GPIO Pin Assignment of the Prototype

Function	GPIO Pin	Physical Pin
White LED Control	GPIO17	Pin 11
UV LED Control	GPIO27	Pin 13
Shutdown Button	GPIO3	Pin 5
HX711 DOUT	GPIO22	Pin 15
HX711 SCK	GPIO23	Pin 16

Table 4.1 shows the GPIO pin assignment used in the MICROTECT prototype with the assigned functions, GPIO pins and the corresponding physical pin numbers on the controller board. Proper GPIO allocation is needed to ensure stable communication, efficient operation, and compatibility between the hardware components of the system.

To turn on the illumination for sample view and image acquisition, the White LED Control was mapped to GPIO17 (Physical Pin 11). GPIO17 was chosen as it is a stable general-purpose output pin that can be used for reliable digital signal control, and is suitable for switching the white LED on and off without interfering with other communication protocols employed in the system.

UV LED Control was assigned to GPIO27 (Physical Pin 13). GPIO27 is used as an active light source to detect possible microplastic particles in rock salt samples. GPIO27 was selected as it is a dedicated GPIO pin, can provide a stable digital signal output, and is not frequently used for essential communication interfaces, reducing any potential interference and guaranteeing reliable activation of the UV LED.

The Shutdown Button was connected to GPIO3 (Physical Pin 5) to provide safe and convenient system shutdown procedures. GPIO3 was chosen because it has a special function as a serial communication pin (UART RX) but also could be used as a trigger based on input. This pin is commonly used in embedded systems for wake-up or shutdown related functions, so it is a good choice for power management in the MICROTECT prototype.

The HX711 DOUT (Data Output) was connected to GPIO22 (Physical Pin 15) and the HX711 SCK (Serial Clock) was connected to GPIO23 (Physical Pin 16) for weight measurement. The chosen pins both support reliable digital communication and are next to each other, making it easier to organize wiring and reduce signal interference. Their task is to transmit the data from the load cells accurately to the controller, which measures the rock salt samples precisely during the detection process.

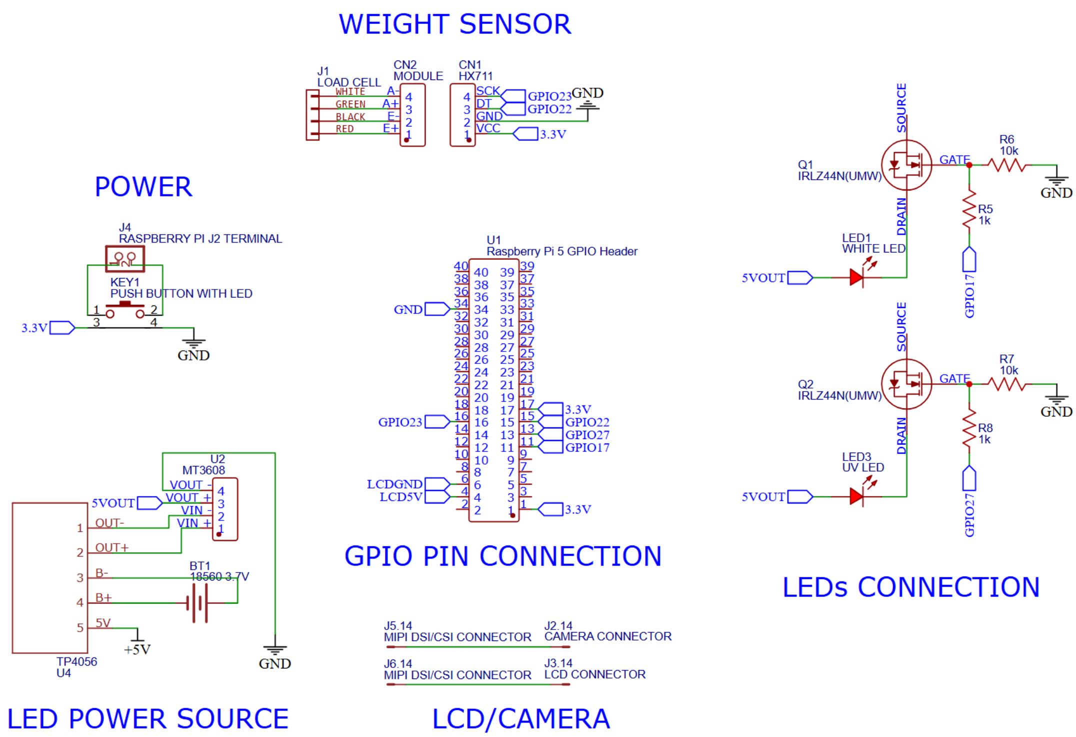

Figure 4.5 Circuit Diagram

Figure 4.5 shows the complete circuit diagram of the MICROTECT prototype, showing the electrical interconnection between the main processing unit and the peripheral components used in the system. The diagram shows the integration of the different hardware modules for automated microplastic detection in rock salt samples by controlled illumination, image acquisition, weight sensing and user operation.

The weight sensor wiring connection is done in the system to measure the weight of rock salt samples before analysis consists of the HX711 module and load cell. The HX711 is a specialized high precision analog to digital converter for load cell applications.

Because the load cell produces very small analog signals, the HX711 amplifies and converts these signals to digital data that can be processed by the Raspberry Pi 5.

For the communication of the HX711, the GPIO22 and GPIO23 pins were used because of their reliable digital input/output feature and ability to send stable signals for accurate sample measurement. Accurate weight measurement is important in MICROTECT as it contributes to determining the concentration level of detected microplastics based on the sample mass.

The LED connection circuit includes both the white LED and UV LED, which are critical illumination components of the detection process. The white LED was used for general lighting for the visibility of the samples and image clarity while the UV LED was used as the major light source to improve the visibility of potential microplastic particles under ultraviolet exposure. As the LEDs might need more current than the Raspberry Pi GPIO pins can safely provide, MOSFET transistors (IRLZ44N) were added to the circuit as switching devices. These MOSFETs are electronic switches that allow the Raspberry Pi to control higher-powered LEDs without putting the GPIO pins at risk. Resistors connected to the gate terminals help to stabilize switching performance and prevent unwanted activation caused by floating signals. We choose GPIO17 and GPIO27 because they are stable digital output pins, which can provide a stable control signal to activate the LED.

Figure 4.6 Actual Photo of the Prototype

On the other hand, the first thing in software development is to test the Raspberry Pi 5 in the laptop by using the command prompt. The software was developed using the Python language. The researchers addressed challenges related to optimizing memory usage for inference, calibrating the camera focus, and synchronizing image acquisition with model inference. Moreover, interface in this project will provide users an easy way to control microplastic detection process, view real time results and monitor statistical data. It connects the hardware and AI system to the user ensuring simple, accurate and efficient operation for retail and domestic use.

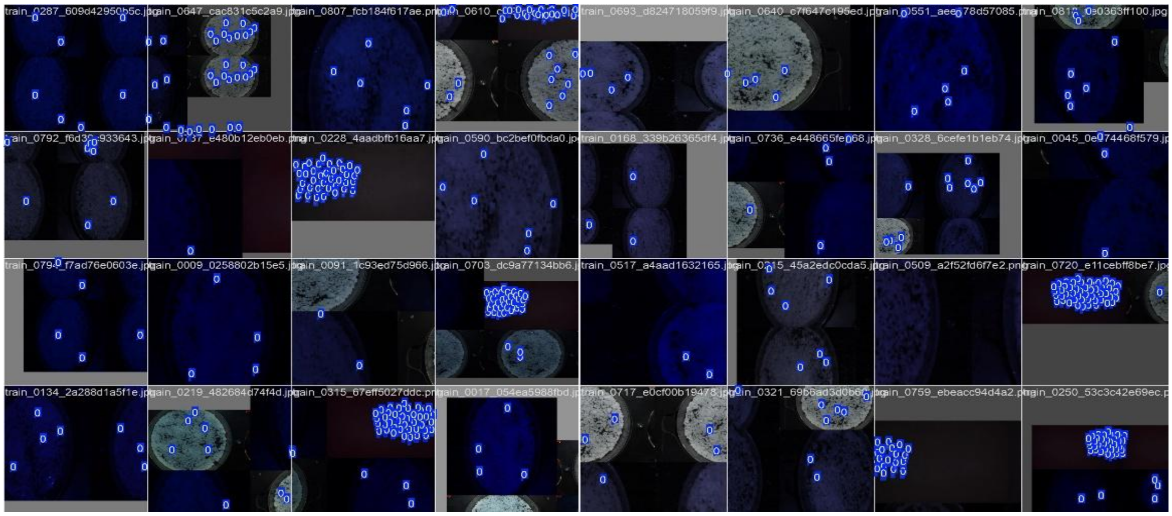

Figure 4.7 Training Object Detection Model

The development of the microplastic detection model focused on creating an AI capable of identifying suspected microplastic particles from captured sample images. The prototype uses a YOLOv11 nano object detection model to trained to identify microplastic or similar visual features from the dataset. In the prototype, white and UV light model paths are supported, letting the prototype to work in the most applicable model for the illumination condition used during testing.

1.1 AI Model Development and Training Performance

YOLOv11 was selected for embedded deployment because it strikes a balance between inference speed and detection accuracy on the Raspberry PI 5's hardware. The system maintains two model weights that was trained individually, one optimized for white light imagery and one for UV light imagery. This enables the application to switch models based on the active LED channel.

Particle Sizing Formula

This was used in each detected bounding box, the MICROTECT software converts the pixel-space measurement into a physical size estimate expressed in micrometres (μm). The conversion follows the formula:

$$d_{\mu\mathrm{m}} = N_{px} \times C \tag{9}$$

Where:

$\boldsymbol{d_{\mu\mathrm{m}}}$ = Estimated particle size in micrometers (μm)

$\boldsymbol{N_{px}}$ = Bounding box diagonal in pixels, computed as

$$N_{px} = \sqrt{w^2 + h^2} \tag{10}$$

$\boldsymbol{w}$ = Width of the Bounding box diagonal (pixels)

$\boldsymbol{h}$ = Height of the Bounding box diagonal (pixels)

$\boldsymbol{C}$ = Pixel to micrometer calibration factor (μm/pixel), determined during the calibration process using a reference object of known physical dimensions.

The calibration factor was determined before testing by imaging a reference object of known dimensions. This calibration allows the pixel measurements to be correctly translated into real world units. The calculated factor was saved in the system and was applied uniformly during all detection processes.

The model was trained on a custom dataset of microplastic images taken under white and UV light. Images were taken from the prepared samples seeded with microplastic fragments of different morphology including fibers, fragments and films. Each image was manually annotated by the researchers with bounding boxes for the single class microplastic. The dataset was divided into training and validation subsets as is done in standard machine learning practice.

Table 4.2 Training Configuration

Parameter	Value
Model architecture	YOLOv11n
Total training epochs	140
Input image size	640x640 px
Training platform	Google Colab (GPU acceleration)
Optimiser	SGD with momentum
Output classes	1 (microplastic)

Model Training Results

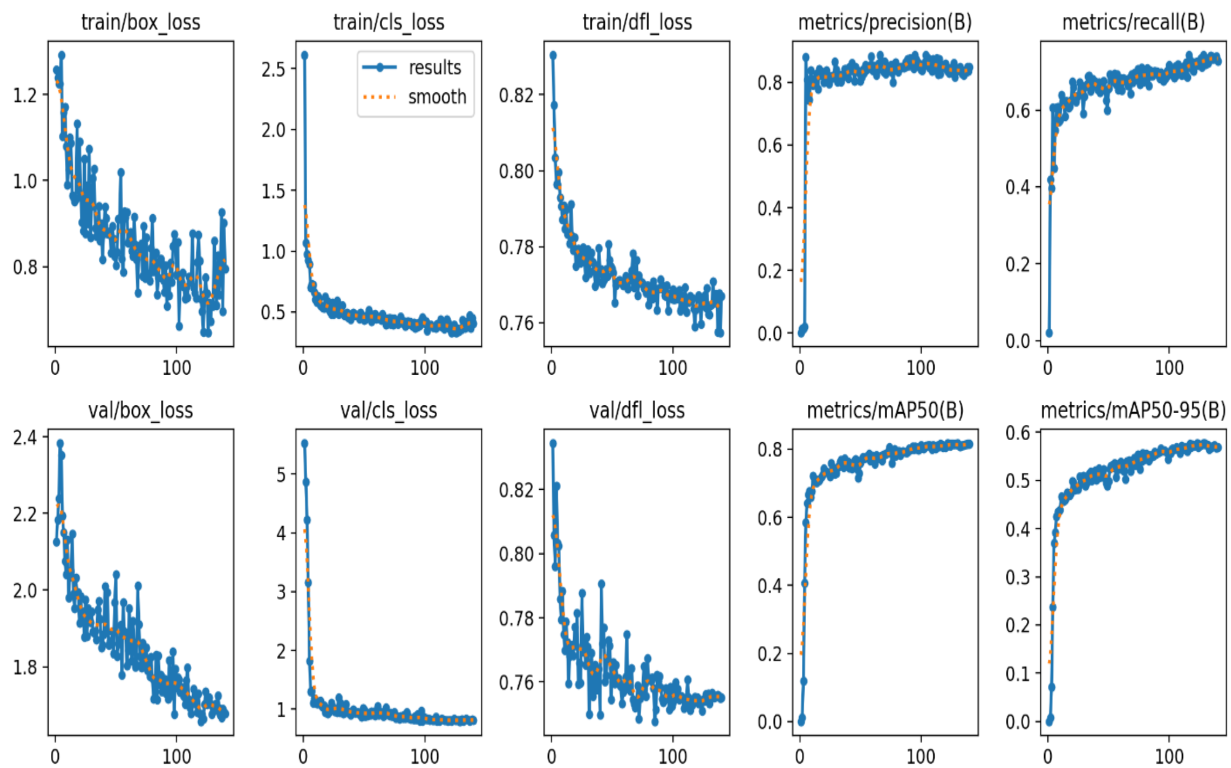

Figure 4.8 Training and Validation Loss and Map Curves

Figure 4.8 shows the loss and mean average precision (mAP) values recorded per epoch during AI training. The box loss, classification loss, and distribution focal loss curves for the both training and validation sets are shown together with the mAP@50 and mAP@50-95 curves. The constant downward trend of all loss curves and the stable rise of the mAP curves prove that the model learned to detect microplastic particles with improving accuracy over the full 140-epoch run.

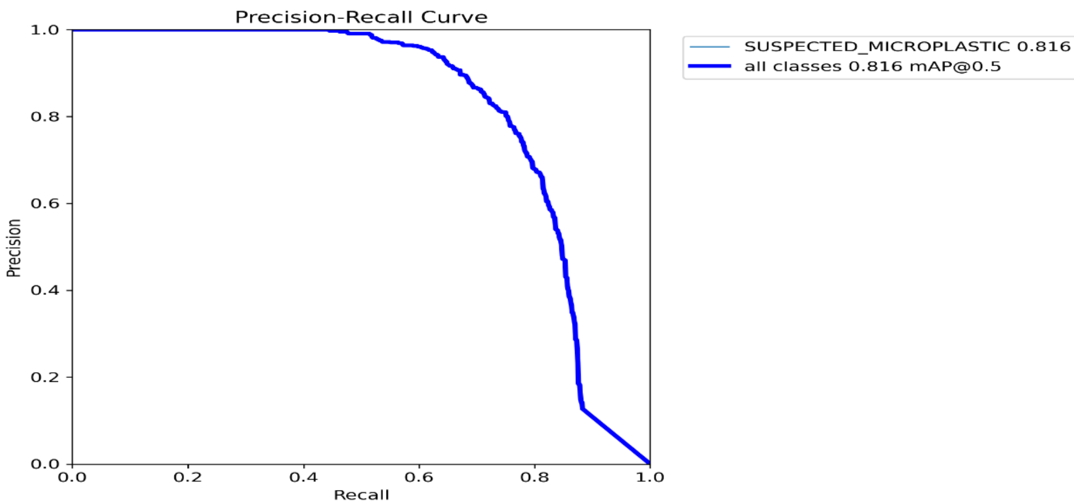

Figure 4.9 Precision-Recall Curve

Figure 4.9 shows the Precision-Recall (PR) curved of the trained model evaluated on the validation set. The area under the curve measures the model's ability to recover true positive detections at different confidence thresholds. The large area under the curve is typical for high detection quality at different operating points.

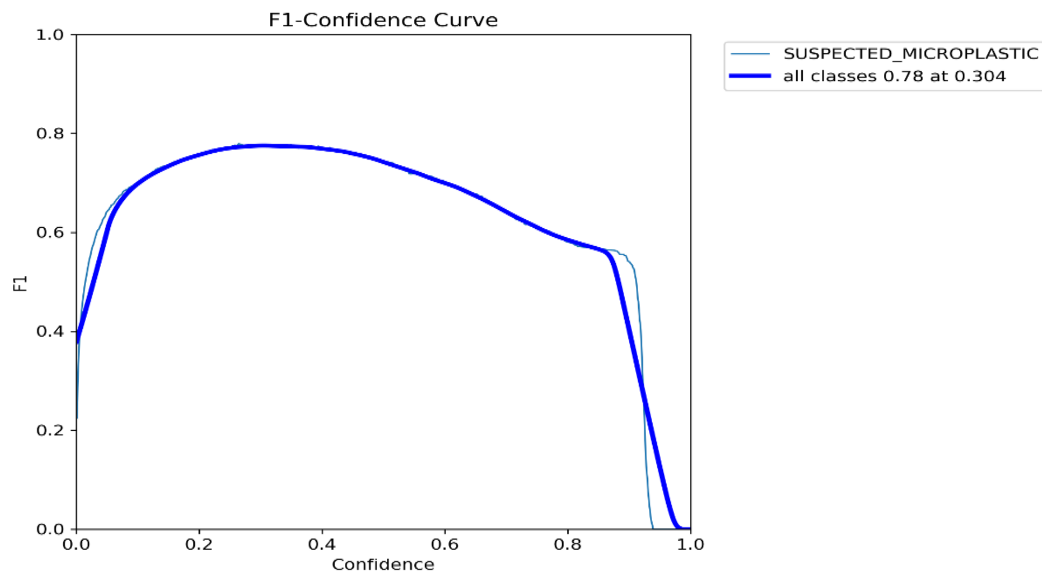

Figure 4.10 F1-Confidence Curve

Figure 4.10 F1 score as a function of the confidence threshold. The F1 score is the harmonic mean of precision and recall. The curve shows the confidence in the operating point that maximizes the balance between precision and recall for the microplastic class, which informs the confidence threshold setting used during live sample detection.

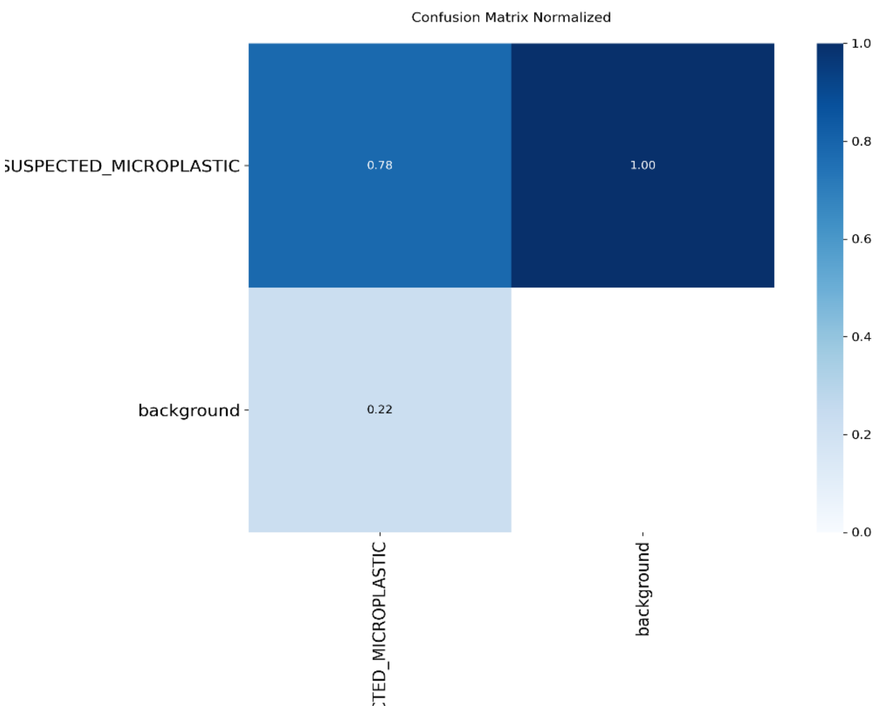

Figure 4.11 Normalized Confusion Matrix

Figure 4.11 shows the normalized confusion matrix produced after the final validation pass. The matrix shows the proportion of true positive and true negative predictions relative to the ground truth, enabling assessment of classification reliability for the microplastic class.

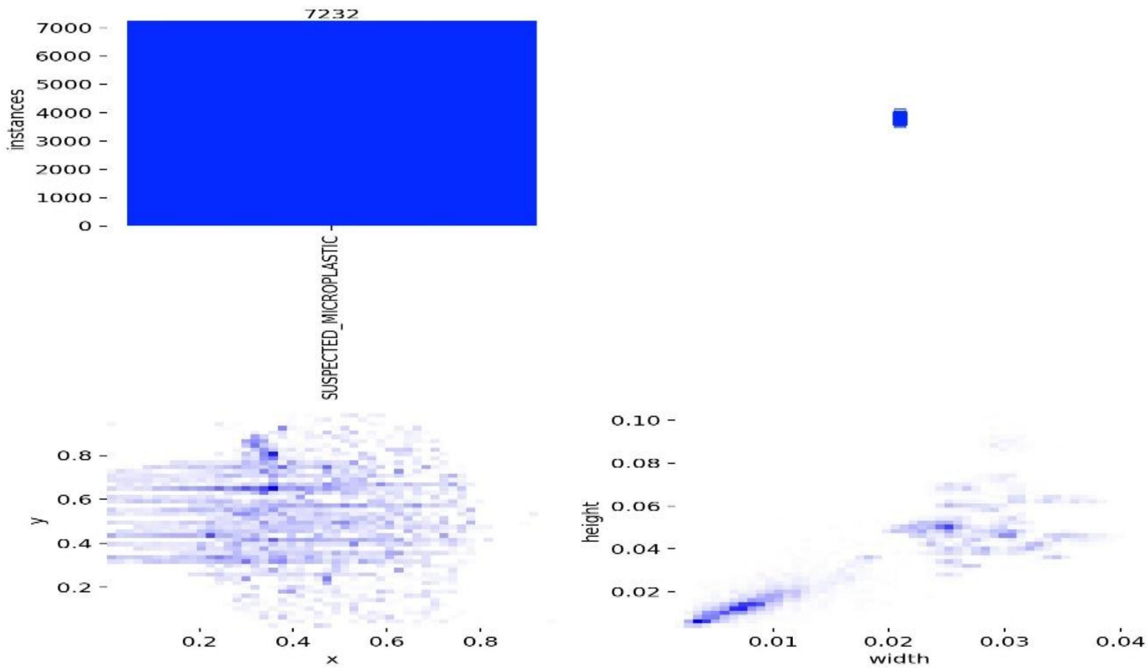

Figure 4.12 Training Label Distribution

Figure 4.12 shows the position and size distribution of the ground truth bounding box annotations in the training dataset. This figure confirms the annotations are well spread among particle sizes and positions in the image frame, allowing strong generalization of the trained model.

Training Results Summary

Training and validation metrics were collected for each epoch. Below is a table summarizing the best checkpoint values recorded over the 140 epoch run.

The dataset used for the YOLOv11n model training consists of images captured under white-light and ultraviolet (UV) illumination conditions. The dataset was divided into training, validation and test sets to ensure proper model development and evaluation.

This split allows the model to learn relevant features, validate the model's performance during training, and perform unbiased testing.

Table 4.3 Training Results Summary

Variable	Value
Precision	82.6%
Recall	73.3%
Mean Average Precision (mAP@0.5)	81.4%
Mean Average Precision (mAP@0.5:0.95)	58.5%
Number of Validation Images	82 Images
Total Detected Instances	813 Instances
Inference Time per Image	31.3 ms

Table 4.3 summarizes the results of the microplastic detection from rock salt samples of the YOLOv11n model. The model achieved a precision of 0.826, meaning that the model correctly identified a high proportion of detected microplastic particles, and there was a low rate of false-positive detection.

The recall value of 0.733 indicates that the model is able to detect a significant number of actual microplastic cases, but still misses some particles. This reflects a balanced trade-off between precision and recall, which is essential in object detection tasks.

The model achieved a mean Average Precision (mAP@0.5) of 0.814, indicating robust detection performance at the standard Intersection over Union (IoU) threshold. However, the mAP@0.5:0.95 value of 0.585 indicates moderate performance with more strict localization criteria, which indicates that there is room for further improvements in bounding box accuracy.

In terms of computational efficiency, the model recorded an average inference time of 31.3 ms per image, showing its potential for near real-time detection. Overall, these results prove the effectiveness and suitability of the developed model for integration into the proposed automated optical microplastic detection system.

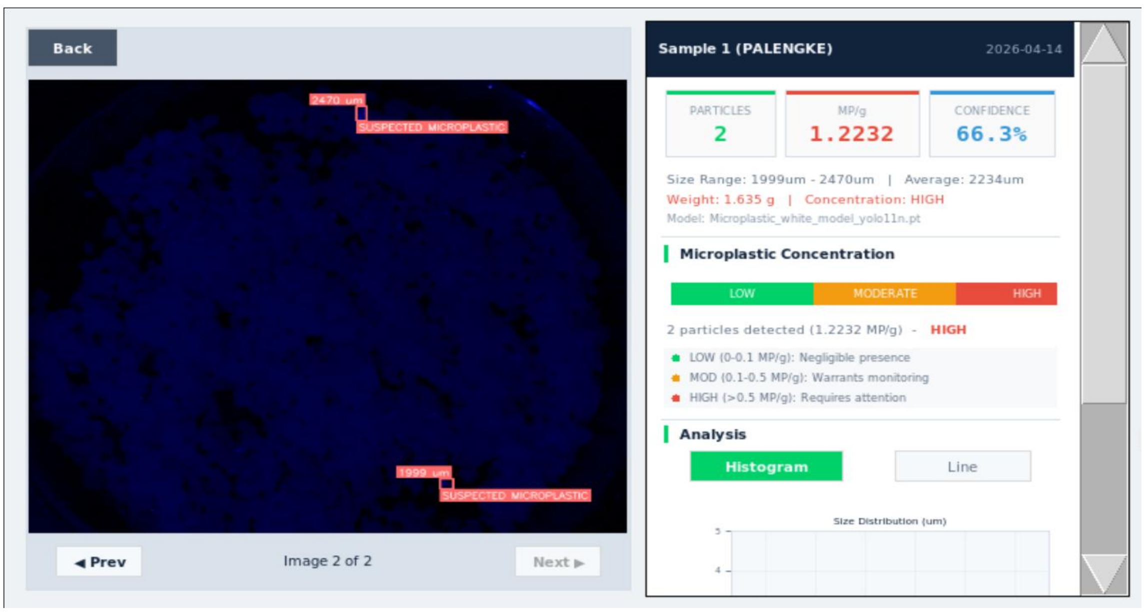

Figure 4.13 Detection Page of the System

This figure 4.13 shows the results of the microplastic analysis using the developed system. The left window displays the captured image of rock salt sample under UV light, where you can see the suspected microplastics are highlighted and labeled with their measured sizes. The right window displaying the summarize results of the analysis, that includes the total number of particles detected, microplastic concentration, and confidence level of the AI. Also, it provides other data like size range, average particle size, and sample weight. It also contains the classification of contamination level as low, moderate, or high based on the set thresholds. Moreover, the interface includes a graphical representation to visualize the size distribution of detected particles. Overall, this page allows users to easily interpret the detection results and assess the level of contamination of rock salt sample.

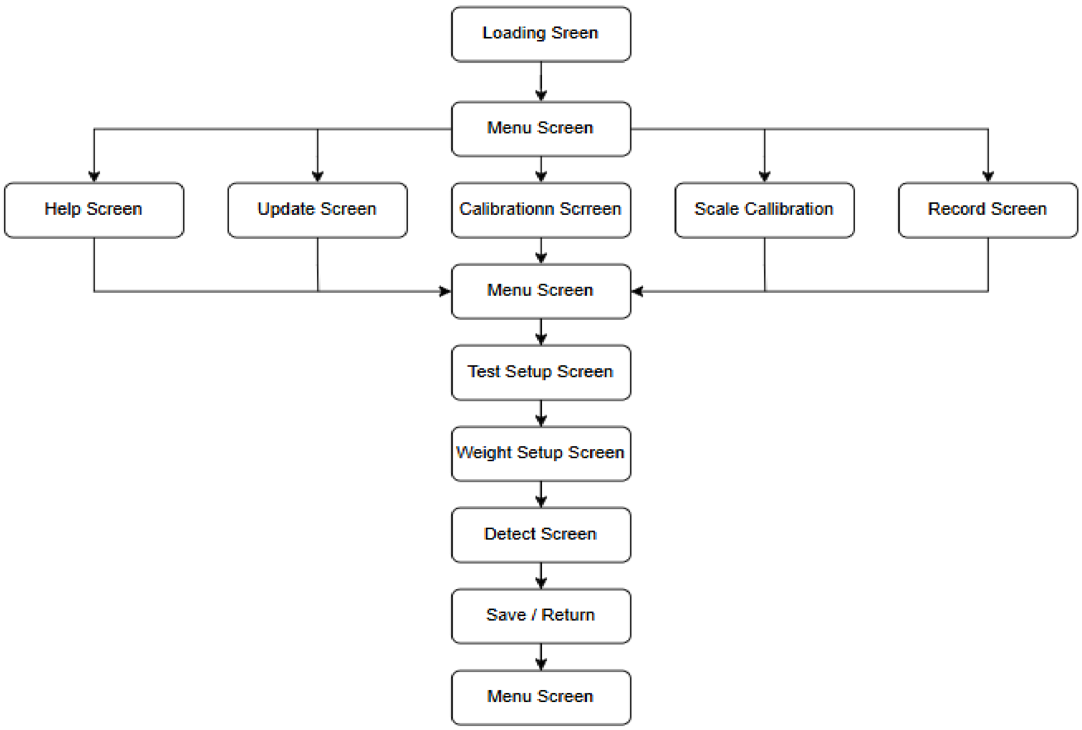

Figure 4.14 Transition Diagram

The Figure 4.14 shows the system begins when the prototype is powered on and the application loads the main system. Following the loading screen, the system opens the main menu. From that, the user may begin the test, view records, access calibrations, open the scale monitor, update screen, or read the help section. The testing workflow follows the sequence of test setup, weight setup, and live detection, come after the result saving and return to the main menu.

The Graphical User Interface (GUI) was executed using Tkinter and layout as a full screen touchscreen environment fitted for kiosk-style deployment. The interface of the "MICROTECT" system was developed as a user-friendly touchscreen-based application for Raspberry Pi 5 using Python language, designed for easy and guided operation for the users. It follows a complete workflow where the user begins at the Main Menu and selects "Begin Test", it will walk you through to a test setup screen where sample information is entered, next is the weight confirmation screen that displays the measured rock salt weight using load cell.

Once the user clicks "Begin Detection", the system opens a two window, one is for a live camera showing a real-time particle detection, counts, and size measurements, on the other hand, the other window is where you can see the data you can be obtained like numbers of particle, graphs, and classification.

Additionally, users can control capture, lighting (white and UV LED), and settings and watch the results. Once completed, clicking "End Test" displays a summary screen where results can be reviewed and saved. The users can automatically generate pdf records and reports. The interface allows a smooth and automated process from entering data to the final analysis output.

2. System Functionality and Performance Testing

In this section, the functionality and performance testing results of the developed system are presented and the operational capability of the system is evaluated according to the quality characteristics of ISO/IEC 25010.

Table 4.4 Camera and Illumination Functionality Test Summary

Component	Test Performed	Expected Output	Actual Output	Measured Value	Result
Camera Module (IMX477)	Live preview at 1920×1080	Real-time video at 30 FPS	Stable live video displayed	~28–30 FPS	Passed
White LED Module	GPIO17 toggle ON/OFF	Uniform white illumination	Even illumination across sample	~400–600 lux (estimated)	Passed
UV LED Module	GPIO27 toggle ON/OFF	UV illumination of sample	Fluorescent particles visible under UV	365 nm wavelength	Passed
Detection Preview	YOLOv11n inference on live feed	Real-time bounding box detection	Bounding boxes displayed correctly	~31 ms inference time (~30 FPS)	Passed

The table 4.4 shows the functional and performance evaluation of the camera, illumination and detection subsystems based on commonly accepted real-time vision system performance criteria and illumination standard reported in the industry practices.

. Every component was evaluated for functional fitness and performance efficiency to ensure that the developed system meets the operational requirements.

The HQ camera module showed stable real-time video acquisition at approximately 28–30 frames per second, 24 frames per second is widely recognized as the minimum for continuous motion perception in video systems, while 30 frames per second is the standard for real-time computer vision applications, satisfying the required performance efficiency for continuous monitoring. The white LED module provided uniform illumination across the area of the sample with an estimated intensity sufficient for clear image acquisition, supporting its functional suitability. The selected range of illumination is based on a standard indoor and laboratory lighting conditions used for visual inspection tasks, which are generally between 300 and 1000 lux to allow sufficient visibility without overexposure.

The UV LED module was successfully used to visualize fluorescent particles under ultraviolet light. Its operating wavelength range of 365-395 nm is consistent with known fluorescence characteristics of some microplastics including forensic and material analysis applications as it effectively reacts to fluorescing particles such as microplastics. Moreover, the addition of an auto shut-off feature enhances the safety of operation by preventing heat accumulation.

Real-time detection systems typically require inference times below 50 milliseconds to maintain interactive performance, with 30 milliseconds corresponding to approximately 30 frames per second.

Overall, the results indicate that the developed system satisfies key ISO/IEC 25010 quality attributes, particularly in terms of functional suitability and performance efficiency, validating its readiness for deployment in microplastic detection applications.

Table 4.5 Load Cell and Sample Weight Accuracy Test

Trial No.	Reference Weight (g)	Measured Weight (g)	Absolute Difference (g)	Percentage Error (%)	Remarks
1	5.00	4.95	0.05	1.00%	Within acceptable range
2	10.00	9.98	0.02	0.20%	High accuracy
3	20.00	19.96	0.04	0.20%	High accuracy
4	50.00	49.95	0.05	0.10%	High accuracy

. The accuracy of the load cell measurements was evaluated by calculating the absolute difference and percentage error using the formulas as shown in Chapter 3.

The table 4.6 presents the accuracy evaluation of the load cell used in measuring the sample weight for microplastic detection. The results show a small difference between the reference and measured weights in a number of trials.

The computed percentage error ranges from 0.10% to 1.00%, indicating a high level of measurement accuracy.

. The results confirm that the load cell provides reliable and reproducible weight measurements, which are essential for standardization of sample preparation.

The results confirm that the load cell provides accurate and reliable measurements, which are important for consistent microplastic concentration analysis.

Phase 4: Implement

In the implementation phase, unit testing was performed to assess the performance of the AI-based optical detection system developed. The system was tested using multiple rock salt samples, and the results were recorded for analysis.

The microplastic detection results were obtained using the developed system. The values were computed using the formulas presented in Chapter 3, including particle size estimation and microplastic concentration.

2.1 Microplastic Detection Results Using Image Processing and AI

This section presents the recorded data obtained from the developed system, including microplastic count and particle size.

Data Gathering for Rock Salt Samples

Table 4.6a. Microplastic Count Detected per Sample Source

Source	Location	No. of Tests (n)	Mean MP Count	Standard Deviation	Minimum	Maximum
Pangasinan	A	50	1.86	1.46	0	8
Pangasinan	B	50	0.46	0.71	0	2
Mindoro	A	50	2.78	1.45	1	7
Mindoro	B	50	1.50	1.23	0	5
Palengke	A	50	4.12	2.23	0	9
Palengke	B	50	1.08	1.01	0	3

In Table 4.6a presents the microplastic count detected per sample source across different locations. Among all sources, Palengke Location A had the highest mean microplastic count with a value of 4.12, showing that there is greater presence of microplastics in the collected samples. In contrast, Pangasinan Location B has the lower mean microplastic count of 0.46. The result also shows that all sample sources had varying levels of microplastic contamination, as reflected in standard deviation and range of minimum to maximum values. Moreover, Palengke Location A showed the highest maximum count of 9, while Pangasinan Location B had the lowest maximum of 2. The findings suggest that the level of microplastic contamination fluctuate depending on the sample source and location.

Table 4.6b. Average Microplastic Size Detected

Source	Location	No. of Tests (n)	Mean Size (µm)	Standard Deviation	Minimum	Maximum
Pangasinan	A	50	1561.42	469.69	927	3195
Pangasinan	B	50	2227.26	2142.75	927	9072
Mindoro	A	50	1448.03	482.49	721	3814
Mindoro	B	50	1527.66	634.57	927	5463
Palengke	A	50	1510.25	475.96	824	3092
Palengke	B	50	1599.43	999.29	824	7938

In Table 4.6b presents the average microplastic size detected from different sample sources and locations. Out of all samples, Pangasinan Location B recorded the highest mean microplastic size at 2227.26 µm, while Mindoro Location A obtained the lowest mean microplastic size of 1448.03 µm. The results indicate variations in microplastic sizes across all sample sources. Pangasinan Location B showed the highest maximum size of 9072 μm, suggesting the presence of larger microplastic particles in some rock salt samples. Meanwhile, the minimum detected sizes ranges from 721 μm to 927 μm across all different sample location. These findings indicated that the detected microplastic sizes vary depending on the sample source and location.

Table 4.6c. Concentration Level of Microplastics

Source	Location	No. of Tests (n)	Mean Concentration (MP/g)	Standard Deviation	Minimum	Maximum
Pangasinan	A	50	0.1912	0.1435	0.0000	0.7624
Pangasinan	B	50	0.0493	0.0771	0.0000	0.2415
Mindoro	A	50	0.2846	0.1515	0.0941	0.7689
Mindoro	B	50	0.1705	0.1824	0.0000	1.0936
Palengke	A	50	0.4326	0.2411	0.0000	0.9697
Palengke	B	50	0.1227	0.1139	0.0000	0.3765

The table 4.6c presents the concentration level of microplastics detected from the collected different sample sources and locations. Based on the results, Palengke Location A has the highest mean concentration level with a value of 0.4326 MP/g, indicating that this sample source contained the greatest amount of microplastics among all tested samples, while, Pangasinan Location B has the lowest mean concentration level at 0.0493 MP/g, suggesting a lower presence of microplastic contamination in the collected samples from that location.

In addition, Palengke Location A had the highest standard deviation of 0.2411, which indicates that the concentration levels among its samples were more dispersed or inconsistent. Meanwhile, Pangasinan Location B had the lowest standard deviation of

0.0771, suggesting that the detected concentration levels were more consistent across the conducted tests.

Moreover, in terms of maximum values, Mindoro Location B recorded the highest maximum concentration level of 1.0936 MP/g, while the other locations had minimum values of 0.0000 MP/g, indicating that some rock salt samples showed no detectable microplastics during the testing. The overall findings indicate that the microplastic concentration level varies with respect to the sample location and source. These variations may be affected by environmental conditions, handling processes, or the different source of the rock salt samples collected during the study.

Distribution of Microplastic Concentration Across 300 Tests by Source/Location

. To further illustrate the distribution and consistency of microplastic contamination across the entire 300 tests, Figure 4.6d shows the concentration values per test by source and location.

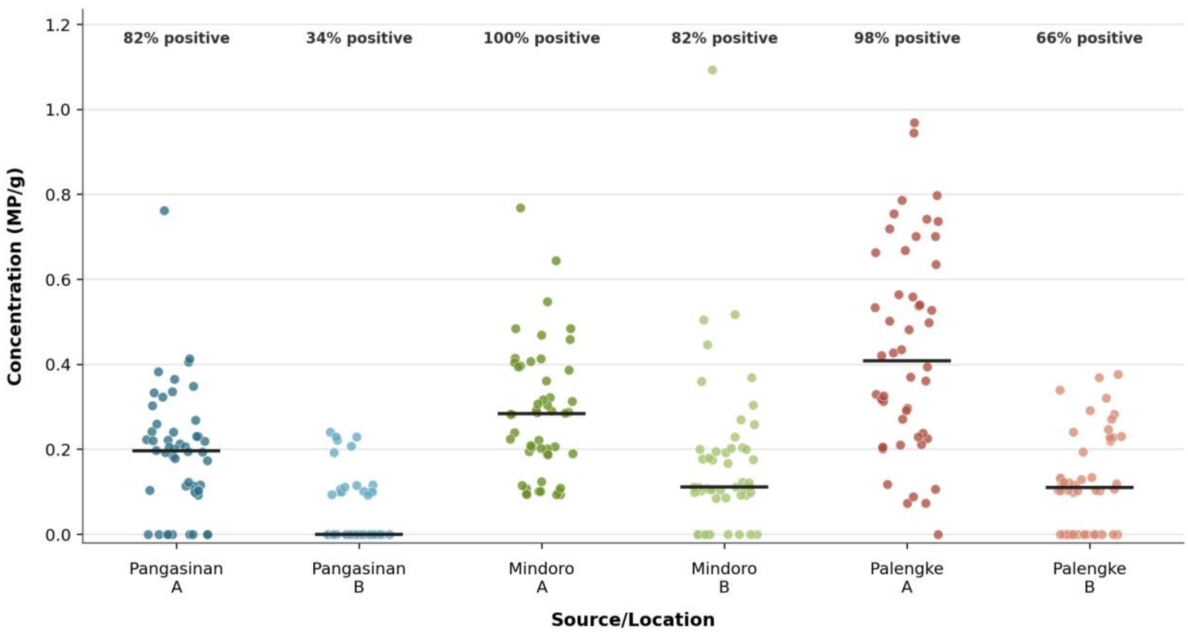

Figure 4.15 Scatter Plot Microplastic Concentration Across 300 Tests

As shown in Figure 4.15, the concentration values of microplastics varied among the different sample sources and locations. Palengke A had the highest overall contamination pattern as evidenced by its generally higher concentration values with the highest mean concentration of 0.4326 MP/g. It also showed a very high positive detection rate of 98%, indicating that microplastics were detected in almost all tested samples from this source. Mindoro A, on the other hand, showed the most consistent presence of contamination with a 100% positive detection rate across all 50 tests and a mean concentration of 0.2846 MP/g. In contrast, Pangasinan B had the lowest contamination pattern, with many tests with zero detected concentration and a positive detection rate of only 34%, consistent with its lowest mean concentration of 0.0493 MP/g. The results indicate that the levels and variability of microplastic contamination in rock salt samples depend on the source and location.

The scatter plot shows the distribution of the concentration values of each group as well. Some sources, such as Palengke A and Mindoro B, showed a wider variation in the recorded levels of concentration. Pangasinan B, however, was concentrated near zero, which results in a lower number of contaminated samples. This figure supports the interpretation of Table 4.6c by showing not only the mean values of concentration but also the actual distribution of contamination in the repeated tests.

3. Comparison of AI-Based Detection and Laboratory Results

This section compares the developed AI-based detection system with conventional laboratory methods in terms of microplastic count, size estimation, and processing time.

Table 4.7 Comparison with Laboratory Results

The results of the developed system were compared with laboratory analysis to evaluate its accuracy. The comparison was performed using percentage error and mean absolute error.

Comparison Results for Microplastic Analysis							
Source	AI-Based Optical Detection System					Laboratory Result	Accuracy Percentage (%)
	Trial No.	Microplastic Count	Detected Size (µm)	Microplastic Concentration (MP/g)	Remarks	Microplastic Count	
Pangasinan	1	4	967 - 1505	0.313	MODERATE	3	88.89%
	2	2	1176 - 1411	0.160	MODERATE		
	3	2	1075 - 1397	0.158	MODERATE		
Mindoro	1	3	1182 - 3333	0.280449	MODERATE	6	55.56%
	2	4	967 - 2795	0.37021	MODERATE		
	3	3	1182 - 2580	0.279293	MODERATE		
Marikina Public Market	1	3	1764 - 2117	0.0731	LOW	5	66.67%
	2	4	1411-1764	0.3160	MODERATE		
	3	3	1058-1529	0.5341	HIGH		
Household Rock Salt	1	7	1411 - 2705	0.9883	HIGH	10	53.33%
	2	5	1294 - 1999	0.4658	MODERATE		

	3	4	1058 - 2705	0.2694	MODERATE		
Random	1	3	1529 - 3999	0.2352	MODERATE	4	75.00%
	2	3	1294 - 1529	0.4274	MODERATE		
	3	3	1058 - 2352	0.3102	MODERATE		
Mean Detection Accuracy Based on Microplastic Count of AI (%)				**67.89%**			

The table 4.7 presents the results of microplastic detection using the AI-based optical detection system across multiple trials for each sample source. For each source, three trials were conducted to assess the system's detection behavior in terms of microplastic count, detected size, processing time, and contamination classification. In addition, it also shows how it differs from the laboratory method in terms of microplastic count.

For the Pangasinan samples, the detected microplastic counts ranged from 2 to 4 particles, with all trials classified under a moderate contamination level, while the detected microplastic count in laboratory is 3 particles. The variation in counts across trials and laboratory resulted an accuracy percentage of 88.89%, with the AI system is closely matching the laboratory, indicating stable detection and reasonably reliable performance.

The detected counts for the Mindoro samples were 3 to 4 particles also consistently classified as moderate contamination but it yielded a lower accuracy percentage of 55.56%, where the AI system is slightly underperformed compared to laboratory count results.

The difference in sizes and numbers detected indicates the difficulty in detecting some particles which may be due to differences in the composition or visibility of the sample.

For Marikina Public Market, the detected microplastic particles ranged from 3 to 4 particles and the levels of detection were from low to high. This variability indicates that the performance of the system is affected by the difference of the size and distribution of the particles, although the detection is reasonably with the lab results with an accuracy percentage of 66.67%.

. The samples from the Household source showed the lowest accuracy of 53.33% among the sources. One of the trials received a high score, but the results indicate some discrepancies, which could suggest that some microplastics are more difficult to identify due to the presence of overlapping particles or less pronounced features.

For the Random source, the samples had a relatively good accuracy of 75% with moderate remarks being consistent across trials. This indicates that the system shows a generally reliable performance under varied and less controlled conditions, keeping a balanced detection capability.

Overall, the calculated mean detection accuracy of 67.89% indicates that the AI-based optical detection system shows generally consistent performance, with variability determined by sample characteristics such as particle size, concentration, and visibility.

Table 4.8 Processing Time Classification

The results of the system processing time were interpreted according to the classification in Table 4.9. These categories provide a common basis for evaluating system responsiveness.

Processing Time (s)	Classification
< 0.010	Very fast
0.010 – 0.030	Fast
> 0.030	Acceptable

Table 4.9 Processing Time Analysis

To evaluate the performance efficiency, the processing time of the developed AI-based optical detection system was evaluated. The time taken to analyze each sample was recorded and compared over multiple trials.

Source	Trial No.	Processing Time (ms)	Remarks
Pangasinan	1	9	Very Fast
	2	9	Very Fast
	3	8	Very Fast
Mindoro	1	23	Fast
	2	23	Fast
	3	36	Acceptable
Mean	—	18 ms	Near real-time detection

The system achieved an average processing time of about 0.018 seconds (18 milliseconds) per sample, indicating near real-time detection capability.

The processing times measured for the various samples show a very little variation indicating a stable performance of the system during operation. Except for the recorded values that are all within a narrow millisecond range, Pangasinan samples had a faster processing time than Mindoro samples.

The processing time was classified using latency categories such as “very fast”, “fast” and “acceptable” for standardization of the interpretation of the system performance. Research shows that humans perceive system response in milliseconds and the faster the response time the better the system performs (PubNub, 2024). Since processing times from the system are between 0.8 and 0.36 seconds, the results are well within real-time performance.

The results indicate that the system can analyze samples efficiently with a minimum of delay, which is suitable for practical applications. Moreover, the accuracy across multiple runs shows that the system shows a stable performance without degradation upon repeated use, confirming its reliability for continuous operation.

The results show that in general the developed system provides significantly faster analysis than the conventional laboratory methods with acceptable accuracy, confirming the efficiency and practical applicability of the developed system.

4. System Evaluation Based on ISO/IEC 25010

This section presents the evaluation of the developed system based on ISO/IEC 25010 quality characteristics, including functionality, usability, performance efficiency, and reliability.

Phase 5: Evaluate

The Evaluations of the Respondents About the “Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use” Based on ISO 25010 Product Quality Standard.

Table 4.10 Likert Scale Guide

Scale	Range	Verbal Interpretation
5	4.50-5:00	Highly Acceptable
4	3.30-4.49	Acceptable
3	2.50-3.29	Moderately Acceptable
2	1.80-2.49	Needs Improvement
1	1.00-1.79	Unacceptable

Fifteen respondents completed the evaluation form and the researchers gathered and analyzed the data to get the mean result and to make a verbal interpretation. The rating scale ranged from 5 which meant Highly Acceptable to 1 the lowest meant Unacceptable. The following table presents information according to the answers of respondents.

Functionality

Table 4.11 Likert Scale for the Functionality

Statement	Mean	Verbal Interpretation
The system accurately detects microplastic particles in rock salt samples.	4.80	Highly Acceptable
The AI-based image processing correctly identifies the microplastic particles.	4.80	Highly Acceptable
The system provides reliable and consistent detection results for retail and domestic users.	4.60	Highly Acceptable
Overall Mean	4.73	Highly Acceptable

The table 4.12 shows the system has a high level of functionality as it scored a Highly Acceptable, which means it performs well on its intended task. The average of the functionality is 4.73 which indicates that the system detects microplastic articles accurately, uses the AI based image processing effectively for the identification and gives reliable and consistent outputs. It also shows consistent performance across multiple trials, producing stable and reliable outputs. Some differences were found based on sample characteristics, but overall the system functioned.

Usability

Table 4.12 Likert Scale for Usability

Statement	Mean	Verbal Interpretation
The system is easy to understand and operate for retail and domestic users.	4.86	Highly Acceptable
The user interface clearly presents detection results and system Information.	4.86	Highly Acceptable
The detection process can be performed with minimal user effort.	4.73	Highly Acceptable
Overall Mean	4.82	Highly Acceptable

In the table 4.13, the system achieved very high mean ratings of 4.82 for usability, indicating that it is user friendly and accessible. The results show that it is Highly Acceptable, i.e. the system is easy to understand and operate, with a clear and informative display that guides the user effectively throughout the detection process. The layout and visual outputs allow users to easily interpret results without confusion. It also allows detection tasks to be performed with little effort and technical expertise, even by non-expert users.

Performance Efficiency

Table 4.13 Likert Scale for Performance Efficiency

Statement	Mean	Verbal Interpretation
The system detects microplastic in rock salt **quickly** using AI-based image processing.	4.93	Highly Acceptable
The response time of the automated optical detector is **acceptable for retail and domestic use.**	4.60	Highly Acceptable
The system performs efficiently **without slowing down or lagging** during continuous operation.	4.40	Acceptable
Overall Mean	4.64	Highly Acceptable

In the table 4.14, the system shows a strong performance efficiency as indicated by the mean scores in detection speed and response time with an overall mean of 4.64. This means that the system is capable of processing and displaying result quickly and responsive supporting real time operation. However, continuous operation shows a slightly lower mean indicating slight performance limitations over longer usage, possibly due to system resource limitations. Overall, the result still shows a responsive and efficient system performance.

Safety and Maintenance

Table 4.14 Likert Scale for Safety and Maintenance

Statement	Mean	Verbal Interpretation
The system is easy to clean, inspect, and maintain for long-term use.	4.86	Highly Acceptable
The system components are durable and protected from damage during normal operation.	4.60	Highly Acceptable
The device provides indicators or alerts in case of system errors or improper operation.	4.53	Highly Acceptable
Overall Mean	4.66	Highly Acceptable

In the table 4.15 showed a high mean rating of 4.66, the results indicate that the prototype is easy to clean and maintain, built with durable components, and equipped with indicators that can ensure proper operation and user safety over time. These results show that the system is feasible for routine application and can sustain performance with minimal maintenance.

The system's overall performance was rated as "Highly Acceptable" in all parameters evaluated, showing that the developed system is effective, reasonably reliable and appropriate for the detection of microplastics in retail and domestic rock salt..

CHAPTER V

SUMMARY FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

This chapter discusses the summary of findings, conclusions, and recommendations of the study. Each answer corresponds to the statement of the problem presented in Chapter 1.

SUMMARY OF FINDINGS

1. **Development of the Automated Optical Detection System**. The successful design and implementation of the system was achieved by integrating hardware and software components. The hardware setup including the Raspberry Pi, camera module, illumination system and sensors complemented the developed AI model and image processing algorithms well. The ADDIE model was used to provide a systematic approach that led the researchers from analysis to evaluation, making sure that all system requirements were met. The successful completion of the prototype shows that the system is operational and able to perform its intended function.
2. **Recorded Data from Image Processing and AI-Based Classification.** The developed system was able to detect microplastic particles in rock salt samples using image processing and AI-based detection. It could produce relevant data such as microplastic count and estimated particle size. Bounding box identification and pixel-to-micrometer conversion were used in the detection process to estimate the particles' physical dimensions. The results validate that

the system can produce meaningful and usable data for microplastic analysis and, therefore, is applicable to retail and domestic use.

3. **Comparison of System Efficiency with Laboratory Procedures.** The system was shown to be able to detect and analyze microplastics in rock salt samples in comparison with conventional laboratory procedures. The mean detection accuracy in terms of microplastic count detection obtained by the system from the comparison result was 67.89 %. Pangasinan had the highest accuracy percentage at 88.89% among the sample sources tested, whereas Household Rock Salt had the lowest at 53.33%. The system was also able to identify particle sizes and concentration levels as well as contamination remarks such as LOW, MODERATE and HIGH. Furthermore, the developed system achieved an average processing time of ~0.018 seconds per sample, demonstrating a faster analysis and confirming its practical use as a preliminary screening tool for microplastic detection.
4. **Effectiveness of the Developed System Based on ISO/IEC 25010.** The evaluation was conducted using the ISO/IEC 25010 quality model and it has revealed that the system is very acceptable based on the respondents' evaluation. Functionally, the camera module, illumination system, detection algorithm and load cell performed according to their intended purpose and produced reliable results. In terms of usability, respondents stated that the system is easy to use and operate. Respondents found the system simple to use and operate. The system showed high performance efficiency with fast

sample processing and analysis with less delay. Lastly, for safety and maintenance, the respondents pointed out that the system is safe to operate, easy to clean and maintain, and constructed with components that support long term operation. The system demonstrated its efficacy and suitability for practical implementation in identifying microplastics in rock salt samples, as evidenced by its overall highly acceptable rating across all evaluated criteria.

CONCLUSIONS

The automated optical detection system was successfully developed and implemented. The study achieved its primary objective of developing a functional automated optical detection system for microplastics in rock salt. The achieved model performance, with a precision of 0.826, recall of 0.733, and mAP@50 of 0.814, indicates that the system can reliably detect microplastic particles with a favorable balance between detection accuracy and false positives, confirming that the system meets its intended design goals.

The system is capable of detecting microplastic particles and generating their microplastic count and size. The system that was developed showed that it could identify microplastic particles and produce important analytical results such as the number of particles and the estimated particle size.

The combination of bounding box detection and pixel to micrometer calibration allowed the system to estimate the physical size of the detected particles. Moreover, the system was able to analyze the contamination level and render the corresponding remarks such as LOW, MODERATE and HIGH. These results show the system as a feasible tool for preliminary detection and assessment in non-laboratory settings.

The developed system demonstrated efficient performance compared with conventional laboratory procedures for preliminary microplastic detection and analysis. The system achieved a mean detection accuracy of 67.89% across all tested sample sources, as determined by the comparative analysis. The highest accuracy was obtained in the samples from the Pangasinan at 88.89% and the lowest accuracy was obtained in the Household Rock Salt samples at 53.33%. While there were differences in accuracy levels, the system was able to consistently identify microplastics in multiple trials and determine the microplastic count, particle size and concentration level. The system also achieved near real time processing with an average analysis time of ~0.018 seconds per sample. These results suggest that the developed system can be used as a reliable preliminary microplastic detection and analysis. But further validation and improvement are still recommended.

The developed system is highly effective and acceptable based on ISO/IEC 25010 evaluation. The evaluation results based on ISO/IEC 25010 quality characteristics indicate that the system meets high standards in terms of functionality, usability, performance efficiency, and safety and maintenance. The system components were operated according to its function and produced efficient processing performance and were considered easy by the respondents. Moreover, it was found that the system was safe to operate and easy to maintain for continuous operation. The positive evaluation for all criteria confirms the effectiveness, reliability, and suitability of the developed system for practical implementation in the preliminary detection of microplastics in rock salt samples.

RECOMMENDATIONS

The following recommendations are based on the findings, conclusions and limitations of the study.

Enhancement of the AI model through expanded and diversified datasets. The system achieved a 67.89% mean detection accuracy, which dropped to 53.33% for household salt due to overlapping particles. Adding more samples to the training dataset, YOLOv11 will improve at identifying complex visual features.

Improvement of imaging and illumination components for better detection accuracy. Detection accuracy fluctuated based on particle visibility and size. Improving camera resolution and efficiency of UV and white LED intensity will help in capturing clear images of smaller transparent microplastics.

Laboratory Validation and Expanded Testing. The system was fast, with 18 milliseconds processing time, but particle counts were sometimes different than what was found in the lab. More rock salt types will be tested and compared to lab data in order to calibrate the detection threshold.

Classification of microplastic morphology. The current system works well for microplastics detection and sizing. We will improve the AI to be able to classify specific shapes like fibers, fragments, and pellets for detailed analytics and to help identify contamination sources.

Integration of cloud-based data storage and monitoring systems. The continuous operation score was slightly lower (4.40) in the ISO 25010 evaluation, due to possible lagging. The data is stored in the cloud system which reduces the processing load of Raspberry Pi 5 and in the long run, it ensures the smooth use of it.

Improvement of hardware design for durability and long-term performance. The physical enclosure must be upgraded for the prototype to be used in retail and domestic settings. Enhanced component protection will safeguard the optical and load cell sensors to ensure operational stability.

APPENDICES

Appendix A. Letter for Adviser

Republic of the Philippines MARIKINA POLYTECHNIC COLLEGE Shoe Ave. cor. Chanyungco St., Sta. Elena, Marikina City www.mpc.edu.ph	*Document Code:* MPC.QAO.F.011 *Revision:* 0
REQUEST LETTER FOR CAPSTONE ADVISER	*Effectivity Date:* May 31, 2022

October 07, 2025
Engr. Kent Bryan Padua
Professor, College of Engineering
Marikina Polytechnic College

Dear Sir:

The undersigned BSECE students are currently conducting study related to the following topics as requirement in **ECE 15 (Design 1/Capstone 1)** course in partial fulfilment for the degree of Bachelor of Science in Electronics Engineering.

Topic: Microplastic Detector in Rocksalt using AI

In this regard, may we humbly request you to be the **ADVISER** of the group.

Thank you very much.

Sincerely:

Antonio, Gabriel Nicolai S.
Intela, Jeffrey N.
Melendez, Eurl A.
Obero, Arnel Joshua Dean
Obja-an, Jasmine R.
Santos, Kyle Nero A.

Endorsed by:

Engr. Philip John L. Salvaloza, PECE
Capstone1 Professor

Conforme:

Date:
October 07, 2025

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Future-Ready MPC, United in Purpose
Page PAGE 1 of NUMPAGES 1

CamScanner

Appendix B. Letter for Critic

Republic of the Philippines **MARIKINA POLYTECHNIC COLLEGE** Shoe Ave. cor. Chanyungco St., Sta. Elena, Marikina City www.mpc.edu.ph	*Document Code:* **MPC.QAO.F.011** *Revision:* **0**
REQUEST LETTER FOR CAPSTONE ADVISER	*Effectivity Date:* **May 31, 2022**

October 22, 2025
Prof. Nelson Arguilles
Professor, College of Engineering
Marikina Polytechnic College

Dear Sir:

The undersigned BSECE students are currently conducting study related to the following topics as requirement in **ECE 15 (Design 1/Capstone 1)** course in partial fulfilment for the degree of Bachelor of Science in Electronics Engineering.

Topic: Microplastic Detector in Rocksalt using AI

In this regard, may we humbly request you to be the **CRITIC** of the group.

Thank you very much.

Sincerely:

Antonio, Gabriel Nicolai S.
Intela, Jeffrey N.
Melendez, Euri A.
Obero, Arnel Joshua Dean
Obja-an, Jasmine R.
Santos, Kyle Nero A.

Endorsed by:

Engr. Philip John L. Salvaloza, PECE
Capstone1 Professor

Conforme:

Prof. Nelson Arguilles
Date:
October 22, 2025

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Future-Ready MPC, United in Purpose   *Page* ***PAGE 1*** *of* ***NUMPAGES 1***

Appendix C. Certificate of Similarity

Republic of the Philippines **MARIKINA POLYTECHNIC COLLEGE** **RESEARCH AND DEVELOPMENT OFFICE** 2 Mayor Chanyungco St. Sta. Elena, Marikina City -1800 Tel. no: 8682-0672 loc 118 || email: researchoffice@mpc.edu.ph	Document No.	**MPC.RDO.C.003**
	Effectivity	2022-03-31
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	Page	1/1
	Issue No.	2026-266
Certificate of Similarity	Date Issue	06-08-2026

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Classification	_x_ Undergraduate __ Graduate _Faculty Others: ___________
Course & Major/ Department/ Institution:	BSECE
Type of Document	UNDERGRADUATE COURSE
Title of Document/ Thesis/Dissertation/ Research	**Development of an Automated Optical Detector for Microplastics in Rock Salt for Retail and Domestic Use**
Name of Co-Researchers (if any)	Euri A. Melendez, Jasmine R. Obja-an, Jeffrey Intela, Gabriel S. Antonio, Arnel Obero
Name of Research Adviser/Head	Engr. Kent Bryan P. Padua
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Endorsed By:

JOANA MAE S. PENALOSA
OIC Director of Research and Development

Appendix D. Interview Questionnaire (Needs Assessment Stage)

Survey Questionnaire: Survey on Salt Buying Practices and Awareness of Microplastics

Instructions: Please indicate your level of agreement with each statement by selecting the corresponding number on the scale:

1 - Strongly Disagree | 2 – Disagree | 3 - Agree | 4 - Strongly Agree

STATEMENT	1	2	3	4
1. I am aware that some salt products may contain microplastics. Alam ko na ang ilang produkto ng asin ay maaaring mayroong microplastics.				
2. I understand what microplastics are and how they might enter salt. Naiintindihan ko kung ano ang microplastics at kung paano ito maaaring mapasama sa asin.				
3. I believe that consuming salt with microplastics may affect our health. Naniniwala ako na ang pagkain ng asin na may microplastics ay maaaring makaapekto sa kalusugan.				
4. I usually buy salt from the same store or supplier. Karaniwan akong bumibili ng asin sa iisang tindahan o nag-aangkat.				
5. I prefer buying salt from well-known or trusted brands. Mas gusto kong bumili ng asin mula sa kilala o pinagkakatiwalaang mga brand.				
6. I often buy salt from local markets or sari-sari stores. Madalas akong bumili ng asin sa palengke o sari-sari store.				
7. I often buy salt from supermarkets or grocery stores. Madalas akong bumili ng asin sa supermarket o grocery store.				
8. I sometimes purchase salt online. Minsan akong bumibili ng asin online.				
9. I check product labels or packaging before buying salt. Tinitingnan ko ang label o pakete ng produkto bago ako bumili ng asin.				

10. I prefer salt products that claim to be purified or refined. **Mas gusto ko ang mga produktong asin na nagsasabing purified o refined.**				

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Survey Questionnaire: Survey on Salt Buying Practices and Awareness of Microplastics

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Survey Questionnaire: Survey on Salt Buying Practices and Awareness of Microplastics

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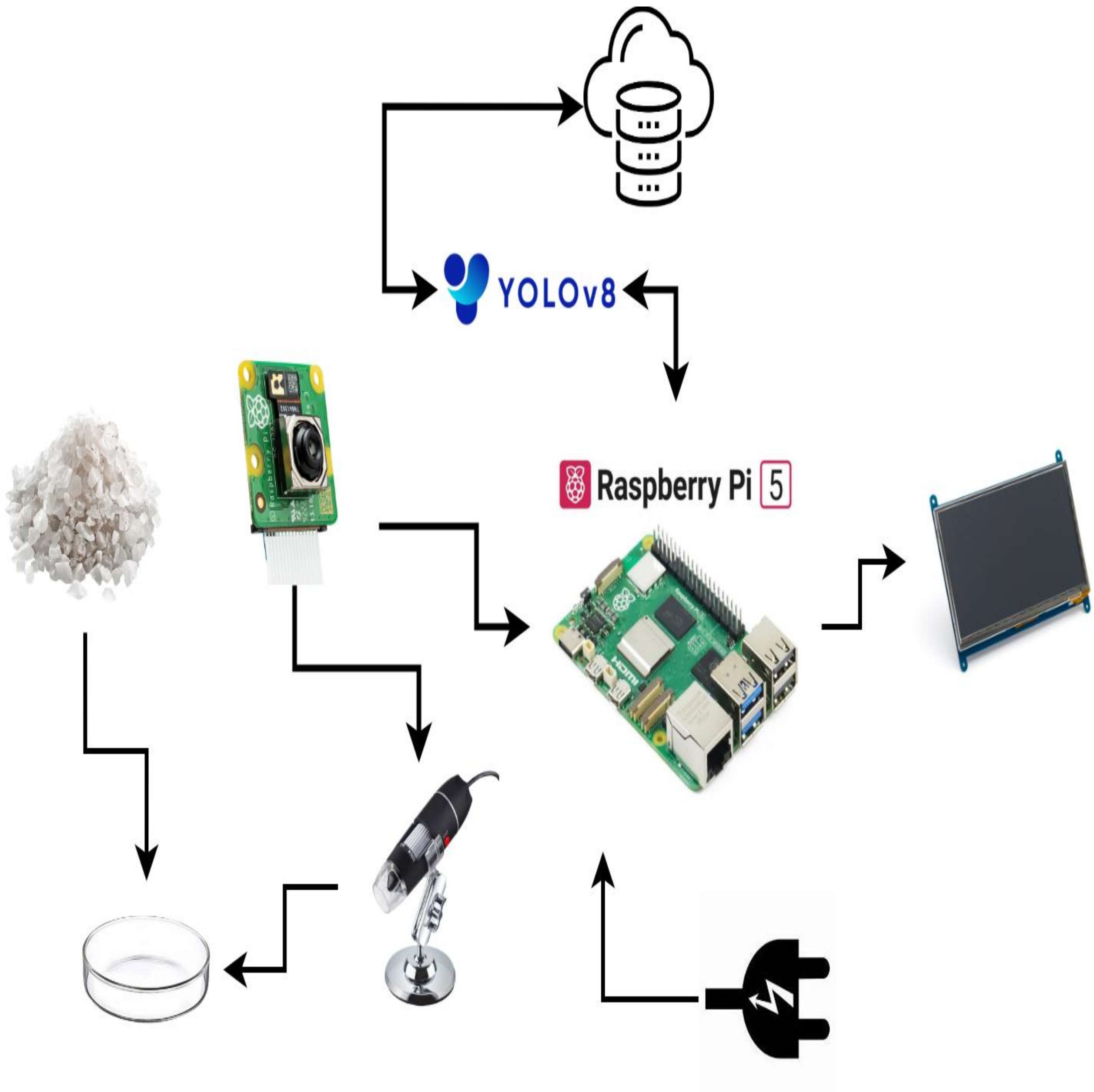
Appendix E. Bill of Materials

Bill of Materials

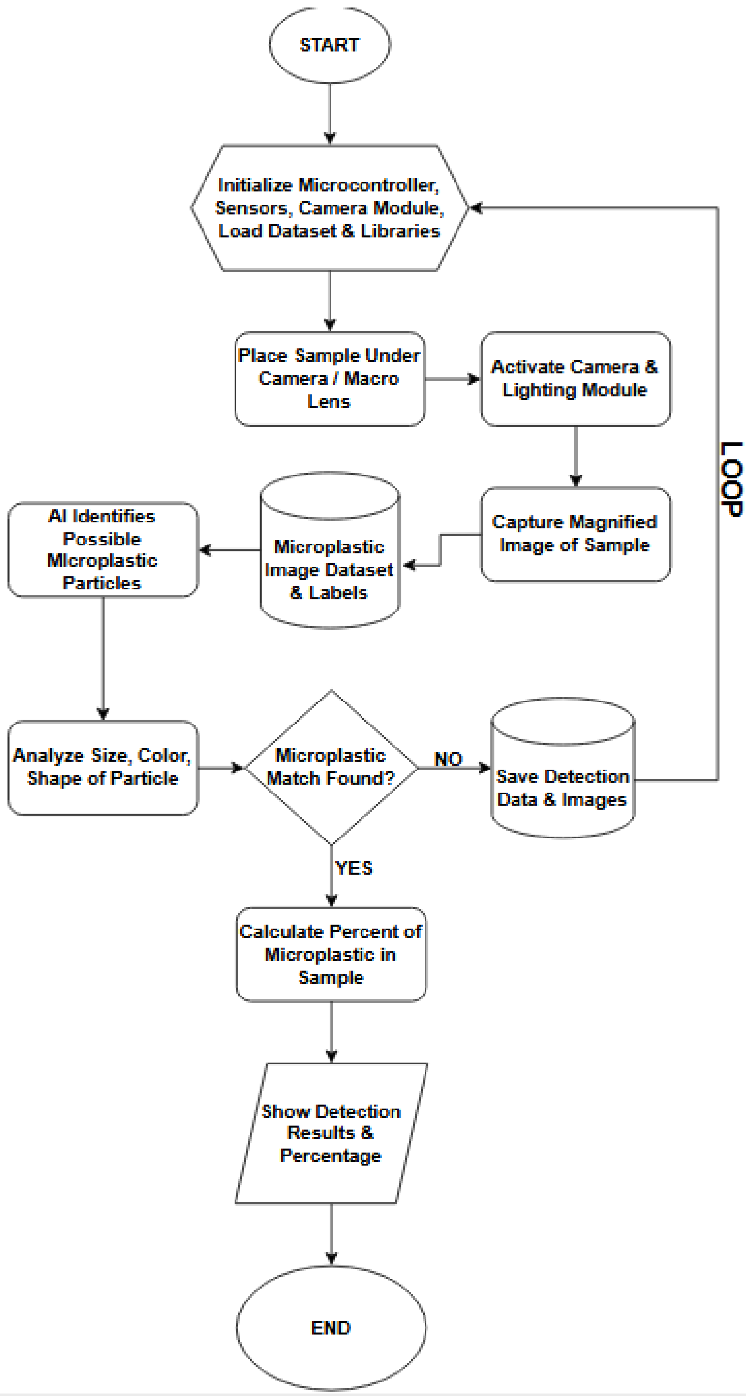
ITEM/COMPONENT	QUANTITY	PRICE RANGE
Raspberry Pi 5 Kit (8gb) Incl. Cable SD Card Power Supply	1	9660 PHP
HQ Camera (Sony IMX477)	1	4330 PHP
Microscope Lens 2MP 6mm	1	2930 PHP
Raspberry Pi Display Cable	1	940 PHP
Camera Cable	1	395 PHP
T-Bolt with Rubber	2	240 PHP
YOLOv11 AI Model		
7'' LCD Monitor	1	1960 PHP
Acrylic	10	200 PHP
Filter Paper	1 BOX	290 PHP
Petri Dish Glass	4	420 PHP
UV 395 NM 19MM	1	100 PHP
UV 395 NM 3W Chip Beads	1	155 PHP
Sample Container	6	100 PHP
Load Cell and Straight Bar	1	165 PHP
Allen Flat Head Bolt/ Allen Capscrew	50	300 PHP
Round Nut 304	3	81 PHP
Transistor and Lithium Battery Charger Module	1	171 PHP
Step Up Circuit Board	1	65 PHP
Universal Protoboard PCB	1	20 PHP
18650 Rechargeable Lithium Battery	1	100 PHP
Filter Paper	10	80 PHP
10ml Glass	20	140 PHP
UV Flashlight	1	300 PHP
Laboratory Fee		3250 PHP

Aluminum Heat Sink	1	30 PHP
Rock Salt Samples		900 PHP
Aluminum Foil Pouch	300	255 PHP
Documentation Material		3000 PHP
Transportation		
TOTAL		30,577 PHP

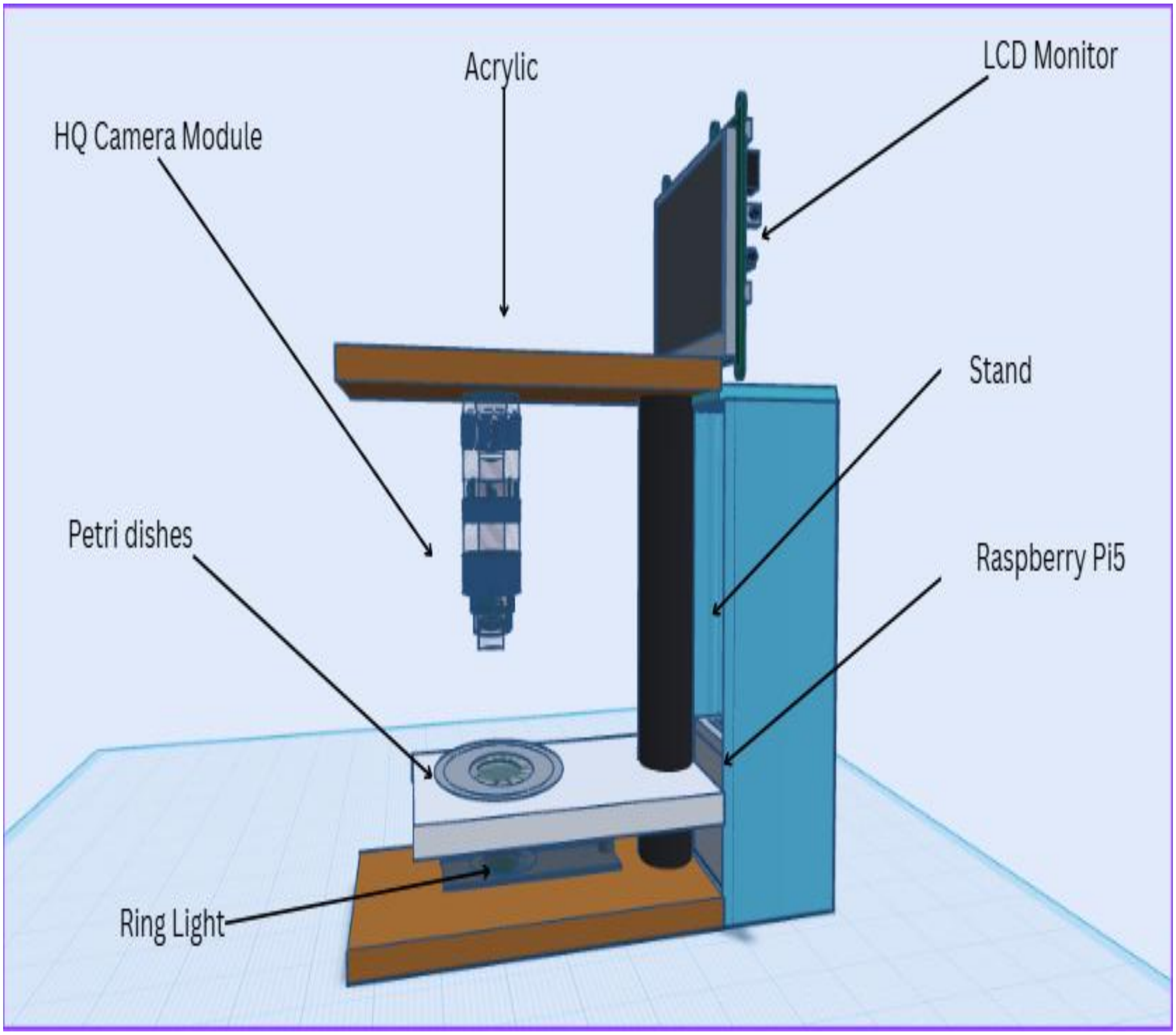
Appendix F. System Architecture

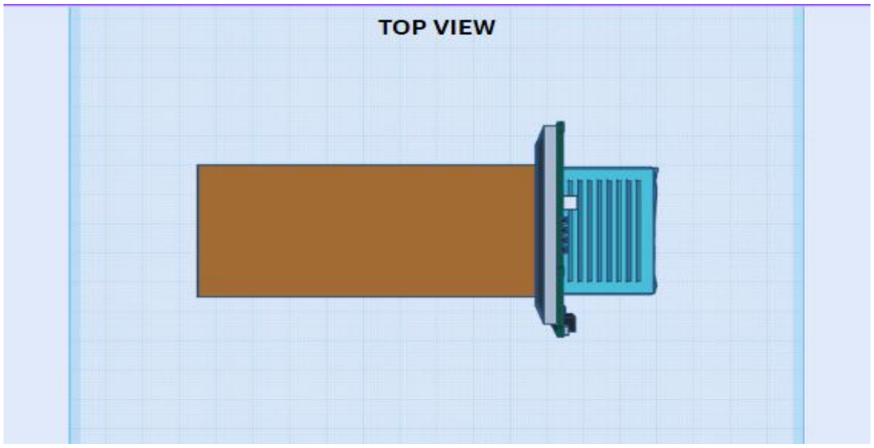


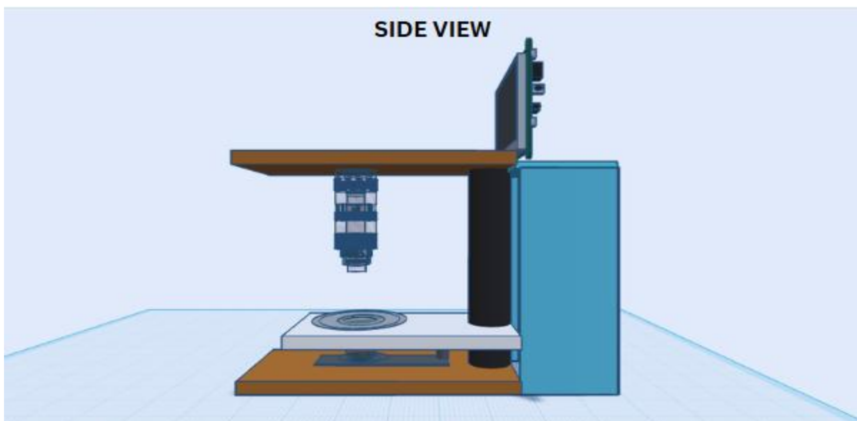
Appendix G. Flowchart

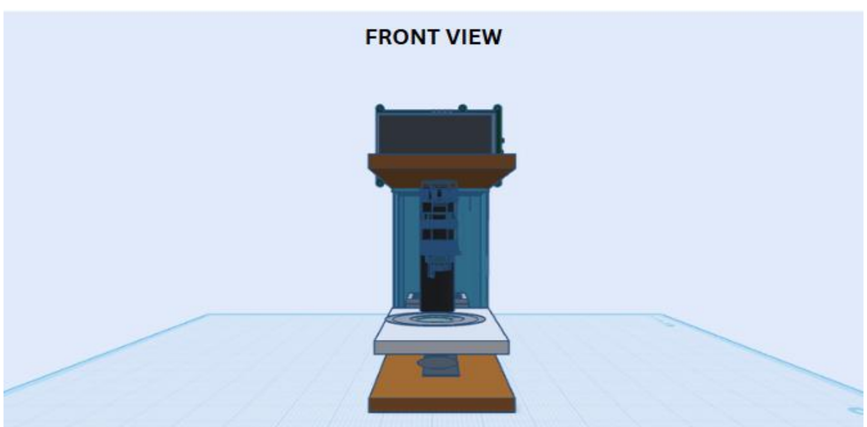


Appendix H. 3D Design / Application Development

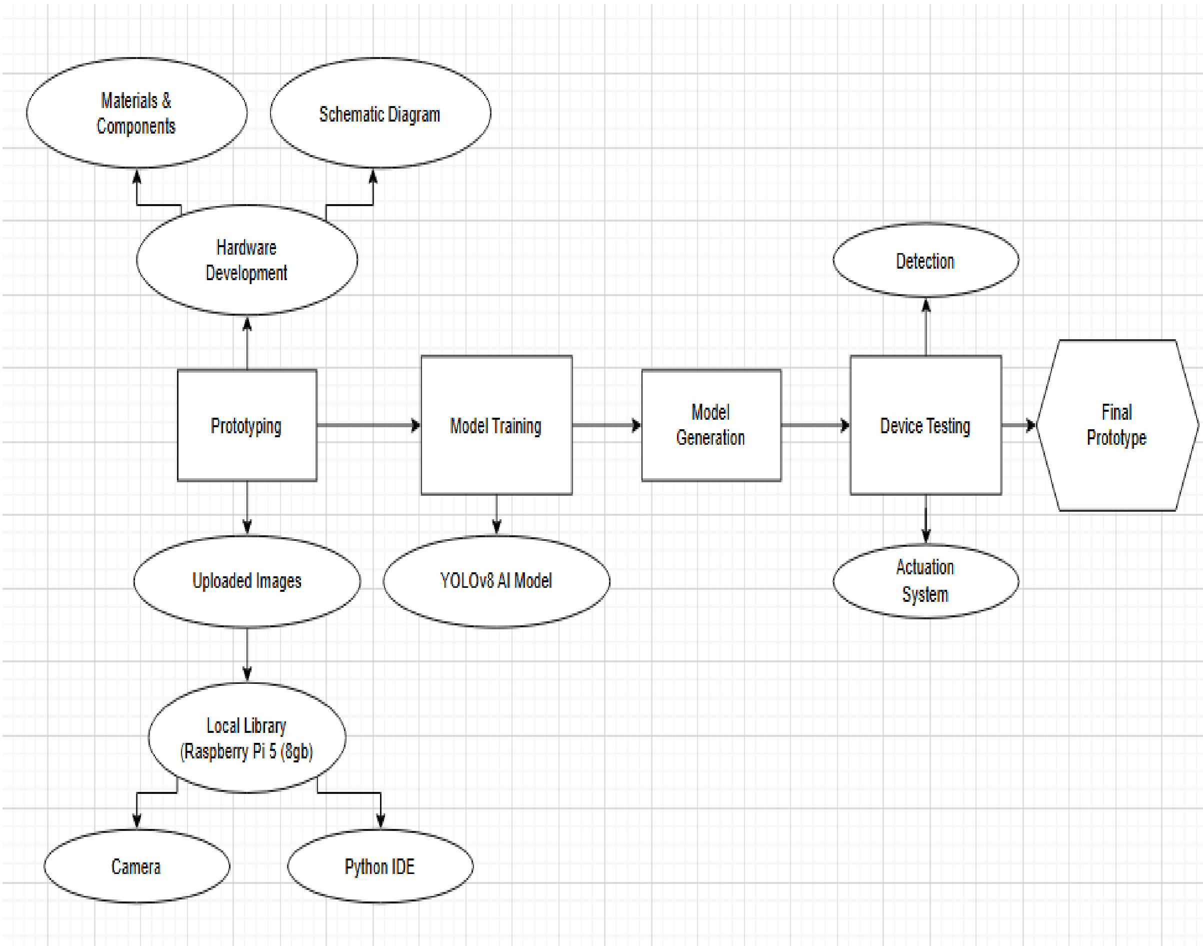



TOP VIEW


SIDE VIEW


FRONT VIEW

Appendix I. System Integration



Appendix J. Prototype and Software Quality Evaluation

PART 1: RESPONDENT'S PROFILE

Name: ____________________ Age: ____________________

Occupation: ____________________ Date: ____________________

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

INSTRUCTIONS:

Please evaluate the quality of the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use** adopted on the ISO 25010 Product Quality Standards. Use the five- point scale provided, where **5 is the highest quality and 1 is the lowest**. Kindly check the box that best reflects your assessment for each criterion.

The following are the description values of the scales used based on ISO-25010:

5 – Highly Acceptable (HA) | **4** – Acceptable (A) | **3** – Moderately Acceptable (MA)

2 – Needs Improvement (NI) | **1**– Unacceptable (UA)

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY– *The ability of the system to accurately detect and Identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.*					
1. The system accurately detects microplastic particles in rock salt samples.					
2. The AI-based image processing correctly identifies the microplastic particles.					
3. The system provides reliable and consistent detection results for retail and domestic users.					

III. **USABILITY** – *How easily users can set up, understand, and operate the system, Including the clarity of the interface and the presentation of detection results, even for non-technical users.*					
1. The system is easy to understand and operate for retail and domestic users.					
2. The user interface clearly presents detection results and system Information.					
3. The detection process can be performed with minimal user effort.					
III. PERFORMANCE EFFICIENCY – *Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.*					
1. The system detects microplastic in rock salt **quickly** using AI-based image processing.					
2. The response time of the automated optical detector is **acceptable for retail and domestic use**					
3. The system performs efficiently **without slowing down or lagging** during continuous operation.					
IV. SAFETY & MAINTENANCE – *Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.*					
1. The system is easy to clean, inspect, and maintain for long-term use.					
2. The system components are durable and protected from damage during normal operation.					
3. The device provides indicators or alerts in case of system errors or improper operation.					

PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.**

Please share your overall feedback on the system, including its strengths and areas for improvement.

____
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____
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Please suggest any additional features or enhancements to improve system performance and user experience.

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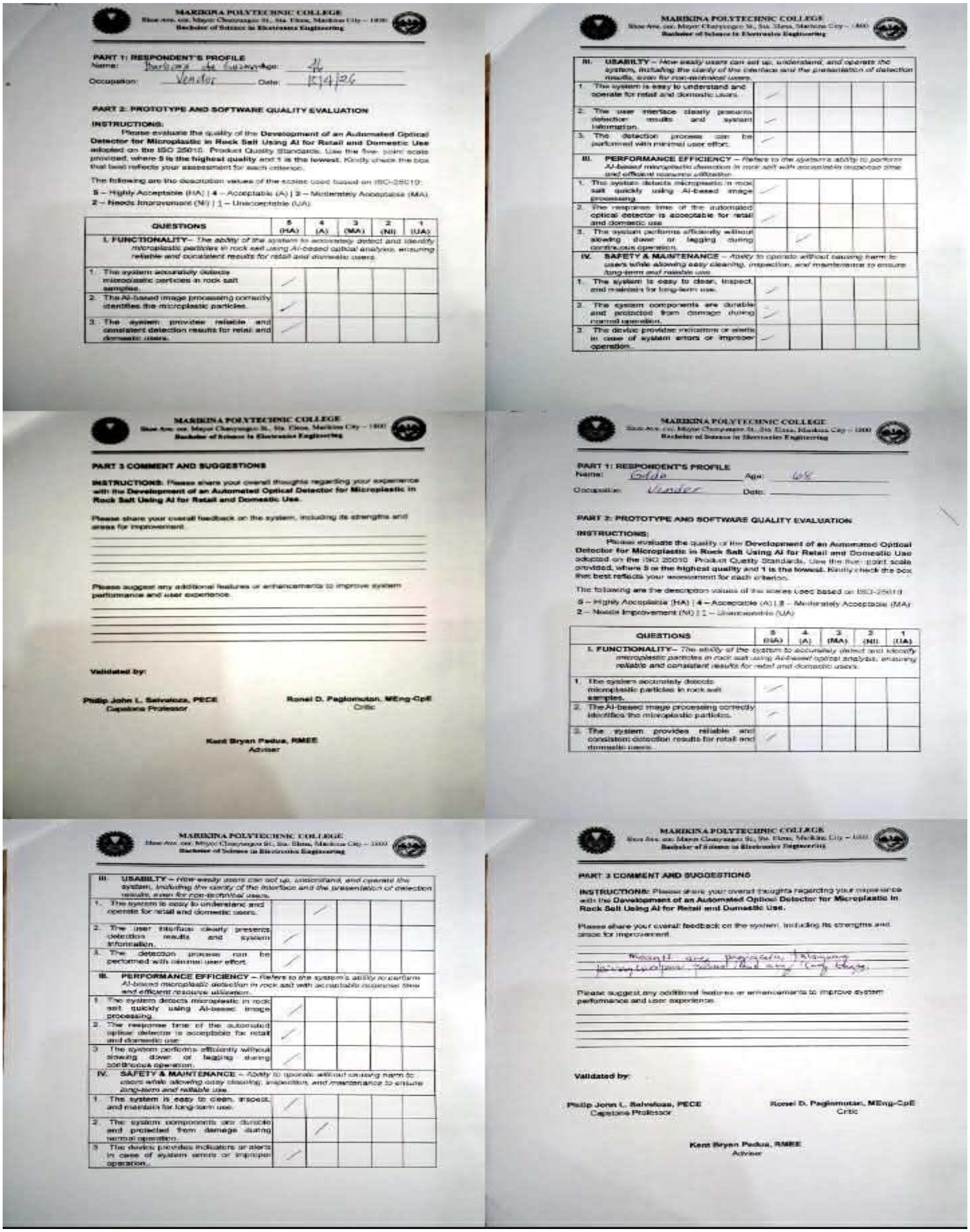

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Bachelor of Science in Electronics Engineering

PART 1: RESPONDENT'S PROFILE
Name: [illegible] Age: 46
Occupation: Vendor Date: 15/4/26

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

INSTRUCTIONS:
Please evaluate the quality of the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use** adopted on the ISO 25010 Product Quality Standards. Use the five-point scale provided, where **5 is the highest quality** and 1 is the **lowest**. Kindly check the box that best reflects your assessment for each criterion.

The following are the description values of the scales used based on ISO-25010:
5 – Highly Acceptable (HA) | **4** – Acceptable (A) | **3** – Moderately Acceptable (MA)
2 – Needs Improvement (NI) | 1 – Unacceptable (UA)

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY– *The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.*					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.	✓				

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III. USABILTY – *How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.*					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
III. PERFORMANCE EFFICIENCY – *Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.*					
1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
2. The response time of the automated optical detector is acceptable for retail and domestic use.	✓				
3. The system performs efficiently without slowing down or lagging during continuous operation.		✓			
IV. SAFETY & MAINTENANCE – *Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.*					
1. The system is easy to clean, inspect, and maintain for long-term use.	✓				
2. The system components are durable and protected from damage during normal operation.	✓				
3. The device provides indicators or alerts in case of system errors or improper operation.	✓				

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PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.**

Please share your overall feedback on the system, including its strengths and areas for improvement.

Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Paglomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser

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Bachelor of Science in Electronics Engineering

PART 1: RESPONDENT'S PROFILE
Name: Gilda Age: 68
Occupation: Vendor Date:

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

INSTRUCTIONS:
Please evaluate the quality of the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use** adopted on the ISO 25010 Product Quality Standards. Use the five-point scale provided, where **5 is the highest quality** and **1 is the lowest**. Kindly check the box that best reflects your assessment for each criterion.

The following are the description values of the scales used based on ISO-25010:
5 – Highly Acceptable (HA) | **4** – Acceptable (A) | **3** – Moderately Acceptable (MA)
2 – Needs Improvement (NI) | 1 – Unacceptable (UA)

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY– *The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.*					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.	✓				

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III. USABILTY – *How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.*					
1. The system is easy to understand and operate for retail and domestic users.		✓			
2. The user interface clearly presents detection results and system information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
III. PERFORMANCE EFFICIENCY – *Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.*					
1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
2. The response time of the automated optical detector is acceptable for retail and domestic use.	✓				
3. The system performs efficiently without slowing down or lagging during continuous operation.	✓				
IV. SAFETY & MAINTENANCE – *Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.*					
1. The system is easy to clean, inspect, and maintain for long-term use.	✓				
2. The system components are durable and protected from damage during normal operation.		✓			
3. The device provides indicators or alerts in case of system errors or improper operation.	✓				

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PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.**

Please share your overall feedback on the system, including its strengths and areas for improvement.

[illegible]

Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Paglomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser

MARIKINA POLYTECHNIC COLLEGE
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PART 1: RESPONDENT'S PROFILE

Name: Vangie F Nogales Age: 45
Occupation: Vendor Date: 04-5-26

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

INSTRUCTIONS:

Please evaluate the quality of the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use** adopted on the ISO 25010 Product Quality Standards. Use the five- point scale provided, where **5 is the highest quality** and **1 is the lowest.** Kindly check the box that best reflects your assessment for each criterion.

The following are the description values of the scales used based on ISO-25010:

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QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY– *The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.*					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.		✓			
3. The system provides reliable and consistent detection results for retail and domestic users.		✓			

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II. USABILITY – *How easily users can set up, understand and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.*					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
III. PERFORMANCE EFFICIENCY – *Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.*					
1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
2. The response time of the automated optical detector is acceptable for retail and domestic use.		✓			
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IV. SAFETY & MAINTENANCE – *Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.*					
1. The system is easy to clean, inspect, and maintain for long-term use.		✓			
2. The system components are durable and protected from damage during normal operation.		✓			
3. The device provides indicators or alerts in case of system errors or improper operation.		✓			

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PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.**

Please share your overall feedback on the system, including its strengths and areas for improvement.

Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Paglomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser

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PART 1: RESPONDENT'S PROFILE

Name: [illegible] Age: [illegible]
Occupation: Driver Date:

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

INSTRUCTIONS:

Please evaluate the quality of the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use** adopted on the ISO 25010 Product Quality Standards. Use the five- point scale provided, where **5 is the highest quality** and **1 is the lowest.** Kindly check the box that best reflects your assessment for each criterion.

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II. USABILITY – *How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.*					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
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1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
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3. The system performs efficiently without slowing down or lagging during continuous operation.		✓			
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3. The device provides indicators or alerts in case of system errors or improper operation.		✓			

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PART 3 COMMENT AND SUGGESTIONS

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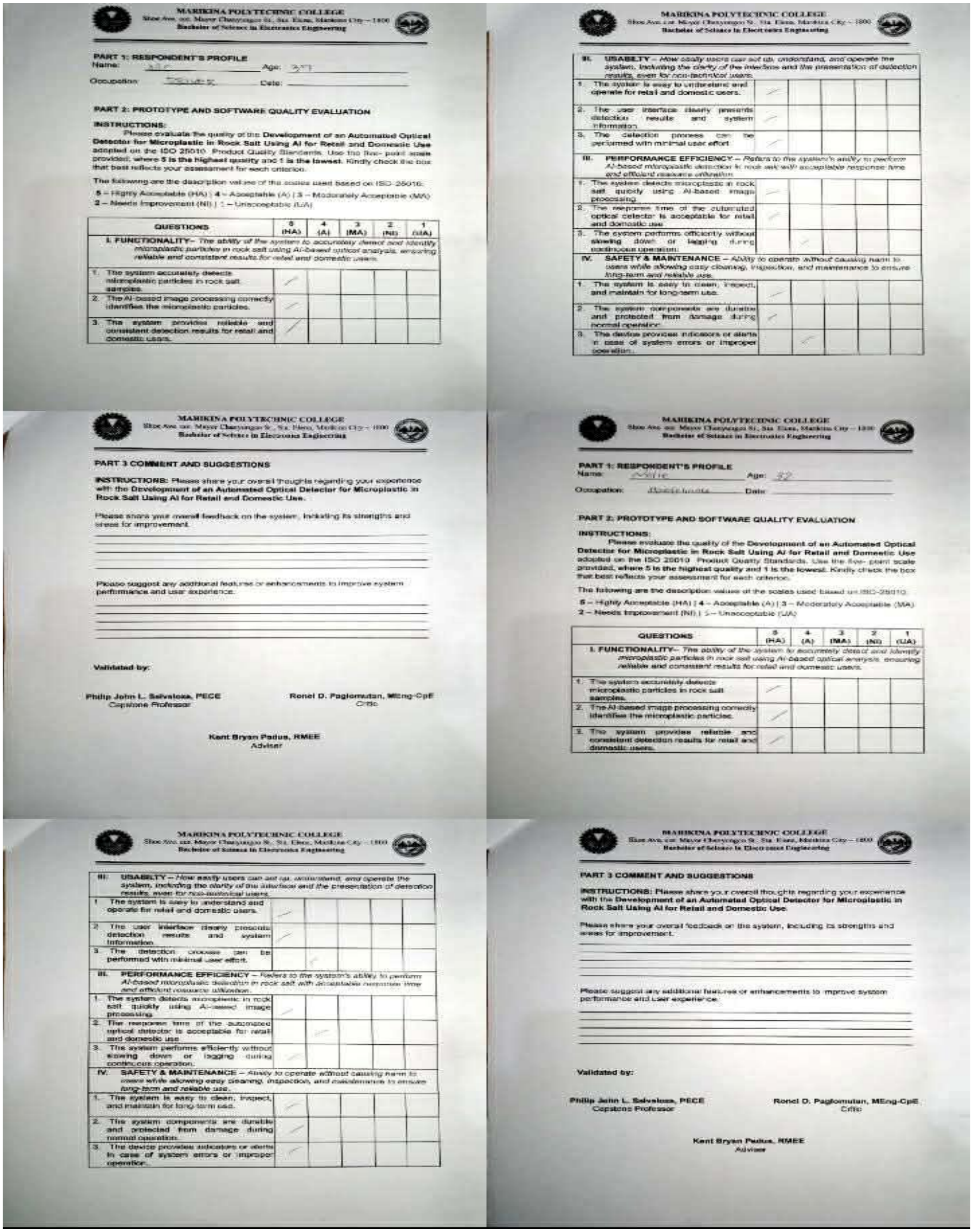
Critic

Kent Bryan Padua, RMEE
Adviser



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PART 1: RESPONDENT'S PROFILE
Name: ____ Age: ____
Occupation: ____ Date: ____

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

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2 – Needs Improvement (NI) | 1 – Unacceptable (UA)

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I. FUNCTIONALITY – The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.					
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2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.	✓				

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
II. USABILITY – How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
III. PERFORMANCE EFFICIENCY – Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.					
1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
2. The response time of the automated optical detector is acceptable for retail and domestic use.	✓				
3. The system performs efficiently without slowing down or lagging during continuous operation.		✓			
IV. SAFETY & MAINTENANCE – Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.					
1. The system is easy to clean, inspect, and maintain for long-term use.	✓				
2. The system components are durable and protected from damage during normal operation.	✓				
3. The device provides indicators or alerts in case of system errors or improper operation.		✓			

PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.**

Please share your overall feedback on the system, including its strengths and areas for improvement.

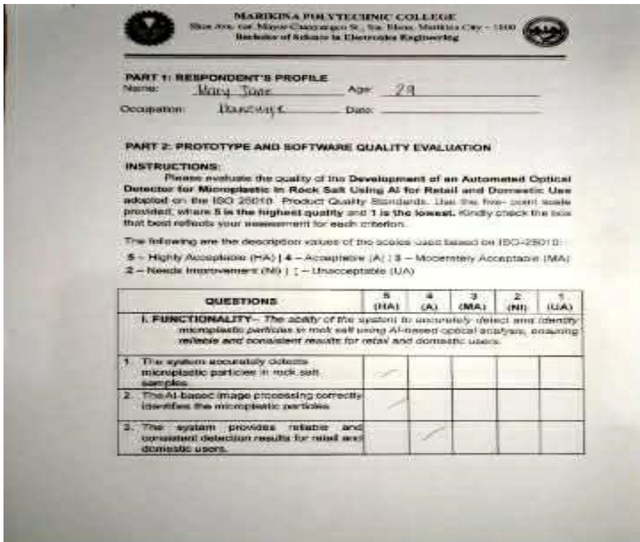
Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Paglomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser



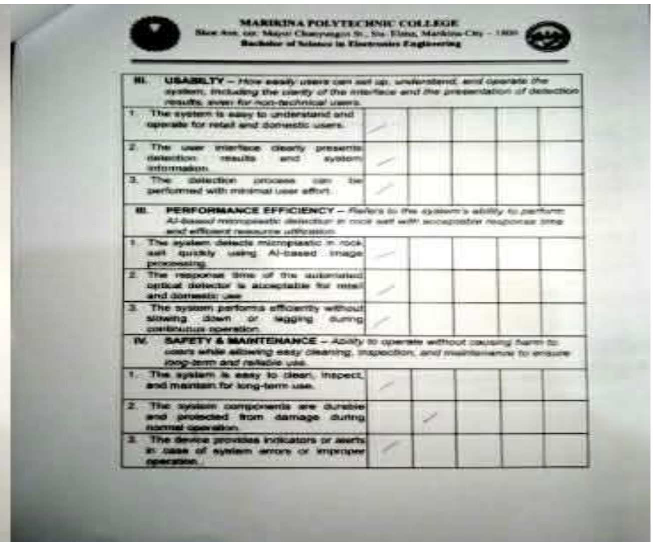
MARIKINA POLYTECHNIC COLLEGE
Shoe Ave. cor. Mayor Chanyungco St., Sta. Elena, Marikina City – 1800
Bachelor of Science in Electronics Engineering

PART 1: RESPONDENT'S PROFILE
Name: Mary Jane Age: 29
Occupation: Housewife Date:

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION
INSTRUCTIONS:
Please evaluate the quality of the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use** adopted on the ISO 25010 Product Quality Standards. Use the five-point scale provided, where **5 is the highest quality** and **1 is the lowest.** Kindly check the box that best reflects your assessment for each criterion.

The following are the description values of the scales used based on ISO-25010:
5 – Highly Acceptable (HA) | 4 – Acceptable (A) | 3 – Moderately Acceptable (MA) | 2 – Needs Improvement (NI) | 1 – Unacceptable (UA)

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY – *The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.*					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.		✓			



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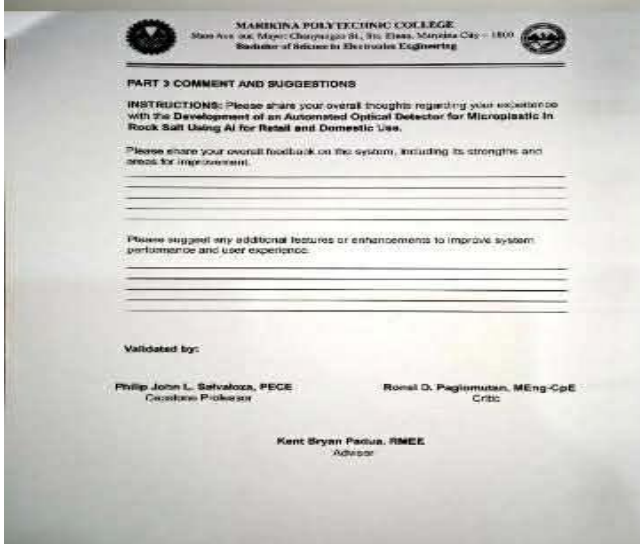
II. USABILITY – *How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.*					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
III. PERFORMANCE EFFICIENCY – *Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.*					
1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
2. The response time of the automated optical detector is acceptable for retail and domestic use.	✓				
3. The system performs efficiently without slowing down or lagging during continuous operation.	✓				
IV. SAFETY & MAINTENANCE – *Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.*					
1. The system is easy to clean, inspect, and maintain for long-term use.	✓				
2. The system components are durable and protected from damage during normal operation.		✓			
3. The device provides indicators or alerts in case of system errors or improper operation.	✓				



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Bachelor of Science in Electronics Engineering

PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.**

Please share your overall feedback on the system, including its strengths and areas for improvement.

Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Pagliomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser

MARIKINA POLYTECHNIC COLLEGE
Shoe Ave. cor. Mayor Chanyungco St., Sta. Elena, Marikina City – 1800
Bachelor of Science in Electronics Engineering

PART 1: RESPONDENT'S PROFILE
Name: Leche Age: 36
Occupation: Housewife Date:

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION
INSTRUCTIONS:
Please evaluate the quality of the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use** adopted on the ISO 25010 Product Quality Standards. Use the five-point scale provided, where **5 is the highest quality** and **1 is the lowest.** Kindly check the box that best reflects your assessment for each criterion.

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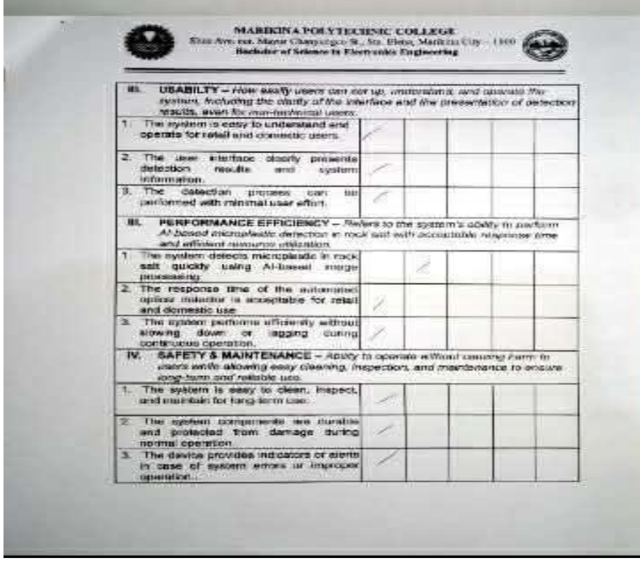
QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY – *The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.*					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.		✓			
3. The system provides reliable and consistent detection results for retail and domestic users.		✓			



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II. USABILITY – *How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.*					
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III. PERFORMANCE EFFICIENCY – *Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.*					
1. The system detects microplastic in rock salt quickly using AI-based image processing.		✓			
2. The response time of the automated optical detector is acceptable for retail and domestic use.	✓				
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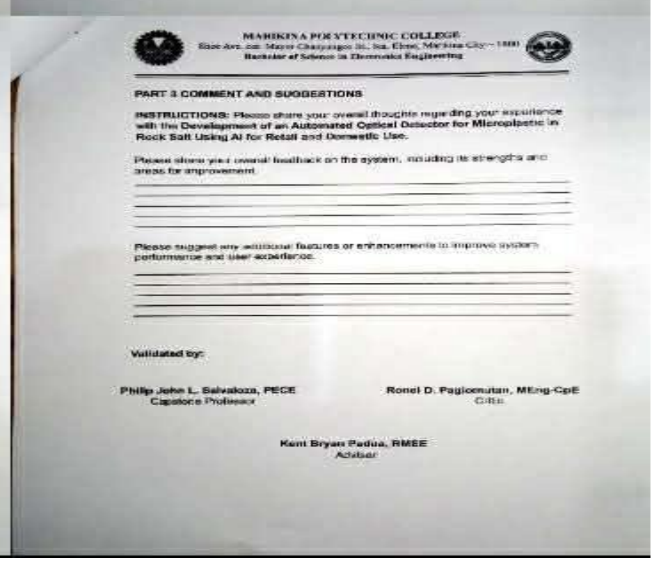
3. The device provides indicators or alerts in case of system errors or improper operation.	✓				



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PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.**

Please share your overall feedback on the system, including its strengths and areas for improvement.

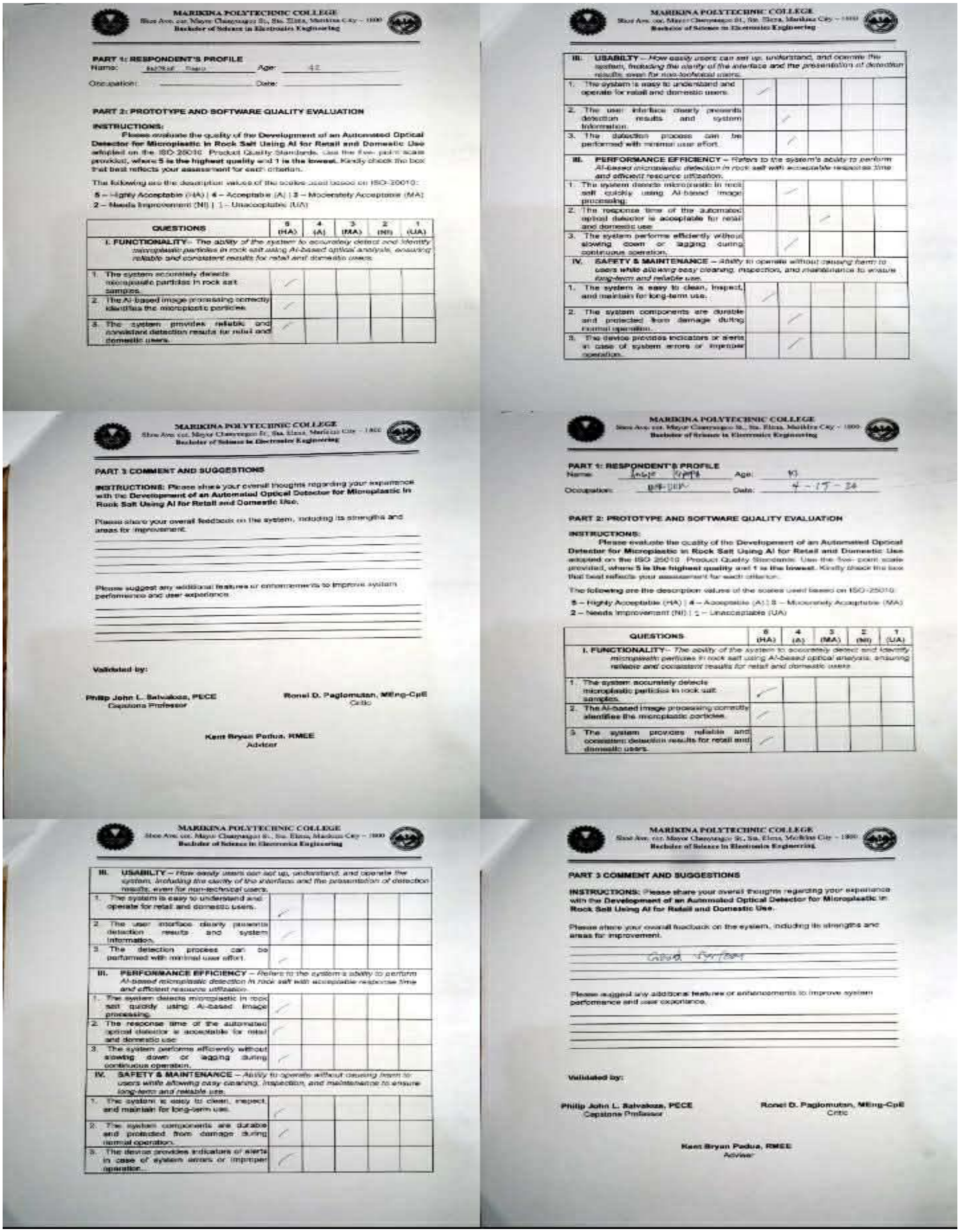
Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Pagliomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser



MARIKINA POLYTECHNIC COLLEGE
Shoe Ave. cor. Mayor Chanyungco St., Sta. Elena, Marikina City – 1800
Bachelor of Science in Electronics Engineering

PART 1: RESPONDENT'S PROFILE
Name: [illegible] Age: 42
Occupation: Date:

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

INSTRUCTIONS:
Please evaluate the quality of the Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use adopted on the ISO 25010 Product Quality Standards. Use the five-point scale provided, where 5 is the highest quality and 1 is the lowest. Kindly check the box that best reflects your assessment for each criterion.

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5 – Highly Acceptable (HA) | 4 – Acceptable (A) | 3 – Moderately Acceptable (MA)
2 – Needs Improvement (NI) | 1 – Unacceptable (UA)

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY – The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.	✓				

II. USABILITY – How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.		✓			
3. The detection process can be performed with minimal user effort.		✓			
III. PERFORMANCE EFFICIENCY – Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.					
1. The system detects microplastic in rock salt quickly using AI-based image processing.		✓			
2. The response time of the automated optical detector is acceptable for retail and domestic use.			✓		
3. The system performs efficiently without slowing down or lagging during continuous operation.		✓			
IV. SAFETY & MAINTENANCE – Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.					
1. The system is easy to clean, inspect, and maintain for long-term use.	✓				
2. The system components are durable and protected from damage during normal operation.		✓			
3. The device provides indicators or alerts in case of system errors or improper operation.		✓			

PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.

Please share your overall feedback on the system, including its strengths and areas for improvement.

Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Paglomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser

PART 1: RESPONDENT'S PROFILE
Name: [illegible] Age: 43
Occupation: [illegible] Date: 4 – 17 – 26

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.	✓				
II. USABILITY					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
III. PERFORMANCE EFFICIENCY					
1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
2. The response time of the automated optical detector is acceptable for retail and domestic use.	✓				
3. The system performs efficiently without slowing down or lagging during continuous operation.	✓				
IV. SAFETY & MAINTENANCE					
1. The system is easy to clean, inspect, and maintain for long-term use.	✓				
2. The system components are durable and protected from damage during normal operation.	✓				
3. The device provides indicators or alerts in case of system errors or improper operation.	✓				

PART 3 COMMENT AND SUGGESTIONS

Please share your overall feedback on the system, including its strengths and areas for improvement.
Good System

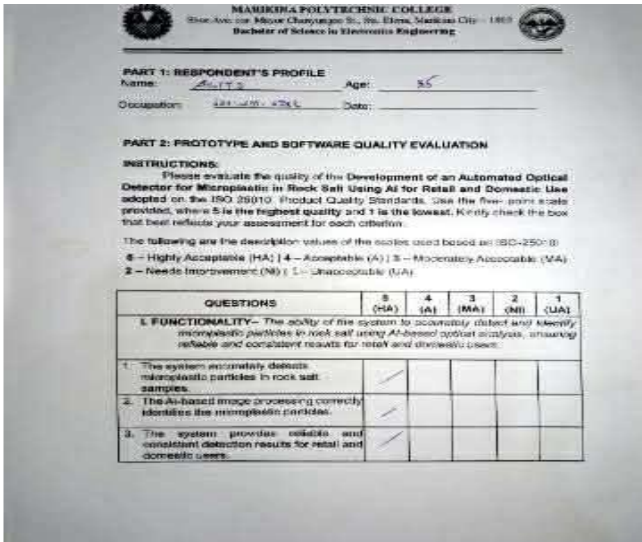
Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Paglomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser



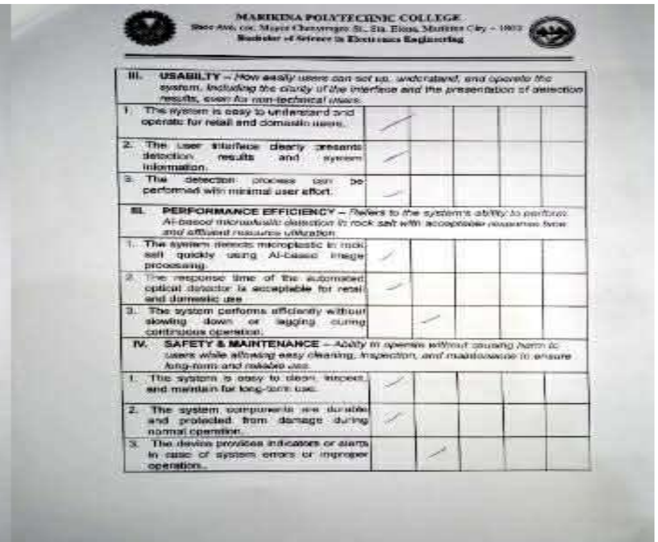
MARIKINA POLYTECHNIC COLLEGE
Shoe Ave. cor. Mayor Chanyungco St., Sta. Elena, Marikina City – 1800
Bachelor of Science in Electronics Engineering

PART 1: RESPONDENT'S PROFILE
Name: [illegible] Age: 55
Occupation: [illegible] Date:

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION
INSTRUCTIONS:
Please evaluate the quality of the Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use adopted on the ISO 25010 Product Quality Standards. Use the five-point scale provided, where 5 is the highest quality and 1 is the lowest. Kindly check the box that best reflects your assessment for each criterion.

The following are the description values of the scales used based on ISO-25010:
5 – Highly Acceptable (HA) | 4 – Acceptable (A) | 3 – Moderately Acceptable (MA)
2 – Needs Improvement (NI) | 1 – Unacceptable (UA)

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY – The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.	✓				



MARIKINA POLYTECHNIC COLLEGE
Shoe Ave. cor. Mayor Chanyungco St., Sta. Elena, Marikina City – 1800
Bachelor of Science in Electronics Engineering

III. USABILITY – How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
III. PERFORMANCE EFFICIENCY – Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.					
1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
2. The response time of the automated optical detector is acceptable for retail and domestic use.	✓				
3. The system performs efficiently without slowing down or lagging during continuous operation.		✓			
IV. SAFETY & MAINTENANCE – Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.					
1. The system is easy to clean, inspect, and maintain for long-term use.	✓				
2. The system components are durable and protected from damage during normal operation.	✓				
3. The device provides indicators or alerts in case of system errors or improper operation.		✓			

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Bachelor of Science in Electronics Engineering

PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.

Please share your overall feedback on the system, including its strengths and areas for improvement.
Very Good

Please suggest any additional features or enhancements to improve system performance and user experience.
NO COMMENT

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Pagiomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser

MARIKINA POLYTECHNIC COLLEGE
Shoe Ave. cor. Mayor Chanyungco St., Sta. Elena, Marikina City – 1800
Bachelor of Science in Electronics Engineering

PART 1: RESPONDENT'S PROFILE
Name: [illegible] Age: 49
Occupation: VENDOR Date:

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION
INSTRUCTIONS:
Please evaluate the quality of the Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use adopted on the ISO 25010 Product Quality Standards. Use the five-point scale provided, where 5 is the highest quality and 1 is the lowest. Kindly check the box that best reflects your assessment for each criterion.

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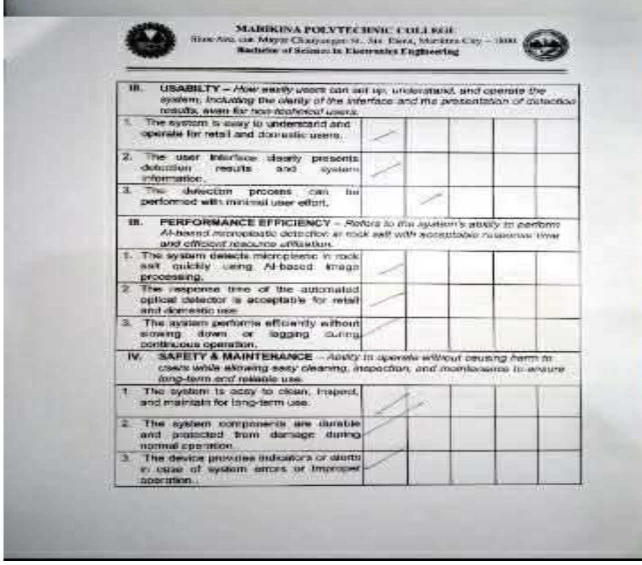
QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY – The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.					
1. The system accurately detects microplastic particles in rock salt samples.		✓			
2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.	✓				



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III. USABILITY – How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.	✓				
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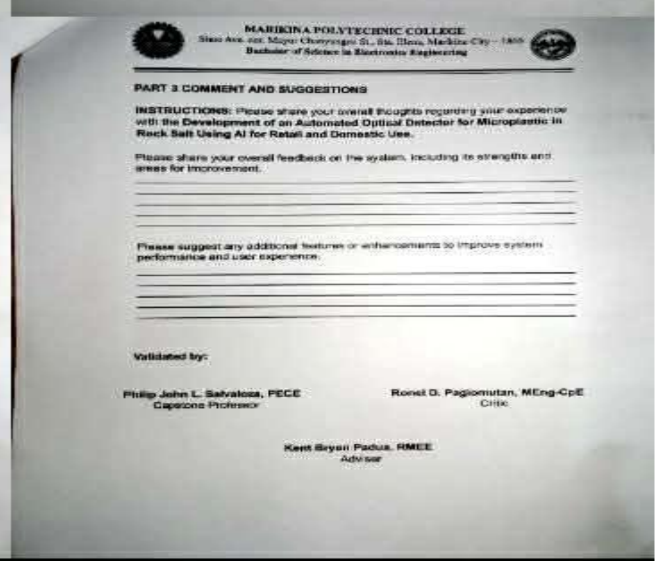

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Bachelor of Science in Electronics Engineering

PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.

Please share your overall feedback on the system, including its strengths and areas for improvement.

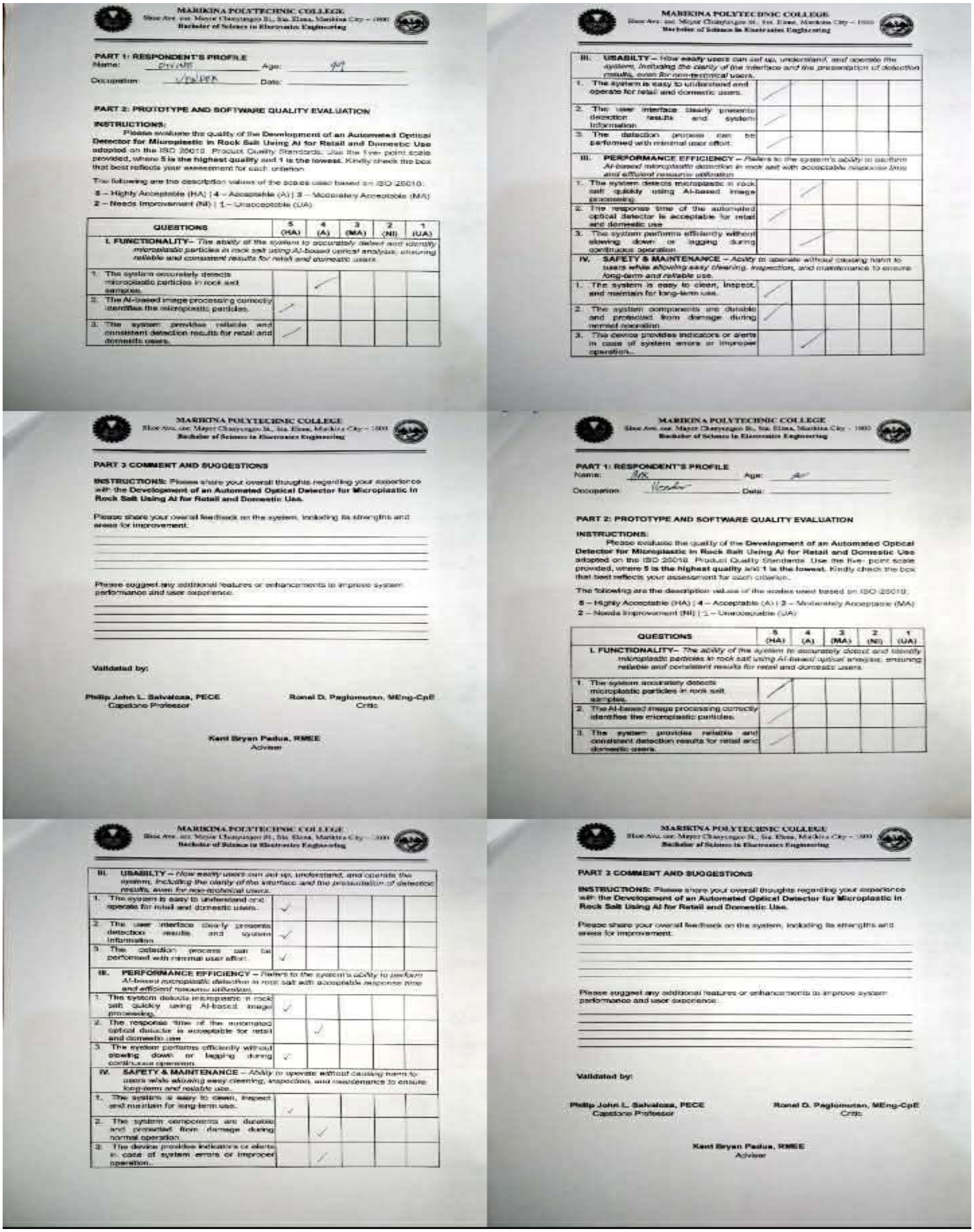
Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Pagiomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser



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PART 1: RESPONDENT'S PROFILE
Name: [illegible] Age: 49
Occupation: Vendor Date:

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

INSTRUCTIONS:
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QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
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PART 3 COMMENT AND SUGGESTIONS

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Validated by:

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Capstone Professor

Ronel D. Paglomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser

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PART 1: RESPONDENT'S PROFILE
Name: [illegible] Age: [illegible]
Occupation: Vendor Date:

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

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2 – Needs Improvement (NI) | 1 – Unacceptable (UA)

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY – *The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.*					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.	✓				

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QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
II. USABILITY – *How easily users can set up, understand, and operate the system, including the clarity of the interface and the presentation of detection results, even for non-technical users.*					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
III. PERFORMANCE EFFICIENCY – *Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.*					
1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
2. The response time of the automated optical detector is acceptable for retail and domestic use.		✓			
3. The system performs efficiently without slowing down or lagging during continuous operation.	✓				
IV. SAFETY & MAINTENANCE – *Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.*					
1. The system is easy to clean, inspect, and maintain for long-term use.	✓				
2. The system components are durable and protected from damage during normal operation.		✓			
3. The device provides indicators or alerts in case of system errors or improper operation.		✓			

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PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.**

Please share your overall feedback on the system, including its strengths and areas for improvement.

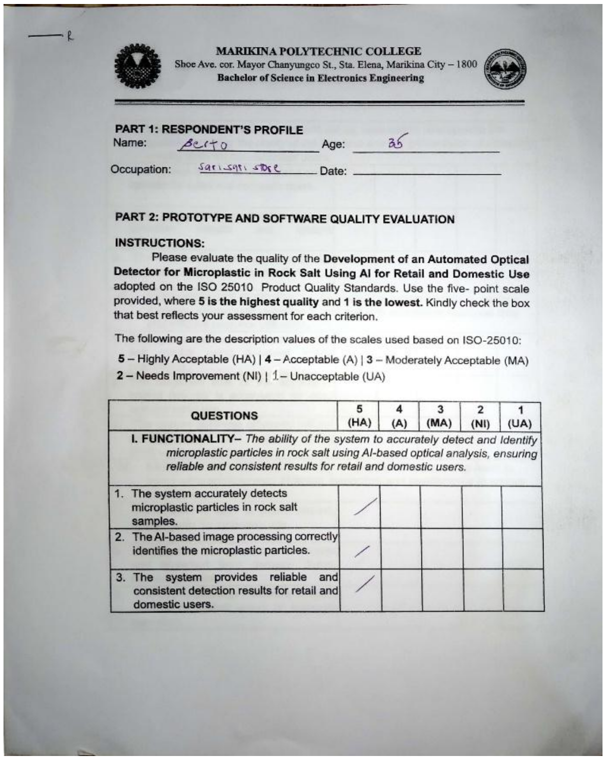
Please suggest any additional features or enhancements to improve system performance and user experience.

Validated by:

Philip John L. Salvaloza, PECE
Capstone Professor

Ronel D. Paglomutan, MEng-CpE
Critic

Kent Bryan Padua, RMEE
Adviser



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PART 1: RESPONDENT'S PROFILE

Name: Berto Age: 35

Occupation: Sari-sari store Date: ____

PART 2: PROTOTYPE AND SOFTWARE QUALITY EVALUATION

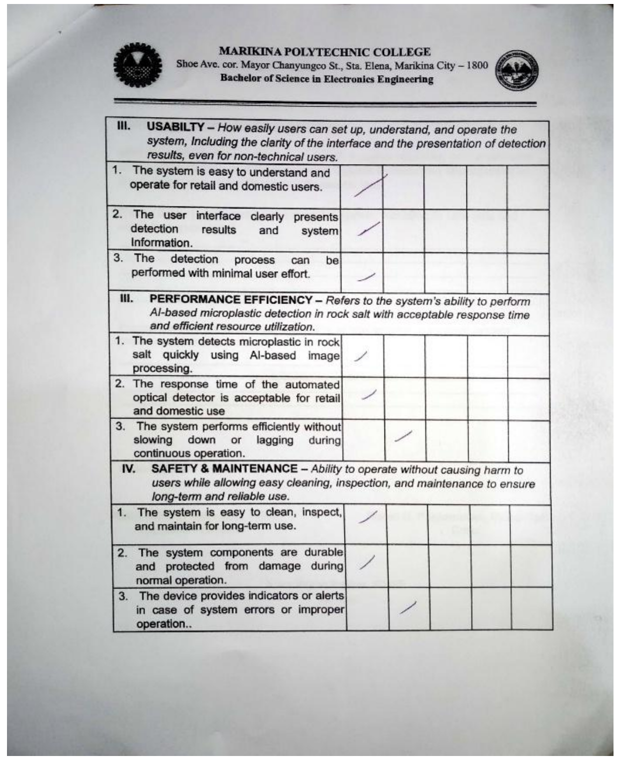
INSTRUCTIONS:

Please evaluate the quality of the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use** adopted on the ISO 25010 Product Quality Standards. Use the five- point scale provided, where **5 is the highest quality** and **1 is the lowest.** Kindly check the box that best reflects your assessment for each criterion.

The following are the description values of the scales used based on ISO-25010:

5 – Highly Acceptable (HA) | **4** – Acceptable (A) | **3** – Moderately Acceptable (MA)
2 – Needs Improvement (NI) | 1 – Unacceptable (UA)

QUESTIONS	5 (HA)	4 (A)	3 (MA)	2 (NI)	1 (UA)
I. FUNCTIONALITY– *The ability of the system to accurately detect and identify microplastic particles in rock salt using AI-based optical analysis, ensuring reliable and consistent results for retail and domestic users.*					
1. The system accurately detects microplastic particles in rock salt samples.	✓				
2. The AI-based image processing correctly identifies the microplastic particles.	✓				
3. The system provides reliable and consistent detection results for retail and domestic users.	✓				



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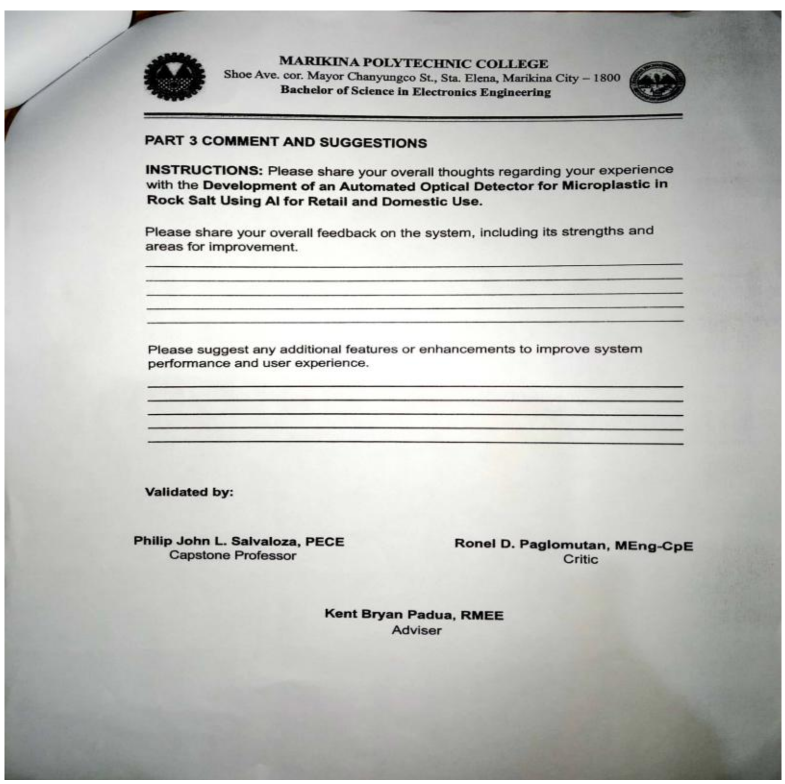
III. USABILTY – *How easily users can set up, understand, and operate the system, Including the clarity of the interface and the presentation of detection results, even for non-technical users.*					
1. The system is easy to understand and operate for retail and domestic users.	✓				
2. The user interface clearly presents detection results and system Information.	✓				
3. The detection process can be performed with minimal user effort.	✓				
III. PERFORMANCE EFFICIENCY – *Refers to the system's ability to perform AI-based microplastic detection in rock salt with acceptable response time and efficient resource utilization.*					
1. The system detects microplastic in rock salt quickly using AI-based image processing.	✓				
2. The response time of the automated optical detector is acceptable for retail and domestic use	✓				
3. The system performs efficiently without slowing down or lagging during continuous operation.		✓			
IV. SAFETY & MAINTENANCE – *Ability to operate without causing harm to users while allowing easy cleaning, inspection, and maintenance to ensure long-term and reliable use.*					
1. The system is easy to clean, inspect, and maintain for long-term use.	✓				
2. The system components are durable and protected from damage during normal operation.	✓				
3. The device provides indicators or alerts in case of system errors or improper operation..		✓			



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PART 3 COMMENT AND SUGGESTIONS

INSTRUCTIONS: Please share your overall thoughts regarding your experience with the **Development of an Automated Optical Detector for Microplastic in Rock Salt Using AI for Retail and Domestic Use.**

Please share your overall feedback on the system, including its strengths and areas for improvement.

Please suggest any additional features or enhancements to improve system performance and user experience.

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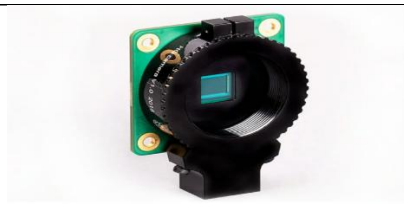
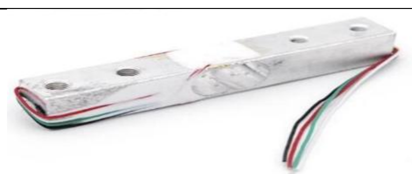
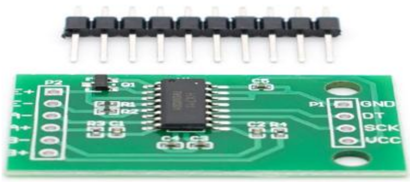
Adviser

Appendix K. Hardware Components of the Prototype

No.	Image	Component	Function in the System
1		Raspberry Pi 5	Serve as the main processing and control unit of the prototype.
2		Raspberry Pi HQ Camera (IMX477)	Captures the sample images for real-time viewing and detection
3		7-Inch LCD Touchscreen	Provides the graphical user interface for user interaction
4		White LED	Provides visible light illumination during sample observation
5		UV LED	Provides ultraviolet illumination during sample observation
6		Load Cell Sensor	Measures the mass of the sample

7		HX711 Amplifier Module	Amplifies and digitizes the load-cell signal.
8		Shutdown Push Button	Allows safe shutdown of the system
9		Power Supply	Supplies the power to the Raspberry Pi and peripherals

Appendix L. Raw Data Tables by Source

This appendix presents the complete raw data gathered from all 300 rock salt sample tests used in the study. The dataset is grouped into three separate tables based on source to make the records easier to review.

Table L.1 Raw Data for Pangasinan Samples

Location	Sample ID	Microplastic Count	Size (μm)	Concentration (MP/g)	Weight (g)
A	PGN A#1 (Live)	1	1546	0.1055	9.481
A	PGN A#2 (Live)	0		0.0000	8.754
A	PGN A#3 (Live)	1	1752	0.1169	8.556
A	PGN A#4 (Live)	0		0.0000	10.395
A	PGN A#5 (Live)	1	1237	0.1035	9.658
A	PGN A#6 (Live)	0		0.0000	9.131
A	PGN A#7 (Live)	1	2680	0.1145	8.734
A	PGN A#8 (Live)	3	927,1340,2783	0.2694	11.137
A	PGN A#9 (Live)	2	1443,1030	0.2423	8.255
A	PGN A#10 (Live)	2	1443,1030	0.2411	8.296
A	PGN A#11 (Live)	2	1340,927	0.2220	9.011
A	PGN A#12 (Live)	2	1855,1237	0.2196	9.109
A	PGN A#13 (Live)	1	1134	0.1145	8.733
A	PGN A#14 (Live)	1	1958	0.1048	9.545
A	PGN A#15 (Live)	2	1546,2164	0.1839	10.874
A	PGN A#16 (Live)	0		0.0000	9.158
A	PGN A#17 (Live)	2	1649,1340	0.2134	9.372
A	PGN A#18 (Live)	0		0.0000	9.713
A	PGN A#19 (Live)	1	1134	0.0931	10.741
A	PGN A#20 (Live)	2	927,2061	0.2072	9.653
A	PGN A#21 (Live)	3	1030,1649,1855	0.3491	8.593
A	PGN A#22 (Live)	0		0.0000	8.712
A	PGN A#23 (Live)	0		0.0000	8.874
A	PGN A#24 (Live)	2	1340,1443	0.1940	10.307
A	PGN A#25 (Live)	1	2474	0.0992	10.077
A	PGN A#26 (Live)	3	1443,927,1134	0.2604	11.521
A	PGN A#27 (Live)	4	1546,1340,1030,1443	0.3652	10.952
A	PGN A#28 (Live)	2	1237,2164	0.2237	8.940
A	PGN A#29 (Live)	3	1340,2164,2164	0.3338	8.987
A	PGN A#30 (Live)	4	1340,927,1443,1546	0.4055	9.865
A	PGN A#31 (Live)	0		0.0000	8.551
A	PGN A#32 (Live)	2	1237,1855	0.1740	11.491
A	PGN A#33 (Live)	2	1958,1030	0.1927	10.376
A	PGN A#34 (Live)	0		0.0000	9.658
A	PGN A#35 (Live)	2	2164,1030	0.2011	9.945
A	PGN A#36 (Live)	2	1443,1030	0.1975	10.127
A	PGN A#37 (Live)	3	2164,1237,1546	0.3032	9.893
A	PGN A#38 (Live)	2	2061,1134	0.1795	11.140
A	PGN A#39 (Live)	4	1134,1855,2268,1030	0.3832	10.439
A	PGN A#40 (Live)	2	1649,1030	0.1952	10.245
A	PGN A#41 (Live)	3	2371,1340,1546	0.3358	8.935
A	PGN A#42 (Live)	1	1134	0.1025	9.758
A	PGN A#43 (Live)	4	1958,1030,2268,1237	0.4140	9.661
A	PGN A#44 (Live)	8	1649,1443,1958,1237,2268,1340,1649,1237	0.7624	10.494
A	PGN A#45 (Live)	2	1546,3195	0.2306	8.673
A	PGN A#46 (Live)	2	1649,1546	0.2310	8.658

Location	Sample ID	Microplastic Count	Size (μm)	Concentration (MP/g)	Weight (g)
A	PGN A#47 (Live)	2	1855,2164	0.2055	9.734
A	PGN A#48 (Live)	3	1649,1030,1752	0.3229	9.291
A	PGN A#49 (Live)	1	1443	0.1231	8.124
A	PGN A#50 (Live)	2	1752,1649	0.2209	9.055
B	PGN B#1 (Live)	1	2680	0.1004	9.956
B	PGN B#2 (Live)	0		0.0000	10.441
B	PGN B#3 (Live)	0		0.0000	9.368
B	PGN B#4 (Live)	0		0.0000	9.896
B	PGN B#5 (Live)	1	927	0.1173	8.523
B	PGN B#6 (Live)	0		0.0000	10.922
B	PGN B#7 (Live)	1	1855	0.1151	8.685
B	PGN B#8 (Live)	0		0.0000	8.161
B	PGN B#9 (Live)	0		0.0000	8.068
B	PGN B#10 (Live)	2	9072,8247	0.1928	10.372
B	PGN B#11 (Live)	1	1340	0.1007	9.935
B	PGN B#12 (Live)	0		0.0000	11.153
B	PGN B#13 (Live)	1	1546	0.1015	9.854
B	PGN B#14 (Live)	0		0.0000	9.427
B	PGN B#15 (Live)	0		0.0000	11.356
B	PGN B#16 (Live)	0		0.0000	9.929
B	PGN B#17 (Live)	0		0.0000	11.239
B	PGN B#18 (Live)	0		0.0000	10.244
B	PGN B#19 (Live)	0		0.0000	9.985
B	PGN B#20 (Live)	0		0.0000	9.055
B	PGN B#21 (Live)	1	3608	0.1069	9.352
B	PGN B#22 (Live)	0		0.0000	7.968
B	PGN B#23 (Live)	0		0.0000	8.534
B	PGN B#24 (Live)	1	1030	0.0999	10.012
B	PGN B#25 (Live)	2	1237,2680	0.2415	8.280
B	PGN B#26 (Live)	1	1030	0.1118	8.943
B	PGN B#27 (Live)	0		0.0000	8.705
B	PGN B#28 (Live)	0		0.0000	8.466
B	PGN B#29 (Live)	0		0.0000	11.609
B	PGN B#30 (Live)	0		0.0000	9.215
B	PGN B#31 (Live)	0		0.0000	9.561
B	PGN B#32 (Live)	0		0.0000	8.882
B	PGN B#33 (Live)	0		0.0000	9.028
B	PGN B#34 (Live)	2	1443,1134	0.2221	9.004
B	PGN B#35 (Live)	0		0.0000	9.675
B	PGN B#36 (Live)	0		0.0000	8.056
B	PGN B#37 (Live)	0		0.0000	9.431
B	PGN B#38 (Live)	2	1340,2268	0.2293	8.721
B	PGN B#39 (Live)	0		0.0000	9.309
B	PGN B#40 (Live)	0		0.0000	9.539
B	PGN B#41 (Live)	2	927,1134	0.2301	8.693
B	PGN B#42 (Live)	1	2061	0.1005	9.948
B	PGN B#43 (Live)	0		0.0000	8.707

Location	Sample ID	Microplastic Count	Size (μm)	Concentration (MP/g)	Weight (g)
B	PGN B#44 (Live)	2	1030,1546	0.2077	9.631
B	PGN B#45 (Live)	0		0.0000	8.954
B	PGN B#46 (Live)	1	1134	0.0932	10.729
B	PGN B#47 (Live)	0		0.0000	10.371
B	PGN B#48 (Live)	1	1958	0.0937	10.677
B	PGN B#49 (Live)	0		0.0000	9.232
B	PGN B#50 (Live)	0		0.0000	9.688

Table L.2 Raw Data for Mindoro Samples

Location	Sample ID	Microplastic Count	Size (µm)	Concentration (MP/g)	Weight (g)
A	MND #A01 (Live)	4	3505,1752,1237,1443	0.3870	10.335
A	MND #A02 (Live)	2	1546, 1958	0.2064	9.691
A	MND #A03 (Live)	1	1443	0.1084	9.226
A	MND #A04 (Live)	2	927,1134	0.1901	10.519
A	MND #A05 (Live)	1	1546	0.0944	10.595
A	MND #A06 (Live)	6	1855,1649,1443,1237,1134,1237	0.6440	9.316
A	MND #A07 (Live)	1	1134	0.1013	9.876
A	MND #A08 (Live)	1	1649	0.0943	10.603
A	MND #A09 (Live)	4	1958,1443,824,1649	0.4149	9.641
A	MND #A10 (Live)	3	2061,1237,1443	0.2874	10.439
A	MND #A11 (Live)	2	1649,1030	0.2223	8.998
A	MND #A12 (Live)	1	1237	0.1150	8.692
A	MND #A13 (Live)	2	1237,1134	0.1952	10.245
A	MND #A14 (Live)	2	1855,927	0.1907	10.485
A	MND #A15 (Live)	4	1443,1134,1443,2061	0.3976	10.060
A	MND #A16 (Live)	3	2061,824,927	0.2857	10.502
A	MND #A17 (Live)	1	1134	0.1018	9.822
A	MND #A18 (Live)	1	1340	0.0941	10.623
A	MND #A19 (Live)	3	1855,1546,1030	0.2872	10.446
A	MND #A20 (Live)	3	1030,1030,1752	0.3226	9.299
A	MND #A21 (Live)	2	1340,927	0.2023	9.888
A	MND #A22 (Live)	3	1030,1237,1649	0.2909	10.312
A	MND #A23 (Live)	4	1958,1134,1030,1134	0.4042	9.895
A	MND #A24 (Live)	3	1030,927,1546	0.2914	10.293
A	MND #A25 (Live)	3	1030,1030,824	0.2826	10.615
A	MND #A26 (Live)	5	2061,1443,1237,1340,1443	0.4693	10.653
A	MND #A27 (Live)	4	1237,1649,1443,1030	0.4074	9.818
A	MND #A28 (Live)	4	1855,1546,1340,1340	0.3947	10.133
A	MND #A29 (Live)	5	1340,1649,1855,1649,2268	0.4850	10.310
A	MND #A30 (Live)	5	2061,3092,1134,1443,1030	0.5474	9.134
A	MND #A31 (Live)	7	1855,1237,1030,1855,1958,1030,1340	0.7689	9.104
A	MND #A32 (Live)	5	1340,1443,1237,1237,927	0.4594	10.884
A	MND A#33 (Live)	3	1237,1855,3814	0.3046	9.849
A	MND A#34 (Live)	2	721,1443	0.2051	9.750
A	MND A#35 (Live)	2	1546,1340	0.1872	10.682
A	MND A#36 (Live)	3	1134,1649,1546	0.2824	10.623
A	MND A#37 (Live)	3	2577,1134,1443	0.3128	9.590
A	MND A#38 (Live)	4	2268,1958,927,1134	0.4137	9.668
A	MND A#39 (Live)	1	1752	0.1097	9.119
A	MND A#40 (Live)	2	1855,1134	0.2395	8.351
A	MND A#41 (Live)	2	927,1030	0.2026	9.870
A	MND A#42 (Live)	1	1958	0.1240	8.066
A	MND A#43 (Live)	4	1134,927,927,1134	0.4851	8.246
A	MND A#44 (Live)	3	927,1649,1340	0.3620	8.288
A	MND A#45 (Live)	1	1443	0.1021	9.792
A	MND A#46 (Live)	1	2474	0.0955	10.475
A	MND A#47 (Live)	2	1958,1546	0.2089	9.576
A	MND A#48 (Live)	3	1237,1752,1649	0.3177	9.444
A	MND A#49 (Live)	3	1134,1030,1030	0.3075	9.757
A	MND A#50 (Live)	2	1443,1237	0.2240	8.930
B	MND B#1 (Live)	1	927	0.0932	10.732
B	MND B#2 (Live)	1	927	0.1102	9.071
B	MND B#3 (Live)	1	1340	0.1030	9.706
B	MND B#4 (Live)	2	2061,1134	0.1673	11.952
B	MND B#5 (Live)	0		0.0000	8.604
B	MND B#6 (Live)	1	1752	0.1119	8.934
B	MND B#7 (Live)	0		0.0000	8.203

Location	Sample ID	Microplastic Count	Size (µm)	Concentration (MP/g)	Weight (g)
B	MND B#8 (Live)	2	1649,1237	0.2292	8.725
B	MND B#9 (Live)	0		0.0000	8.322
B	MND B#10 (Live)	1	1237	0.1226	8.158
B	MND B#11 (Live)	1	1237	0.1099	9.102
B	MND B#12 (Live)	3	1443,1237,1340	0.3696	8.117
B	MND B#13 (Live)	1	1134	0.1077	9.287
B	MND B#14 (Live)	1	1443	0.1119	8.933
B	MND B#15 (Live)	0		0.0000	8.925
B	MND B#16 (Live)	2	2577,1340	0.1960	10.205
B	MND B#17 (Live)	1	1134	0.1126	8.877
B	MND B#18 (Live)	1	1134	0.1041	9.610
B	MND B#19 (Live)	1	1237	0.1067	9.376
B	MND B#20 (Live)	3	1443,1237,1340	0.2585	11.604
B	MND B#21 (Live)	1	2164	0.1058	9.450
B	MND B#22 (Live)	2	1443,1340	0.1923	10.399
B	MND B#23 (Live)	0	0	0.0000	10.368
B	MND B#24 (Live)	3	1134,1134,927	0.3039	9.870
B	MND B#25 (Live)	2	1030,1134	0.1775	11.268
B	MND B#26 (Live)	1	1340	0.0993	10.066
B	MND B#27 (Live)	1	2268	0.1035	9.664
B	MND B#28 (Live)	0	0	0.0000	10.111
B	MND B#29 (Live)	1	1546	0.0984	10.165
B	MND B#30 (Live)	1	1958	0.0922	10.851
B	MND B#31 (Live)	1	1443	0.1120	8.932
B	MND B#32 (Live)	0	0	0.0000	8.880
B	MND B#33 (Live)	1	2371	0.0863	11.583
B	MND B#34 (Live)	2	1237, 2268	0.1754	11.401
B	MND B#35 (Live)	2	1958, 2061	0.2040	9.804
B	MND B#36 (Live)	5	1237, 1752, 1958	0.5179	9.655
B	MND B#37 (Live)	1	1649	0.0856	11.688
B	MND B#38 (Live)	0	0	0.0000	11.103
B	MND B#39 (Live)	3	1134, 2474, 5463	0.2707	11.082
B	MND B#40 (Live)	2	927, 1546	0.1800	11.114
B	MND B#41 (Live)	2	1546, 1855	0.1761	11.360
B	MND B#42 (Live)	4	1030, 1134, 1443, 1237	0.4460	8.969
B	MND B#43 (Live)	2	1134, 1340	0.2006	9.968
B	MND B#44 (Live)	2	1649, 2061	0.1999	10.003
B	MND B#45 (Live)	4	927, 1134, 1546, 1958	0.3601	11.107
B	MND B#46 (Live)	5	1030, 1237, 1546, 2061, 2577,	0.5047	9.907
B	MND B#47 (Live)	2	1134, 1443	0.2025	9.875
B	MND B#48 (Live)	1	1134	0.1216	8.226
B	MND B#49 (Live)	0	0	0.0000	10.120
B	MND B#50 (Live)	1	927	1.0936	10.679

Table L.3 Raw Data for Palengke Samples

Location	Sample ID	Microplastic Count	Size (μm)	Concentration (MP/g)	Weight (g)
A	PLK A#1 (Live)	2	1882	0.0731	7.331
A	PLK A#2 (Live)	0	0	0.0000	8.122
A	PLK A#3 (Live)	2	1764, 2117	0.0731	7.331
A	PLK A#4 (Live)	3	1411, 1647, 1764	0.3160	9.463
A	PLK A#5 (Live)	4	1058, 1294, 1411, 1529	0.5341	7.464
A	PLK A#6 (Live)	7	2577,1443,1958,1237,1030,1134,1443	0.7195	9.730
A	PLK A#7 (Live)	2	1958, 2268	0.2059	9.712
A	PLK A#8 (Live)	5	927, 1030, 1232, 2061, 2268	0.5375	9.302
A	PLK A#9 (Live)	2	1443,1443	0.2014	9.930
A	PLK A#10 (Live)	3	1649,3092,1340	0.2964	10.123
A	PLK A#11 (Live)	4	1340,1030,1443,927	0.5410	7.394
A	PLK A#12 (Live)	4	1030,1752,1546,1237	0.5595	7.150
A	PLK A#13 (Live)	1	1340	0.1176	8.505
A	PLK A#14 (Live)	2	1340,2061	0.2258	8.859
A	PLK A#15 (Live)	2	1134,1546	0.2117	9.448
A	PLK A#16 (Live)	3	1958,1340,1855	0.2391	12.546
A	PLK A#17 (Live)	3	1340,1752,1752	0.3137	9.562
A	PLK A#18 (Live)	2	1237,1134	0.2059	9.713
A	PLK A#19 (Live)	1	1752	0.1071	9.337
A	PLK A#20 (Live)	2	2886,1134	0.2102	9.515
A	PLK A#21 (Live)	7	1443,2061,1134,1134,1134,1030,1134	0.7550	9.272
A	PLK A#22 (Live)	3	2061,1443,1134	0.2919	10.276
A	PLK A#23 (Live)	2	1958,1237	0.2295	8.714
A	PLK A#24 (Live)	7	2061,1443,1030,1237,1030,1546,927	0.7980	8.771
A	PLK A#25 (Live)	4	1030,1649,927,1134	0.4279	9.347
A	PLK A#26 (Live)	6	2577,1649,2164,1649,1134,1030	0.7014	8.554
A	PLK A#27 (Live)	3	1546,1340,1134	0.3294	9.108
A	PLK A#28 (Live)	4	1340,927,1134,1340	0.3702	10.805
A	PLK A#29 (Live)	7	2061,1134,1134,1030,1134,1030,1237	0.7011	9.984
A	PLK A#30 (Live)	3	1134,1340,1546	0.3183	9.424
A	PLK A#31 (Live)	3	1958,1340,1752	0.3257	9.210
A	PLK A#32 (Live)	8	1546,1340,1237,2061,927,2268,1134,927	0.7859	10.179
A	PLK A#33 (Live)	8	1649,1030,1134,1649,1237,1134,1958,1237	0.7366	10.861
A	PLK A#34 (Live)	1	1030	0.0883	11.322
A	PLK A#35 (Live)	6	1030,1443,1134,927,1237,1855	0.6359	9.436
A	PLK A#36 (Live)	3	1030,1649,1443	0.3943	7.608
A	PLK A#37 (Live)	6	1443,1649,1443,1134,1546,1340	0.6689	8.970
A	PLK A#38 (Live)	3	1134,1546,824	0.3621	8.285
A	PLK A#39 (Live)	7	1134,1546,1237,927,2164,1546,2164	0.6635	10.550
A	PLK A#40 (Live)	4	2577,1134,2061,1237	0.4214	9.492
A	PLK A#41 (Live)	5	2061,1546,1134,1030,824	0.4982	10.037
A	PLK A#42 (Live)	7	2371,1752,2061,1855,1237,1752,1030	0.7419	9.436
A	PLK A#43 (Live)	5	1855,1443,1134,1134,1134	0.5025	9.950
A	PLK A#44 (Live)	5	2783,2061,2164,2577,2268	0.4817	10.380
A	PLK A#45 (Live)	9	1649,1340,1649,1443,1134,1752,1443,3092,1030	0.9697	9.281
A	PLK A#46 (Live)	5	2164,1546,1546,1237,1134	0.5279	9.471
A	PLK A#47 (Live)	9	1546,1340,1752,1855,1134,1134,1855,1237,3092	0.9446	9.528
A	PLK A#48 (Live)	4	1752,1030,1237,1134	0.4347	9.201
A	PLK A#49 (Live)	5	1958,1340,1855,2164,3092	0.5640	8.866
A	PLK A#50 (Live)	3	1237,1134,1340	0.2721	11.026
B	PLK B#1 (Live)	1	1546	0.1034	9.671
B	PLK B#2 (Live)	1	7938	0.1061	9.424
B	PLK B#3 (Live)	0		0.0000	8.858
B	PLK B#4 (Live)	1	1134	0.0991	10.094
B	PLK B#5 (Live)	0		0.0000	8.206
B	PLK B#6 (Live)	3	1546,1030,1443	0.3209	9.349
B	PLK B#7 (Live)	2	2268,927	0.2198	9.100

Location	Sample ID	Microplastic Count	Size (µm)	Concentration (MP/g)	Weight (g)
B	PLK B#8 (Live)	0		0.0000	9.617
B	PLK B#9 (Live)	1	1649	0.1139	8.782
B	PLK B#10 (Live)	0		0.0000	8.766
B	PLK B#11 (Live)	0		0.0000	8.797
B	PLK B#12 (Live)	2	1340,1340	0.2231	8.963
B	PLK B#13 (Live)	0		0.0000	9.655
B	PLK B#14 (Live)	0		0.0000	10.814
B	PLK B#15 (Live)	0		0.0000	11.455
B	PLK B#16 (Live)	1	1030	0.1036	9.655
B	PLK B#17 (Live)	1	1958	0.1163	8.600
B	PLK B#18 (Live)	0		0.0000	8.780
B	PLK B#19 (Live)	0		0.0000	9.020
B	PLK B#20 (Live)	1	1030	0.1043	9.583
B	PLK B#21 (Live)	1	1237	0.1029	9.722
B	PLK B#22 (Live)	0		0.0000	9.437
B	PLK B#23 (Live)	0		0.0000	10.778
B	PLK B#24 (Live)	1	1134	0.1043	9.589
B	PLK B#25 (Live)	0		0.0000	10.323
B	PLK B#26 (Live)	2	2886,1237	0.1939	10.312
B	PLK B#27 (Live)	1	1134	0.1029	9.716
B	PLK B#28 (Live)	3	1855,1030,1134	0.2922	10.266
B	PLK B#29 (Live)	3	1237,2061,1855	0.2824	10.624
B	PLK B#30 (Live)	0		0.0000	9.660
B	PLK B#31 (Live)	3	2164,1237,1237	0.3399	8.827
B	PLK B#32 (Live)	1	1237	0.1222	8.185
B	PLK B#33 (Live)	0		0.0000	9.416
B	PLK B#34 (Live)	2	1134,1649	0.2414	8.285
B	PLK B#35 (Live)	0		0.0000	7.057
B	PLK B#36 (Live)	0		0.0000	8.021
B	PLK B#37 (Live)	1	1340	0.1296	7.718
B	PLK B#38 (Live)	1	1134	0.1294	7.731
B	PLK B#39 (Live)	1	1546	0.1348	7.416
B	PLK B#40 (Live)	1	1958	0.1236	8.091
B	PLK B#41 (Live)	2	1546,824	0.2720	7.354
B	PLK B#42 (Live)	1	2164	0.1328	7.532
B	PLK B#43 (Live)	3	1649,1237,1030	0.3765	7.968
B	PLK B#44 (Live)	1	1134	0.1231	8.123
B	PLK B#45 (Live)	2	1134,2061	0.2289	8.737
B	PLK B#46 (Live)	1	2371	0.1192	8.392
B	PLK B#47 (Live)	2	2577,927	0.2308	8.667
B	PLK B#48 (Live)	2	1752,1134	0.2281	8.770
B	PLK B#49 (Live)	2	927,1752	0.2471	8.093
B	PLK B#50 (Live)	3	1030,1237,2268	0.3689	8.133

Appendix M. Graphical Representation of Data

Mean Microplastic Count

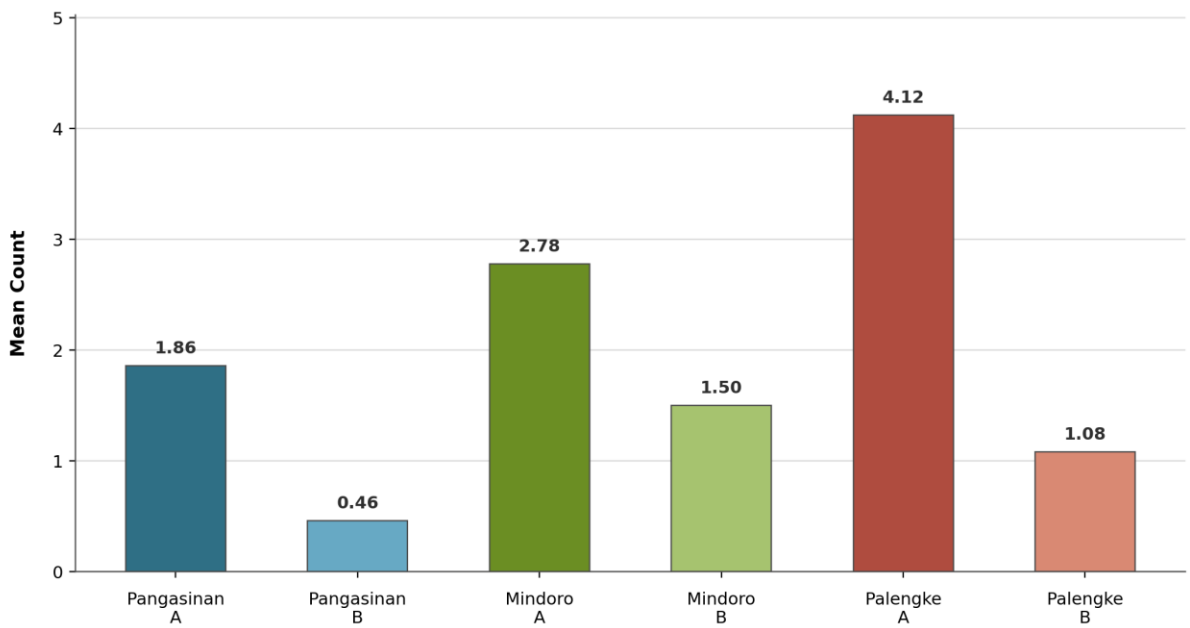


Figure 4.16 Mean Microplastic Count

Mean Microplastic Size

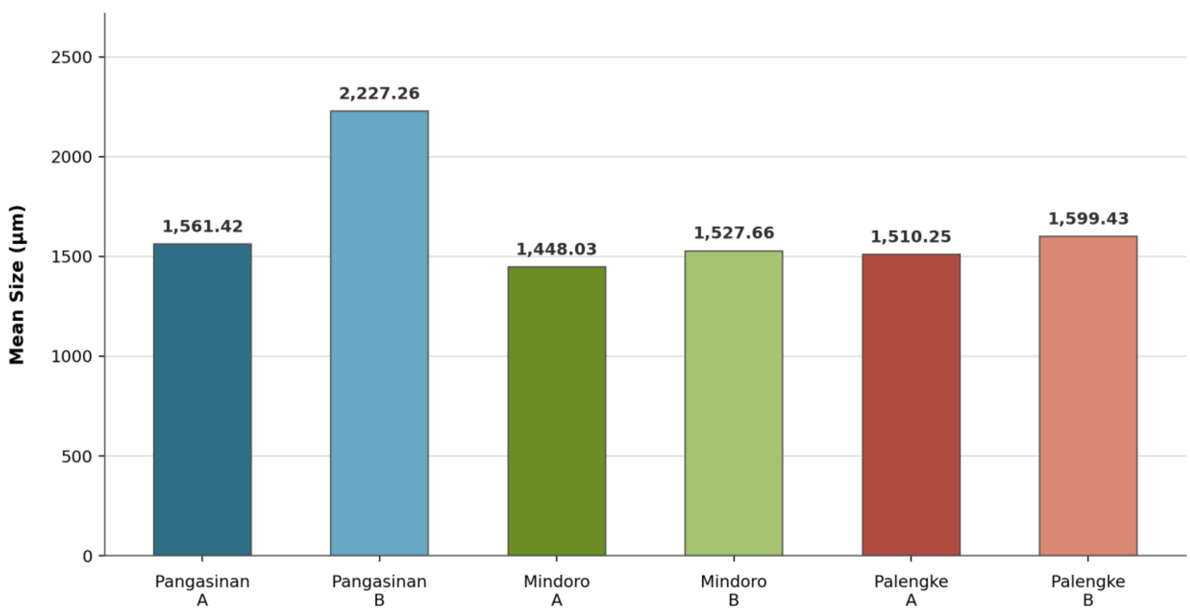


Figure 4.17 Mean Microplastic Size

Mean Microplastic Concentration

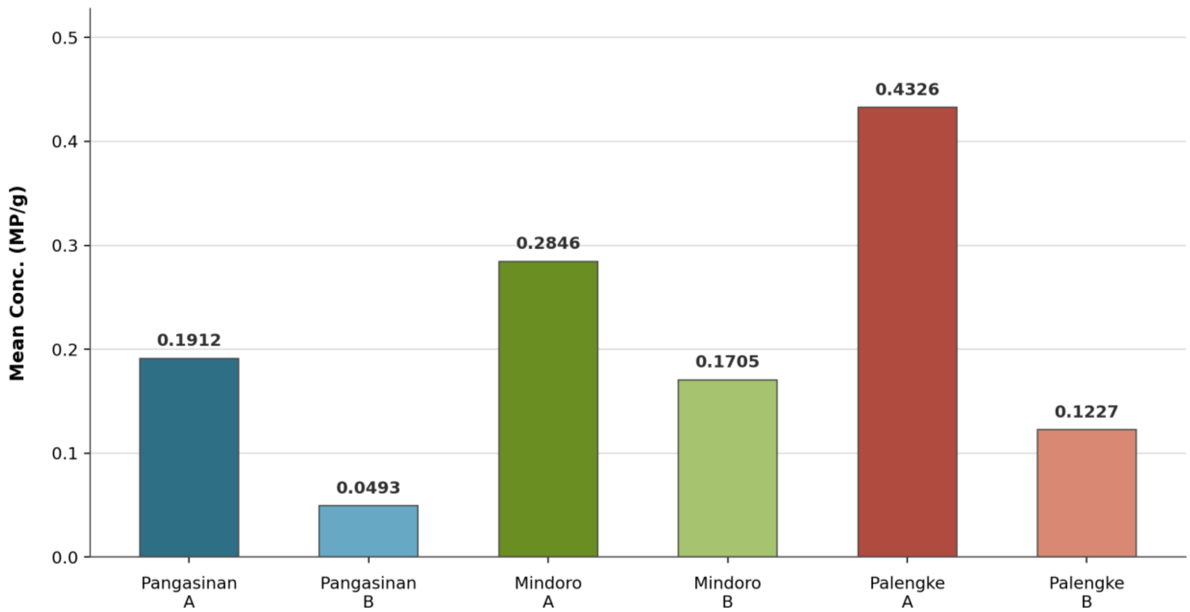


Figure 4.18 Mean Microplastic Concentration

ISO 25010 Evaluation for Functionality

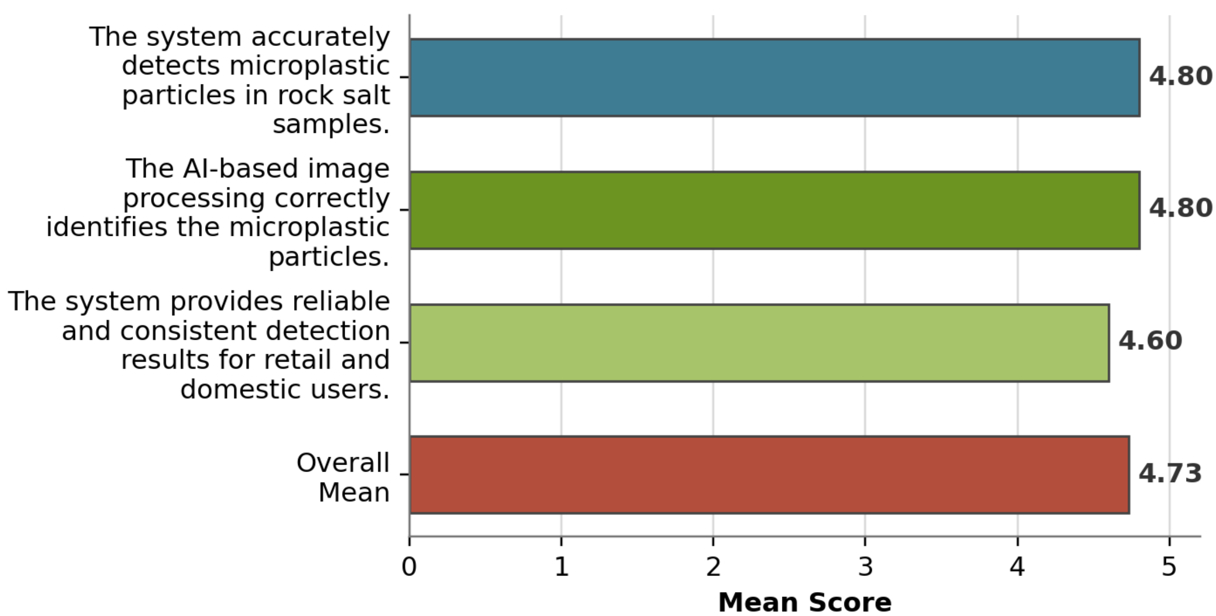


Figure 4.19 ISO 25010 Evaluation for Functionality

ISO 25010 Evaluation for Usability

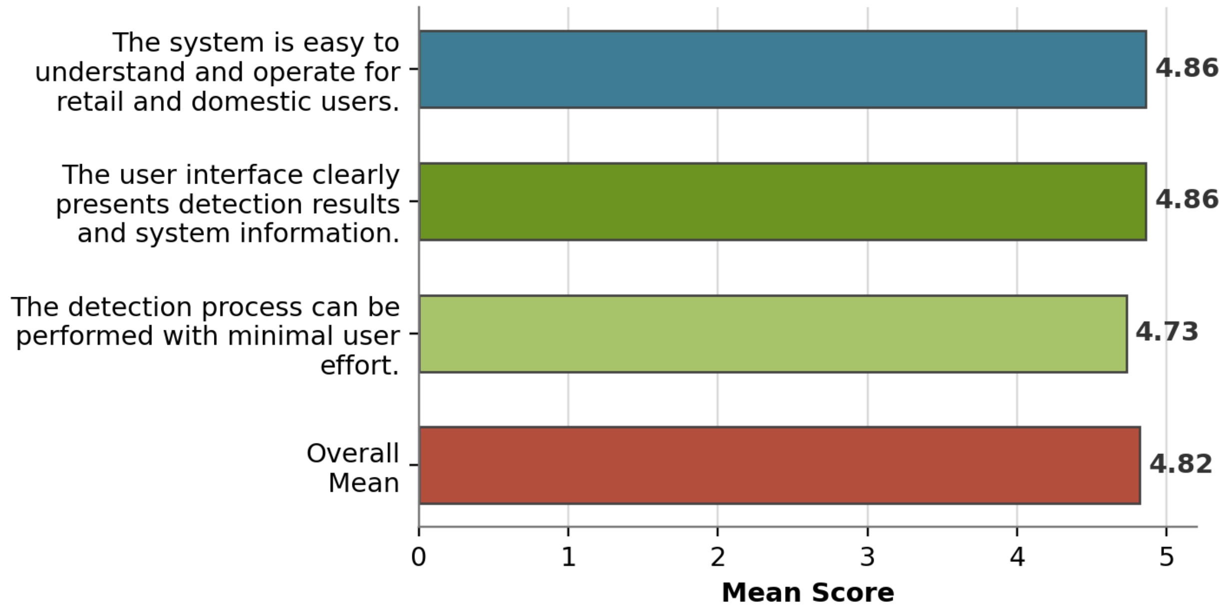


Figure 4.20 ISO 25010 Evaluation for Usability

ISO 25010 Evaluation for Performance Efficiency

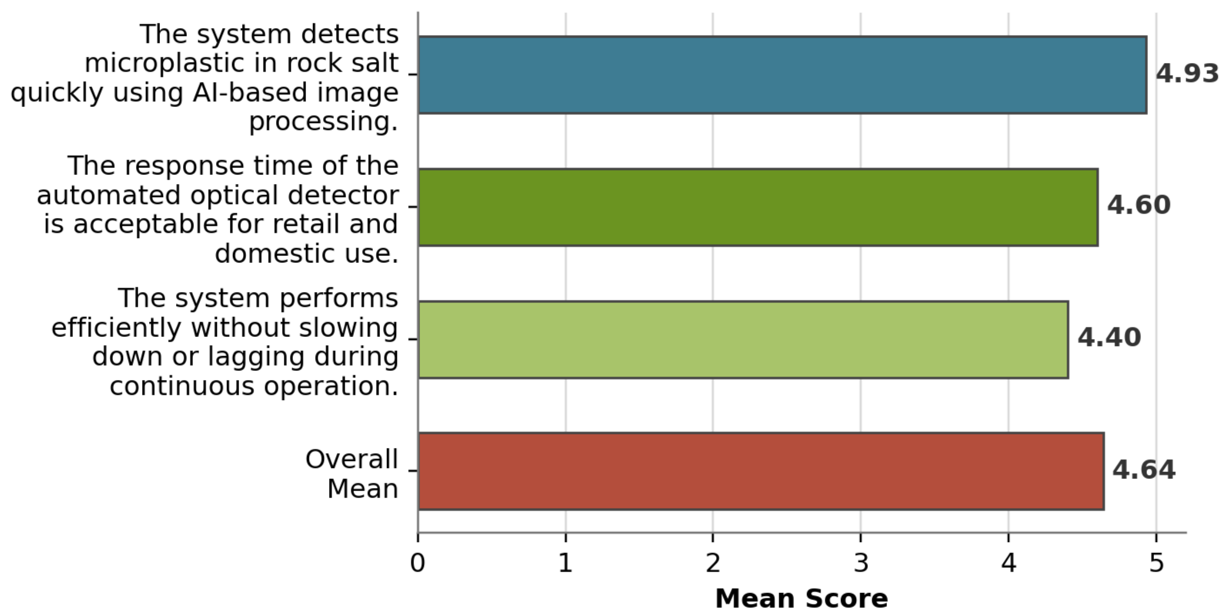


Figure 4.21 ISO 25010 Evaluation for Performance Efficiency

ISO 25010 Evaluation for Safety and Maintenance

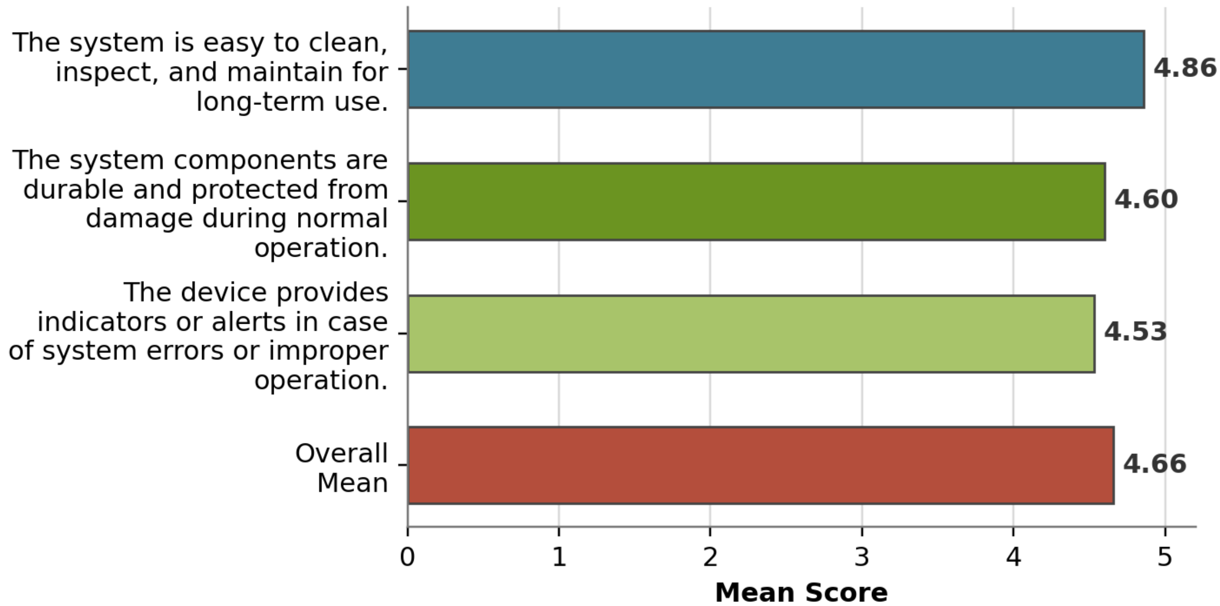


Figure 4.22 ISO 25010 Evaluation for Safety and Maintenance

Appendix N. Laboratory Analysis

Microplastics Research and Training Facility

HOLCIM

Report No. MRTF-2026-0012
Client Name: Euri A. Melendez
Service Requested: Microplastic Counting and Morphological Classification
Date Issued: 29 April 2026

REPORT OF ANALYTICAL RESULTS

A. Executive Summary

This report presents the results of microplastic morphological analysis performed on the submitted samples.

A total of 28 microplastic particles were identified across nine samples. Based on total particle counts, fragment-type particles were the dominant morphology (75.0%), followed by lines/fibers (25.0%) and pellets/beads (0.0%). Sample exhibited the highest total particle count RS-HH-SALT (10 particles), suggesting elevated microplastic presence relative to the other submitted samples.

B. Scope of Analysis

The analysis included:

- Nile red staining and Fluorescence-assisted visualization
- AI-assisted particle counting and morphological classification

C. Results of the Analysis

Table 1. Microplastic count by morphology per sample.

Sample no.	Sample ID	Fragment	Line/Fiber	Pellet/bead	Total
1	RS-PNG-RAW	2	1	0	3
2	RS-MND-RAW	6	0	0	6
3	RS-MRKN-PM	3	2	0	5
4	RS-HH-SALT	7	3	0	10
5	RS-RDM-SALT	3	1	0	4
all		21	7	0	28

**Samples stained as received; values reported reflect digitally enhanced images.*

Table 2. Overall Morphological Distribution.

Morphology	Count	Percentage (%)
Fragment	21	75.0%
Line/fiber	7	25.0%
Pellet/bead	0	0.0%

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Total	28	100%

Figure 1. Representative fluorescence image of microplastics in Sample RS-PNG-RAW.

Figure 2. Representative fluorescence image of microplastics in Sample RS-MND-RAW.

Figure 3. Representative fluorescence image of microplastics in Sample RS-MRKN-PM.

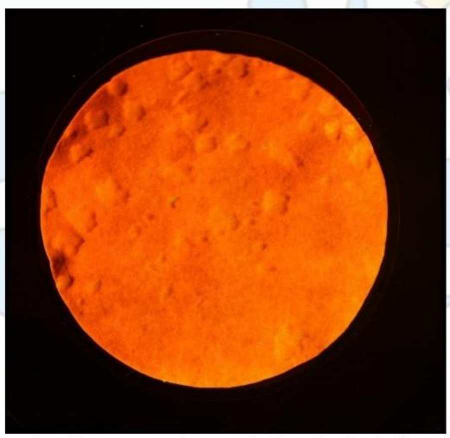

Figure 4. Representative fluorescence image of microplastics in Sample RS-HH-SALT.

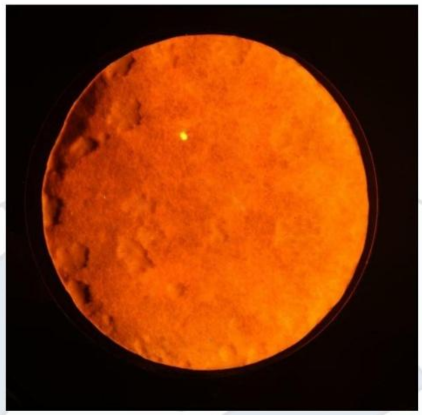

Figure 5. Representative fluorescence image of microplastics in Sample RS-RDM-SALT.

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Samples underwent Nile red staining and visualized under blue light. Digital images were then obtained and analyzed for particle count and morphological classification using an in-house AI software. Generated images and raw files can be accessed in the provided raw file folder.

It is important to note that performing AI-assisted particle counting and morphological classification are still subject to the inherent limitations of the software, including potential inaccuracies in particle detection, counting, and classification. As such, the results should be interpreted with consideration of these technical constraints.

D. Attachments
   a. Record of Sample Receipt

Figure 6. Receipt of samples.

Comments:

Samples on filter paper were placed into large petri dishes.

b. Raw data

All raw image files, AI-generated particle count outputs are provided in the accompanying digital folder:

T2-Melendez_MRTF_RawData_Apr2026

E. Recommended Citation and Acknowledgment

If results from this report are used in publications, presentations, or technical documents, please cite as:

UP MSI - IAEA - HOLCIM - DOST Microplastics Research and Training Facility (MRTF). (YYYY). Report of Analytical Results. Report No. MRTF-YYYY-XXX.

For acknowledgment sections (sample):

The authors acknowledge the UP MSI - IAEA - HOLCIM - DOST Microplastics Research and Training Facility (MRTF) for conducting Nile red-based microplastic analysis and morphological classification.

Analyzed by:

QUEENIE ROSE C. BAGUNAS
MRTF Lab Manager and Analyst

The results of the analyses are strictly for research purposes only as there is currently no available standard method for microplastic extraction and analysis in the Philippines. The MRTF assumes no liability for any losses, damages, or consequences arising from the interpretation or use of the results stated in the Report of Analytical Results, nor for any legal purpose for which the report may be used.

Appendix O. Computation of Detection Accuracy of AI

Formula used:

$$Percentage\ Error = \left(\frac{AI\ count - Laboratory\ Result}{Laboratory\ Result}\right) \times 100$$

$$Mean\ AI\ Count = \left(\frac{Trial\ 1 + Trial\ 2 + Trial\ 3}{Total\ number\ of\ trials}\right)$$

Pangasinan

$$Mean\ AI\ Count = \left(\frac{4 + 2 + 2}{3}\right)$$

$$Mean\ AI\ Count = 2.6667$$

$$Percentage\ Error = \left(\frac{2.6667 - 3}{3}\right) \times 100$$

$$Percentage\ Error = 11.11\%$$

$$Accuracy\ Percentage = 100\% - 11.11\%$$

$$Accuracy\ Percentage = 88.89\%$$

Mindoro

$$Mean\ AI\ Count = \left(\frac{3 + 4 + 3}{3}\right)$$

$$Mean\ AI\ Count = 3.3333$$

$$Percentage\ Error = \left(\frac{3.3333 - 6}{6}\right) \times 100$$

$$Percentage\ Error = 44.44\%$$

$$Accuracy\ Percentage = 100\% - 44.44\%$$

$$Accuracy\ Percentage = 56.56\%$$

Marikina Public Market

$$Mean\ AI\ Count = \left(\frac{3+4+3}{3}\right)$$

$$Mean\ AI\ Count = 3.3333$$

$$Percentage\ Error = \left(\frac{3.3333-5}{5}\right) \times 100$$

$$Percentage\ Error = 33.33\%$$

$$Accuracy\ Percentage = 100\% - \ 33.33\%$$

$$Accuracy\ Percentage = 66.67\%$$

Household Rock Salt

$$Mean\ AI\ Count = \left(\frac{7+5+4}{3}\right)$$

$$Mean\ AI\ Count = 5.3333$$

$$Percentage\ Error = \left(\frac{5.3333-10}{10}\right) \times 100$$

$$Percentage\ Error = 46.6\%$$

$$Accuracy\ Percentage = 100\% - \ 46.67\%$$

$$Accuracy\ Percentage = 53.33\%$$

Random

$$Mean\ AI\ Count = \left(\frac{3+3+3}{3}\right)$$

$$Mean\ AI\ Count = 3.0000$$

$$Percentage\ Error = \left(\frac{3.0000-4}{4}\right) \times 100$$

$$Percentage\ Error = 25\%$$

$$Accuracy\ Percentage = 100\% - \ 25\%$$

$$Accuracy\ Percentage = 75\%$$

Mean Accuracy

$$Mean\ AI\ Accuracy = \left(\frac{88.89 + 55.56 + 66.67 + 53.33 + 75.00}{5}\right)$$

$$Mean\ AI\ Accuracy = \left(\frac{339.45}{5}\right)$$

$$Mean\ AI\ Accuracy = 67.89\%$$

Appendix P. Business Model

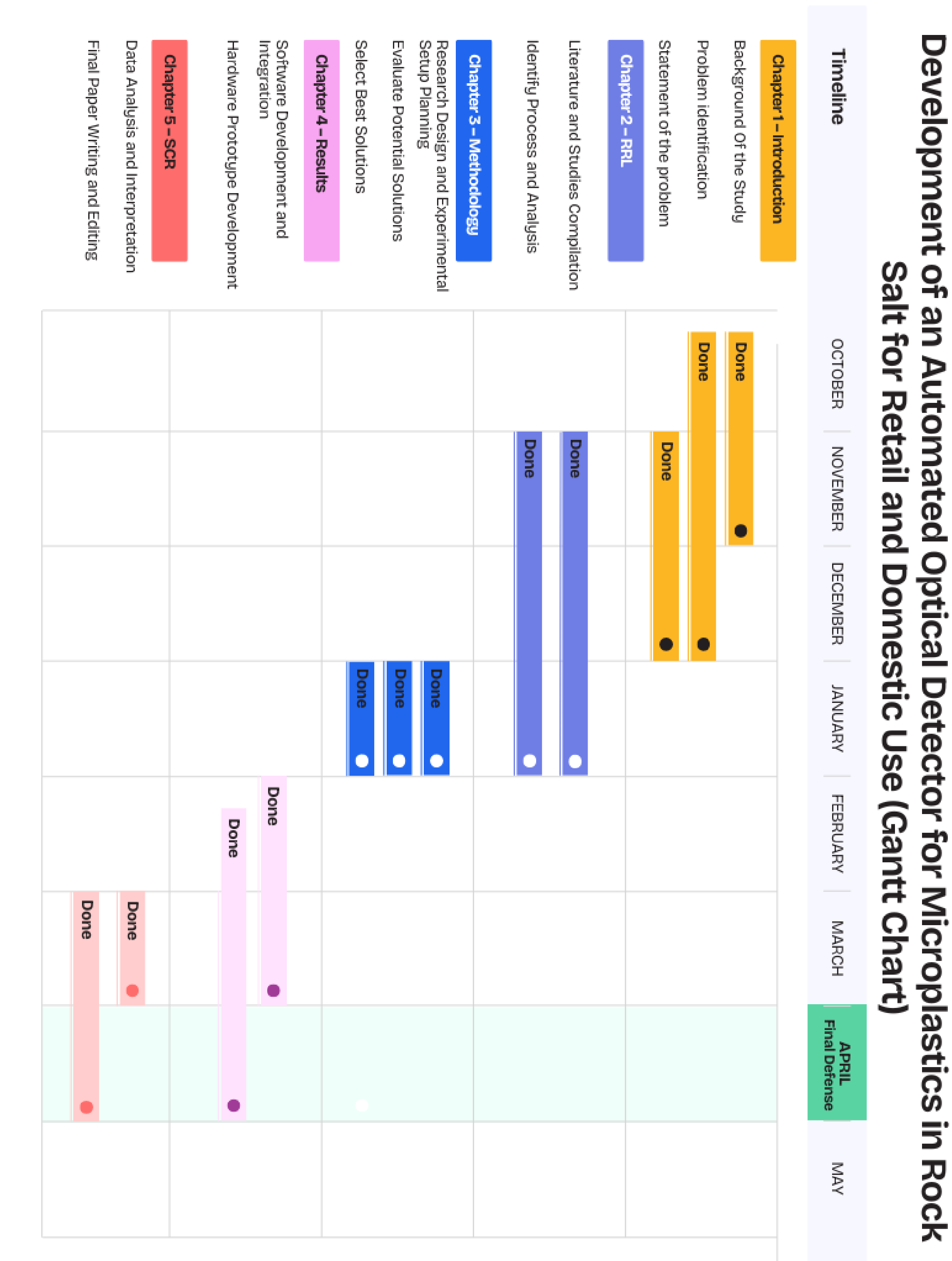
Appendix Q. Source Code

```
import os
import subprocess
import sys
import tkinter as tk
import warnings

from config import SYNC_API_ENABLED
from utils.error_handler import init_error_handler
from utils.gpio_controller import cleanup_gpio
from utils.load_cell import cleanup_load_cell
from utils.theme import Colors, Fonts, style_ttk_widgets

warnings.filterwarnings("ignore", category=UserWarning)

class App(tk.Tk):
    def __init__(self):
        super().__init__()

        # Ensure clean exit
        self.protocol("WM_DELETE_WINDOW", self.on_close)

        # Initialize systems
        init_error_handler(self)

        # Global state
        self.current_test = None
        self.pending_test = None
        self.current_screen_name = None
        self.sync_api_server = None
        self._loading_tap_ready = False

        self.title("MICROTECT")
        self.geometry("800x480")
        self.configure(bg=Colors.BG_DARK)

        try:
            screen_w = self.winfo_screenwidth()
            screen_h = self.winfo_screenheight()
            if screen_w <= 800 or screen_h <= 480:
                # Keep Tk DPI scaling predictable on the Pi 7-inch LCD.
                self.tk.call("tk", "scaling", 1.0)
        except Exception:
            pass
```

```
        style_ttk_widgets()

        if SYNC_API_ENABLED:
            try:
                from utils.sync_api_server import MicrotectSyncApiServer

                self.sync_api_server = MicrotectSyncApiServer()
                self.sync_api_server.start()
            except Exception as error:
                print(f"Failed to start sync API: {error}")

        # True fullscreen, no title bar, no taskbar visible
        self.attributes("-fullscreen", True)

        # Hide the Pi's taskbar (lxpanel) if running on Linux
        self._hide_taskbar()

        self.container = tk.Frame(self, bg=Colors.BG_DARK)
        self.container.pack(fill="both", expand=True)

        # Required: grid weights so stacked screens expand to fill container
        self.container.grid_rowconfigure(0, weight=1)
        self.container.grid_columnconfigure(0, weight=1)

        self.screens = {}

        # Setup built-in on-screen keyboard BEFORE screens load
        try:
            from utils.keyboard_controller import setup_keyboard

            setup_keyboard(self)
        except Exception:
            pass

        # Show loading screen first, then init screens in background
        self._show_loading_screen()
        self._build_temp_overlay()

    def _build_temp_overlay(self):
        """Create a small, global overlay for Raspberry Pi temperature."""
        self.temp_label = tk.Label(
            self,
            text="--°C",
            font=(Fonts.LABEL[0], 10, "bold"),
```

```
            bg=Colors.BG_DARK,
            fg=Colors.TEXT_MUTED,
            padx=4,
            pady=2,
        )
        self.temp_label.place(relx=1.0, rely=0.0, anchor="ne", x=-10, y=10)
        self.temp_label.lift()
        self._update_temp_loop()

    def _update_temp_loop(self):
        try:
            with open("/sys/class/thermal/thermal_zone0/temp", "r", encoding="utf-8") as handle:
                temp_c = int(handle.read().strip()) / 1000.0

            # Pi 5 soft-throttles at 80°C, hard-throttles at 85°C
            if temp_c >= 80:
                fg_color = Colors.ERROR
            elif temp_c >= 70:
                fg_color = Colors.WARNING
            else:
                fg_color = Colors.TEXT_MUTED

            self.temp_label.config(text=f"{temp_c:.1f}°C", fg=fg_color)
        except Exception:
            self.temp_label.config(text="--°C", fg=Colors.TEXT_MUTED)

        if hasattr(self, "temp_label") and self.temp_label.winfo_exists():
            self.temp_label.lift()

        self.after(5000, self._update_temp_loop)

    def _show_loading_screen(self):
        """Display a loading/splash screen while initializing."""
        self.loading_frame = tk.Frame(self, bg=Colors.HERO)
        self.loading_frame.place(relx=0, rely=0, relwidth=1, relheight=1)

center = tk.Frame(self.loading_frame, bg=Colors.HERO)
```

Appendix R. Gantt Chart



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2016 – 2020	**Junior High School**
	Silangan National High School
2010 – 2016	**Elementary School**
	Silangan Elementary School

SKILLS

Basic Embedded Systems Programming

Basic Programming (Python, C++ for Arduino)

Team Collaboration

INTEREST

Prototype development

Artificial Intelligence

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

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2016 – 2020	**Junior High School**
	Canumay National High school
2010 – 2016	**Elementary School**
	San Ysiro Elementary School

SKILLS

Multitasking

Flexibility

Technical Knowledge

INTEREST

Programming

Problem Solving

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

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	Marikina Polytechnic College
2020 – 2022	**Senior High School**
	St. John's Wort Integrated School
2016 – 2020	**Junior High School**
	Dela Paz National High School
2010 – 2016	**Elementary School**
	Dela Paz Elementary School

SKILLS

Reliable

Easy to adapt

Strong Communication Skills

INTEREST

Programming

Problem Solving

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

GABRIEL NICOLAI S. ANTONIO

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EDUCATION

2022 – 2026	**Bachelor of Science in Electronics Engineering**
	Marikina Polytechnic College
2020 – 2022	**Senior High School**
	Pamantasan ng Lungsod ng Marikina San Roque
2016 – 2020	**Junior High School**
	Tanong High School
2010 – 2016	**Elementary School**
	San Nicolas Central School Nueva Ecija

SKILLS

Multitasking

Flexibility

Technical Knowledge

INTEREST

Programming

Problem Solving

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

JASMINE R. OBJA-AN

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EDUCATION

2022 – 2026	**Bachelor of Science in Electronics Engineering** Marikina Polytechnic College
2019 – 2021	**Senior High School** Unida Christian Colleges
2015 – 2019	**Junior High School** Gen. Licerio Topacio National High School
2012 – 2015	**Elementary School** Pasong Buaya II Elementary School

SKILLS

Multitasking

Flexibility

Technical Knowledge

INTEREST

Programming

Problem Solving

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

ARNEL JOSHUA DEAN OBERO

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EDUCATION

2022 – 2026	**Bachelor of Science in Electronics Engineering**
	Marikina Polytechnic College
2019 – 2021	**Senior High School**
	Marikina High School
2015 – 2019	**Junior High School**
	Marikina High School
2009 – 2015	**Elementary School**
	SSS Village Elementary School

SKILLS

Able to work long hours while on feet

Flexibility

INTEREST

Programming

Problem Solving

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

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